

Lesson Summary

When a vehicle is traveling at a constant speed, there is a proportional relationship between the time traveled and the distance traveled. This is true for any person, animal, or object traveling at a constant speed.

For example, imagine someone running 5 meters per second. The table shows the distance they travel over different periods of time. The table also shows their speed, in meters per second.

The last row in the table shows that we can multiply the amount of time, t , with the constant of proportionality, 5, to determine the distance traveled, d .

The equation $d = 5t$ represents this relationship.

Time (sec)	Distance Traveled (m)	Speed (m per sec)
1	5	5
2	10	5
3	15	5
7	35	5
t	$5t$	5

Things to Remember:

Lesson Practice

7.2.05

Name: _____ Date: _____ Period: _____

Problems 1–3: A performer expects to sell 5,000 tickets for an upcoming concert. They plan to make \$311,000 from the sales of these tickets. Assume that all tickets have the same price.

1. What is the price for one ticket? Show or explain your thinking.

\$62.20. Explanations vary. I divided \$311,000 by 5,000 to determine the price for one ticket.

2. Write an equation to represent the relationship between the number of tickets sold, x , and the total amount of money (in dollars) that they make, y .

$y = 62.2x$

3. If they make \$379,420, how many tickets have they sold?

6,100 tickets

Problems 4–5: A car is traveling on a highway at a constant speed. The equation that represents the distance traveled in miles, d , for t hours is $d = 65t$.

4. What does the value 65 represent in this situation?

Responses vary. The value 65 is the speed, or constant of proportionality. The car travels 65 miles every hour.

5. At this rate, how many miles will the car travel in 1.5 hours? Show or explain your thinking.

97.5 miles. Explanations vary.

$d = 65t$

$d = 65(1.5)$

$d = 97.5$

Problems 6–7: On its way from New York to San Diego, a plane flew at a constant speed over Pittsburgh, Saint Louis, Albuquerque, and Phoenix.



6. This table shows the flight time and distance traveled for each segment of the flight. Complete the table.

Segment	Time (hr)	Distance (mi)	Speed (mph)
Pittsburgh to Saint Louis	1	550	550
Saint Louis to Albuquerque	1.7	935	550
Albuquerque to Phoenix	0.6	330	550

7. Let t represent the time in hours and d represent the distance in miles. Write an equation that represents the distance traveled for t hours.

$d = 550t$ (or equivalent)

Lesson Practice

7.2.05

Name: _____ Date: _____ Period: _____

8. A train travels at a constant speed between Springfield and Chicago. The train travels $100\frac{1}{2}$ miles in $\frac{3}{4}$ hours. How far does the train travel in one hour at this same speed?

The train travels 134 miles in one hour.

9. Na'ilah is making a pitcher of a lemon-flavored sports drink. The drink mix container says to mix $\frac{1}{4}$ cups of powdered drink mix with 2 quarts of water. She prefers her sports drink to taste *more* lemony than the recipe on the container. Complete the equation to represent a mixture of cups of drink mix, c , and quarts of water, w , that would be *more* lemony than the original mixture.

Responses where the constant of proportionality is greater than $\frac{1}{8}$ are considered correct.

$$c = \frac{1}{5}w$$

Spiral Review

10. Select *all* the tables that represent a proportional relationship between x and y .

☒ A.

x	0	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$
y	0	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$

☐ B.

x	0	$\frac{1}{2}$	2	3
y	0	$\frac{1}{4}$	4	9

☒ C.

x	0	4	8	12
y	0	2	4	6

☒ D.

x	0	1	2	3
y	0	4	8	12

11. Is 4.5 a solution to the equation $1.5 + x = 6$? Circle one.

Yes

No

Reflection

- Put a heart next to the problem you found most interesting.
- Use this space to ask a question or share something you're proud of.

Lesson Summary

The structure of an equation representing the relationship between two quantities can tell us whether that relationship is proportional. An equation in the form of $y = kx$ has a constant of proportionality, k , which means it represents a proportional relationship.

Equations like $y = 3x + 1$ and $y = x^2$ do not have a constant of proportionality, so they do *not* represent proportional relationships.

Rewriting an equation in another form can help make a proportional relationship easier to see. For example, $y = \frac{x}{3}$ and $y = \frac{1}{3}x$ both represent the same proportional relationship.

Tables can help you determine whether an equation can be rewritten in the form of $y = kx$.

Things to Remember:

Lesson Practice

ACC6.5.06

Name: _____ Date: _____ Period: _____

Problems 1–2: The relationship between a distance in yards, y , and the same distance in miles, x , is represented by the equation $y = 1760x$.

1. Complete the table.

Distance (mi), x	Distance (yd), y
1	1,760
5	8,800
2	3,520
10	17,600

2. Is there a proportional relationship between a distance in yards and the same distance in miles? Circle one.

☒ Yes ☐ No

Explain your thinking.

Explanations vary. The equation is of the form $y = kx$, where $k = 1760$.

Problems 3–6: Determine whether or not each relationship is proportional.

	Proportional	Not Proportional
3. The remaining length, L , of a 120-inch rope after x inches have been cut off: $120 - x = L$	☐	☒
4. The total cost, t , after 8% sales tax is added to an item's price p : $1.08p = t$	☒	☐
5. The number of marbles, x , each sister gets after m marbles are shared equally among four sisters: $x = \frac{m}{4}$	☒	☐
6. The volume, V , of a rectangular prism whose height is 12 centimeters and whose base is a square with side lengths of s centimeters: $V = 12s^2$	☐	☒

Problems 7–8: Determine whether or not each relationship is proportional.

Explain your thinking.

7.

x	y
2	5
3	7.5
6	15

Yes. Explanations vary. It is proportional because $k = \frac{5}{2}$ for each pair of values listed in the table.

8. $y = 3.2x + 5$

No. Explanations vary. It's not proportional because the equation is not of the form $y = kx$.

Lesson Practice

ACC6.5.06

Name: _____ Date: _____ Period: _____

Problems 9–10: Determine whether each relationship is proportional or not proportional. Explain your thinking.

9. The weight of a stack of standard 8.5-by-11-inch paper and the number of sheets of paper

Proportional. Explanations vary. The weight of each sheet of paper is constant.

10. The weight of a stack of different-sized books (where each book weighs a different amount) and the number of books in a stack

Not proportional. Explanations vary. Each book weighs a different amount, so there is no constant weight.

11. Liam and Samar are running a 60-meter race. Each of their distances can be represented by an equation in the form $y = kx$ where y is the distance in meters and x is the time in seconds.

Use this information to complete the table.

- Liam's distance is represented with the equation $y = 6x$.
- At 8 seconds, Liam is 16 meters ahead of Samar.

Time (sec)	Liam's Distance (m)	Samar's Distance (m)
0	0	0
2	12	8
4	24	16
6	36	24
8	48	32

Spiral Review

12. The equation $y = 3.5x$ can be used to determine the total cost, y , in dollars, of x ounces of blueberries. What does the number 3.5 represent in the equation?
- A. The number of blueberries that \$1 can buy
 - B. The number of blueberries in x ounces
 - C.** The cost of 1 ounce of blueberries
 - D. The cost of x ounces of blueberries

Reflection

- Circle the problem that was the most challenging for you.
- Use this space to ask a question or share something you're proud of.

Lesson Summary

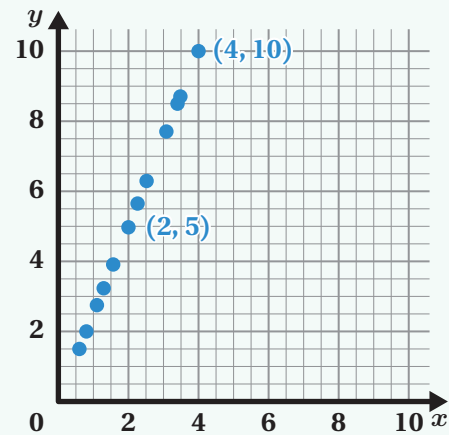
On a *coordinate plane*, if points all fall on a line that passes through $(0, 0)$, the relationship is proportional. The point $(0, 0)$ is known as the origin.

If it's unclear if the points form a line, you can test if the ratios of the coordinates are equivalent.

For example, the coordinates of two points on this line are $(2, 5)$ and $(4, 10)$.

$$5 \div 2 = 2.5 \text{ and } 10 \div 4 = 2.5$$

Since the ratio of the coordinates for both of these points is 2.5, these points are part of a proportional relationship.



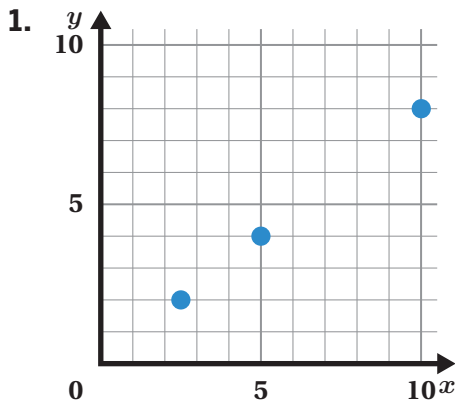
Things to Remember:

Lesson Practice

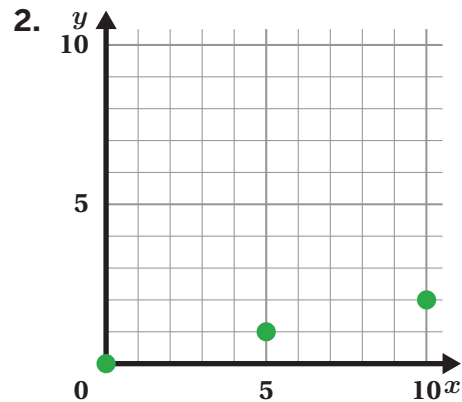
ACC6.5.07

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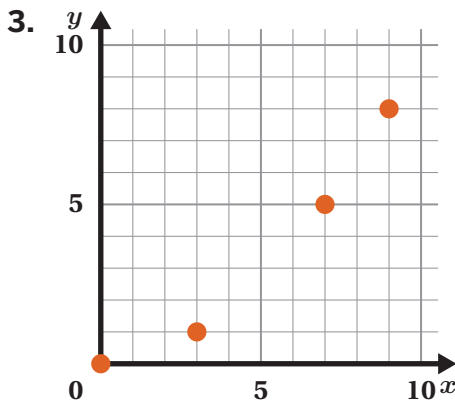
Problems 1–4: Determine if each graph represents a proportional relationship. Explain your thinking.



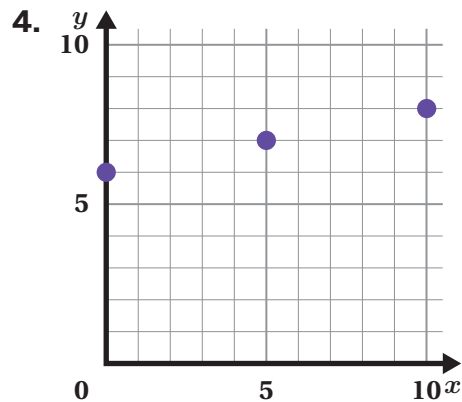
Proportional. *Explanations vary.* The line going through the points passes through the origin.



Proportional. *Explanations vary.* The line going through the points passes through the origin.

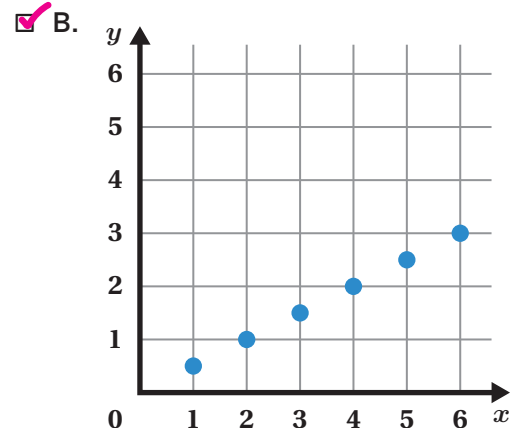
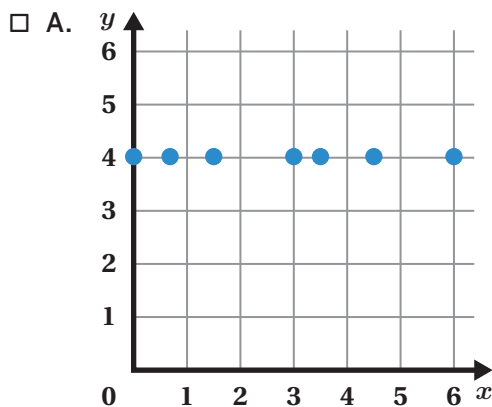


Not proportional. *Explanations vary.* The points lay along a curve.



Not proportional. *Explanations vary.* The line passing through the points does not include the origin.

5. Select *all* the representations that show a proportional relationship.



☐ C.

x	0	2	4	6
y	0	3	9	27

☒ D.

x	0	2	4	6
y	0	12	24	36

Lesson Practice

ACC6.5.07

Name: _____ Date: _____ Period: _____

6. Arturo earns \$33.00 for babysitting for 4 hours. At this rate, how much will Arturo earn if he babysits for 7 hours? Explain your thinking.

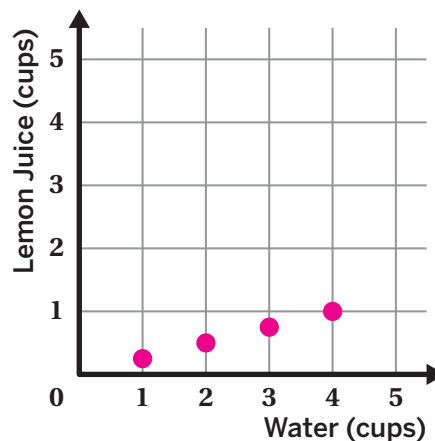
\$57.75. Explanations vary. Arturo's rate is \$8.25 per hour. 7 hours at that rate is \$57.75.

7. A lemonade recipe calls for $\frac{1}{4}$ cups of lemon juice for every cup of water. The table shows different amounts of water and lemon juice you can use to make this recipe. Graph the ordered pairs to determine whether the relationship between water and lemon juice is proportional.

Water (cups)	Lemon Juice (cups)
1	$\frac{1}{4}$
2	$\frac{1}{2}$
3	$\frac{3}{4}$
4	1

Explain your thinking.

The relationship is proportional. Explanations vary. The points fall on a line that passes through the origin.



Spiral Review

8. A turtle is walking away from a rock. x represents the time in minutes that the turtle is walking. y represents the distance in meters between the rock and the turtle. If x and y are in a proportional relationship, select *all* the true statements.
- ☒ A. The equation $y = 3x$ could represent the distance that the turtle walks.
 - ☐ B. The turtle walks for a bit and then stops for a minute before walking again.
 - ☒ C. The turtle walks away from the rock at a constant rate.
 - ☐ D. The equation $y = x + 3$ could represent the distance that the turtle walks.
 - ☐ E. After 6 minutes, the turtle walks 18 meters, and after 10 minutes, the turtle walks 20 meters.

Reflection

- Put a star next to the problem you think is the most important.
- Use this space to ask a question or share something you're proud of.

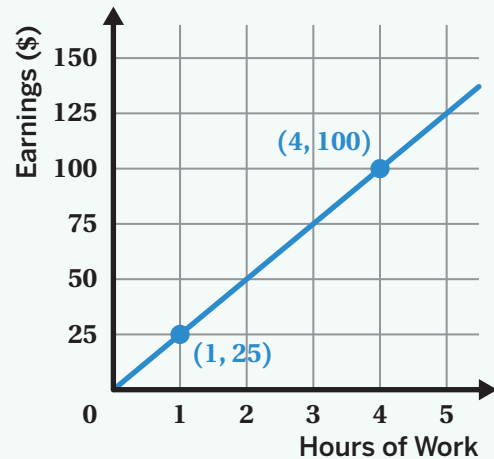
Lesson Summary

Each point on a graph of a proportional relationship tells a story using the quantities represented by x and y . You can determine a constant of proportionality from a graph by using:

- The value of y when x is equal to 1.
- The ratio of $\frac{y}{x}$ for any ordered pair.

For example, this graph shows a proportional relationship between hours worked, x , and money earned in dollars, y . One constant of proportionality is 25 because \$25 is earned for working 1 hour.

The ordered pair (4, 100) shows that \$100 is earned for 4 hours of work, which is an equivalent ratio to earning \$25 per hour.



Things to Remember:

Lesson Practice

ACC6.5.08

Name: _____ Date: _____ Period: _____

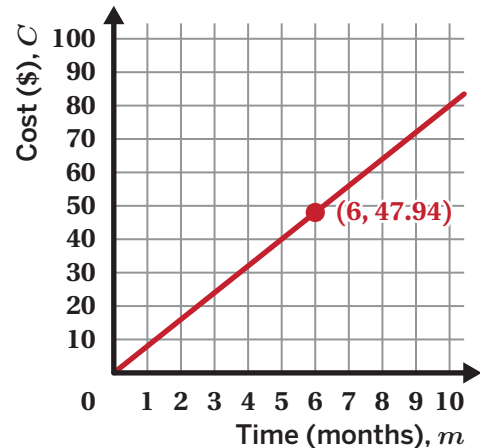
Problems 1–2: Here is a graph that shows a proportional relationship between the number of months Tiara had a streaming service subscription and the total amount of money she paid for the subscription.

After 6 months, Tiara paid \$47.94.

- What is a constant of proportionality in this relationship?
- Write an equation that represents the relationship between C , the total cost of the subscription, and m , the number of months.

7.99 or $\frac{1}{7.99}$ (or equivalent)

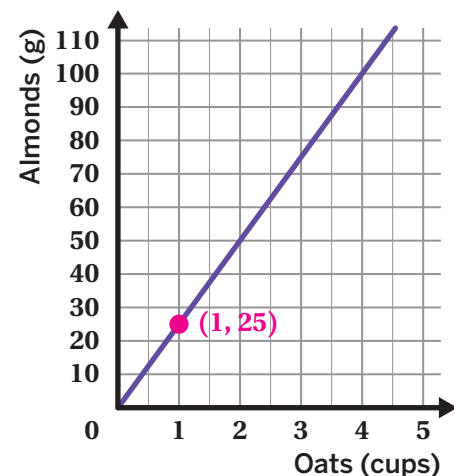
$C = 7.99m$ or $m = \frac{1}{7.99}C$ (or equivalent)



Problems 3–4: A recipe for granola calls for a mix of almonds and oats. The graph shows the amount of almonds, in grams, for different amounts of oats, in cups.

- Determine a constant of proportionality for this relationship, then explain its meaning.
- Label one place you see that constant of proportionality on the graph.

25 or $\frac{1}{25}$. Explanations vary. The graph contains the point (2, 50), which gives the ratio $\frac{50}{2} = 25$. This means 25 grams of almonds are needed for every 1 cup of oats.



- The graph shows the cost for two different varieties of apples. Which variety of apples has a higher cost per pound? Circle one.

Gala

Red Delicious

Explain your thinking.

Explanations vary. A constant of proportionality for Gala apples is 3 and a constant of proportionality for Red Delicious apples is 1. That means Galas have a higher unit price (\$3 per pound) than Red Delicious (\$1 per pound).



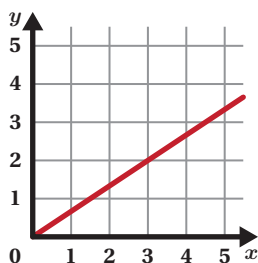
Lesson Practice

ACC6.5.08

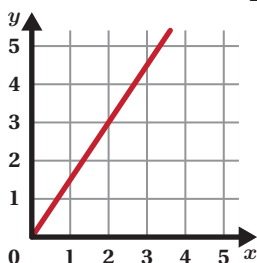
Name: _____ Date: _____ Period: _____

6. Which of these relationships has the same constant of proportionality as the equation $y = \frac{2}{3}x$? Select *all* that apply.

☒ A.



☐ B.



☒ C.

x	y
0	0
3	2
6	4
9	6

☐ D.

x	y
0	0
2	3
4	6
6	9

Spiral Review

7. Write an expression that is equivalent to $8(n + 6)$.

Responses vary. $8n + 48$

Problems 8–10: Haru and Irene were running laps around the track. Their coach recorded their times at the end of Laps 2, 4, and 6.

Haru's Run

Distance (laps)	Time (min)	Minutes Per Lap
2	4	2
4	9	2.25
6	15	2.5

Irene's Run

Distance (laps)	Time (min)	Minutes Per Lap
2	5	2.5
4	10	2.5
6	15	2.5

8. Complete the tables with the minutes per lap for each run.
9. Based on the table, is Haru running at a constant speed? Explain your thinking.
No. Explanations vary. Haru is not running at a constant speed because the number of minutes per lap is not the same for each row in his table.
10. Based on the table, is Irene running at a constant speed? Explain your thinking.
Yes. Explanations vary. Irene may be running at a constant speed because the number of minutes per lap is the same value for each row of her table. To be more certain, I'd also want to know her times for Laps 1, 3, and 5.

Reflection

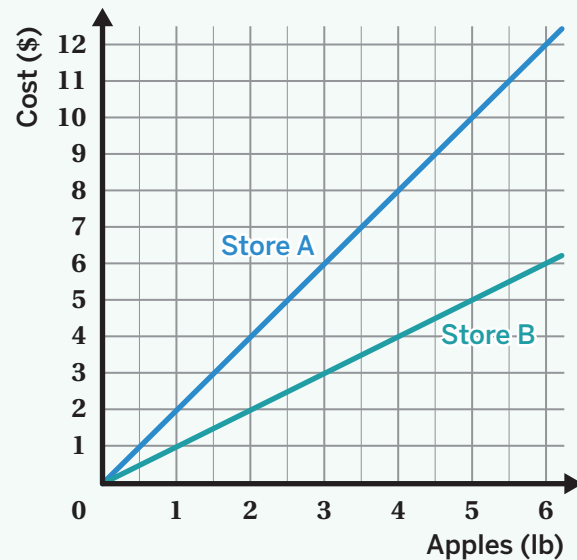
- Put a smiley face next to the problem you learned from most.
- Use this space to ask a question or share something you're proud of.

Lesson Summary

You can compare graphs of proportional relationships when they're on the same coordinate plane. The steeper the line, the greater the constant of proportionality.

For example, you can use this graph to compare the cost of apples at two different stores.

- The line representing Store A is steeper than the line representing Store B, so it has a greater constant of proportionality. This means Store A charges more per pound than Store B.
- Store A charges \$2 for one pound ($k = 2$), while Store B charges \$1 for one pound ($k = 1$). This is another way you can determine that the constant of proportionality at Store A is greater than that at Store B.



Things to Remember:

Lesson Practice

ACC6.5.09

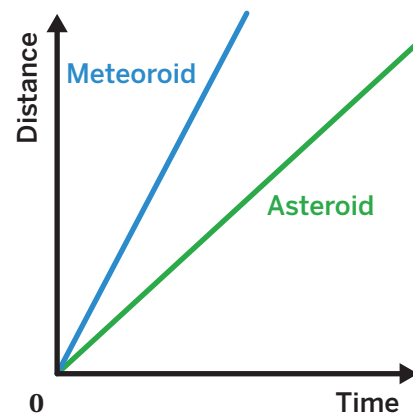
Name: _____ Date: _____ Period: _____

1. A meteoroid and an asteroid travel through the solar system. This graph shows how much distance they travel over time.

Does the asteroid travel faster or slower than the meteoroid? **Slower**

Explain your thinking.

Explanations vary. The asteroid travels slower because its line is less steep. So it takes more time for the asteroid to travel the same distance as the meteoroid.



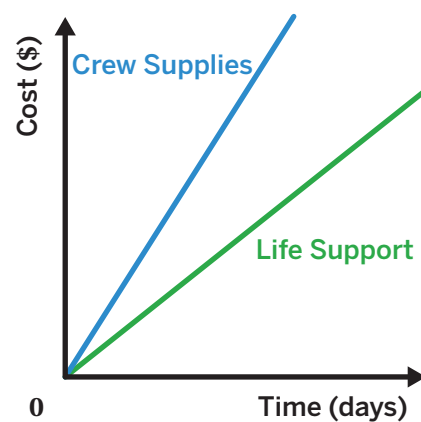
2. Having a crew on the International Space Station requires life support and crew supplies. The graph shows the relationship between the cost of life support and crew supplies, and the number of days spent in space.¹

Which costs less, life support or crew supplies?

Life support

Explain your thinking.

Explanations vary. Life support costs less because its line is less steep than the line for crew supplies. If I draw a vertical line starting from the horizontal axis, it will intersect the life support line before the crew supplies line.



Problems 3–6: At a supermarket, you can fill your own honey container and pay by the ounce. A customer buys 12 ounces of honey for \$5.40.

3. How much does the honey cost per ounce? Show or explain your thinking.

\$0.45 per ounce. Explanations vary. $\frac{5.40}{12} = 0.45$

4. How much honey can be bought per dollar? Show or explain your thinking.

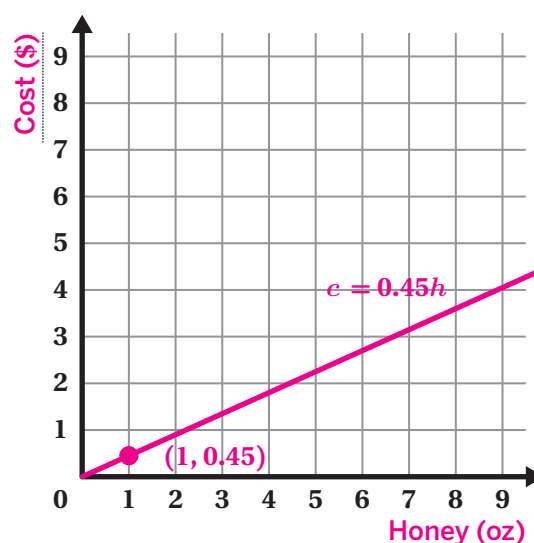
About 2.2 ounces per dollar. Explanations vary. $\frac{12}{5.40} \approx 2.2$

5. Write two different equations representing this situation. Use h for ounces of honey and c for cost in dollars.

$c = 0.45h$ and $h = 2.2c$ (or equivalent)

6. Choose one equation and draw its graph on the coordinate plane. Be sure to label the axes.

Sample shown on graph.



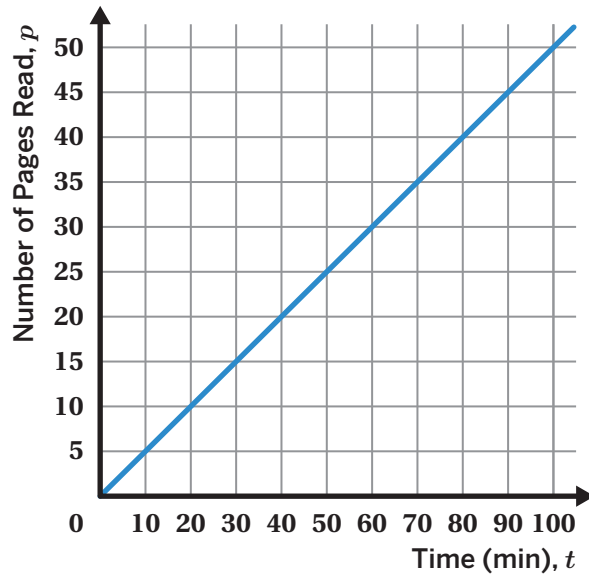
¹ Source: NASA

Lesson Practice

ACC6.5.09

Name: _____ Date: _____ Period: _____

Problems 7–10: The graph shows the relationship between time in minutes, t , and the total number of pages Joel has read, p .



7. Is this relationship proportional?

Explain your thinking.

Yes. Explanations vary. The graph is a straight line that passes through the origin.

8. What is a constant of proportionality in this relationship?

$\frac{1}{2}$ (or equivalent) or 2 (or equivalent)

9. What does this number mean in this situation?

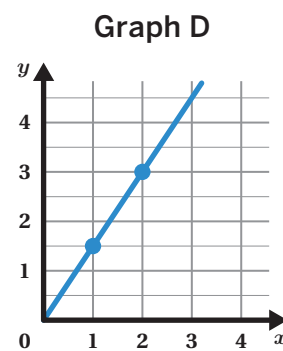
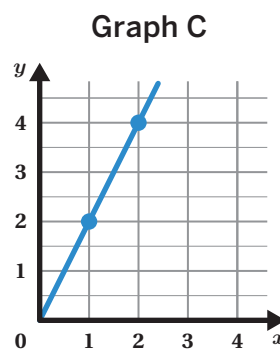
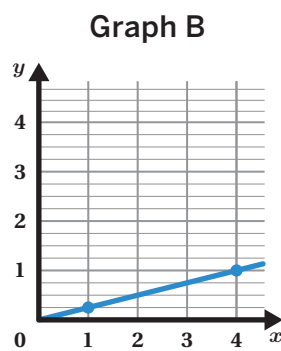
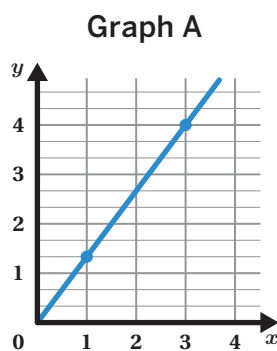
Responses vary. Joel is reading a book at a constant rate of a $\frac{1}{2}$ page per minute.

10. Write an equation that represents the relationship between the time in minutes, t , and the total number of pages Joel has read, p .

$p = \frac{1}{2}t$ or $t = 2p$ (or equivalent)

Spiral Review

Problems 11–14: Match each equation with its graph.



11. $y = \frac{1}{4}x$ **B**

12. $y = \frac{3}{2}x$ **D**

13. $y = 2x$ **C**

14. $y = \frac{4}{3}x$ **A**

Reflection

- Put a heart next to the problem you're most proud of.
- Use this space to ask a question or share something you're proud of.

Lesson Summary

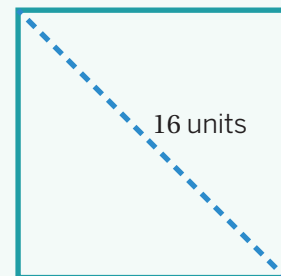
Proportional relationships can appear in geometry when you look at proportional copies of shapes.

Some examples of these proportional relationships include:

- The side lengths of a square and its perimeter.
- The diagonal length of a square and its perimeter.
- The diagonal length of an octagon and its perimeter.

Because the relationship between the diagonal length and the perimeter of a square is proportional, you can determine one of the measurements if you know the other.

For example, to determine the perimeter of a square whose diagonal length is 16 units, you can multiply by the constant of proportionality, approximately 2.83. $16 \cdot 2.83 \approx 45$ units



Things to Remember:

Lesson Practice

ACC6.5.12

Name: _____ Date: _____ Period: _____

Problems 1–3: This graph shows the amount of water in a swimming pool as it fills up.

1. Explain how the graph shows that the relationship between time and the amount of water is proportional.

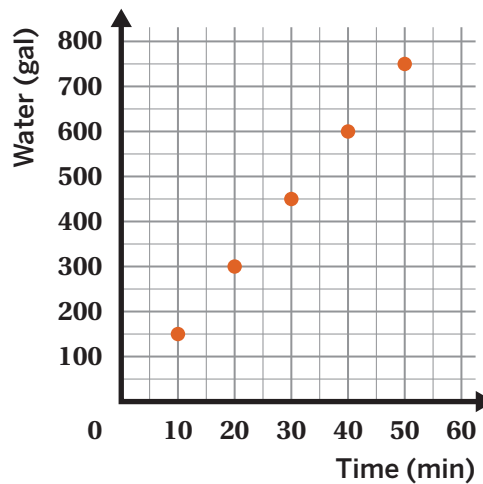
Responses vary. The points all lie on a line that goes through the origin.

2. Calculate a constant of proportionality for this relationship.

15 or $\frac{1}{15}$

3. This pool is safe to swim in when it contains 9,000 gallons of water. How long will it take to fill the pool to this level?

600 minutes



Problems 4–6: An equilateral triangle is a triangle where all three sides have the same length.

4. Complete the table to show the side lengths and perimeters for several equilateral triangles.

Side Length	Perimeter
3	9
10	30
27	81
105	315

5. Explain why the relationship between the perimeter and side length for an equilateral triangle is proportional.

Responses vary. The ratio comparing the side length to the perimeter is always the same. For example, $9 \div 3 = 3$ and $30 \div 10 = 3$.

6. What is a constant of proportionality for the relationship?

3 or $\frac{1}{3}$

7. A lemonade recipe calls for $\frac{1}{4}$ cups of lemon juice, $\frac{1}{2}$ cups of sugar, and 1 cup of water. Complete the table to show the amount of lemon juice for different-sized batches.

Sugar (cups)	Lemon Juice (cups)
2	1
10	5
5	$\frac{5}{2}$

Spiral Review

Problems 8–9: Here are two recipes for making sparkling lemonade. For each recipe, determine how many tablespoons of lemonade mix are required *per cup* of sparkling water. Show or explain your thinking.

8. 4 tablespoons of lemonade mix and 12 cups of sparkling water

$\frac{1}{3}$ tablespoons of lemonade mix per cup of sparkling water. $\frac{4}{12} = \frac{1}{3}$

9. 4 tablespoons of lemonade mix and 6 cups of sparkling water

$\frac{2}{3}$ tablespoons of lemonade mix per cup of sparkling water. $\frac{4}{6} = \frac{2}{3}$

10. Select *all* of the coordinate pairs that could be part of a proportional relationship with a constant of proportionality of 4.

☒ A. (3, 12)

☐ B. (4, 8)

☒ C. (3.5, 14)

☐ D. (6, 10)

☒ E. $\left(\frac{1}{2}, 2\right)$

Problems 11–12: A kangaroo hops 2 kilometers in 3 minutes.

11. How long does it take the kangaroo to hop 5 kilometers?

7.5 minutes

12. How far does the kangaroo hop in 2 minutes?

about 1.33 kilometers

Reflection

1. Put a heart next to a problem you understand well.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

A circle is a shape made out of all the points that are the same distance from a center point. This distance is the radius of the circle.

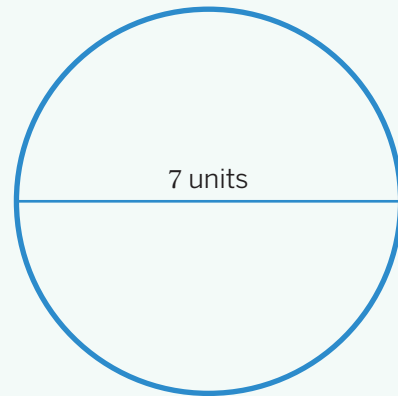
You can also measure a circle with its diameter, which is the distance from one point on a circle through the center to another point on the circle.

The distance around a circle is called the circumference. There is a proportional relationship between the diameter of a circle and its circumference. The constant of proportionality for that relationship is π (pi). π is often approximated as 3.14 or $\frac{22}{7}$.

For any circle, we can calculate the circumference, C , using the equation $C = \pi d$.

For example, if the diameter of a circle is 7 units, the circumference of the circle can be calculated approximately as $7 \cdot 3.14 = 21.98$. More accurately, $C = 7\pi$ units.

If we know the radius of a circle, we can calculate the circumference by first determining the diameter, then using the equation $C = \pi d$.



Things to Remember:

Lesson Practice

ACC6.5.13

Name: _____ Date: _____ Period: _____

1. A student measured the diameter and circumference of several circular objects and recorded their measurements in a table.

Which object's circumference measurement is the least accurate? Explain your thinking.

Flying disc. Explanations vary. The constant of proportionality should be a little more than 3, but $23 \cdot 2.17$ is approximately 50, so the constant of proportionality is less than 3.

Object	Diameter (cm)	Circumference (cm)
Half dollar coin	3	10
Flying disc	23	50
Jar lid	8	25
Flower pot	15	48

2. Complete the table to determine the missing measurements for each object.

Some responses vary as students may use π , 3.14, or $\frac{22}{7}$ in their calculations.

Object	Radius	Diameter	Circumference
Hula hoop	18 inches	36 inches	36π inches
Circular pond	≈ 88.5 feet	≈ 177 feet	556 feet
Magnifying glass	2.6 centimeters	5.2 centimeters	5.2π centimeters
Car tire	≈ 11.4 inches	≈ 22.8 inches	71.6 inches

3. The circumference of a circle is approximately 50.27 inches. Determine the length of the radius of the circle in inches. Round your answer to the nearest whole number.

8 inches

Problems 4–5: The minute hand of a circular clock measures 4 inches. The distance from the end of the minute hand to the outer edge of the clock is 2 inches.

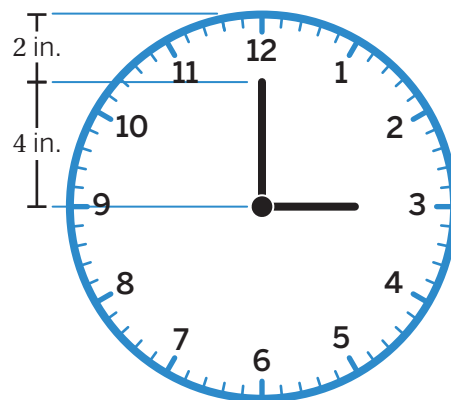
4. What is the length of the radius of the clock?

6 inches

5. What is the circumference of the clock?

Responses vary as students may use π , 3.14, or $\frac{22}{7}$ in their calculations.

- 12π inches
- 37.68 inches
- $\frac{264}{7}$ inches



Lesson Practice

ACC6.5.13

Name: _____ Date: _____ Period: _____

Problems 6–7: The size of a drum affects its pitch. When comparing drums, the one with a larger diameter has a lower pitch because the frequency of its sound waves is slower.

6. If a drummer wants a drum with a lower pitch, should she choose a drum with a circumference of 18π inches or 22.5π inches?

22.5 π inches

7. Determine the diameter of the drum she should choose.

22.5 inches

Problems 8–10: Decide whether the measurement described is the diameter, radius, or circumference. Circle one.

8. The tires on a tractor are 4.5 feet tall. Diameter Radius Circumference

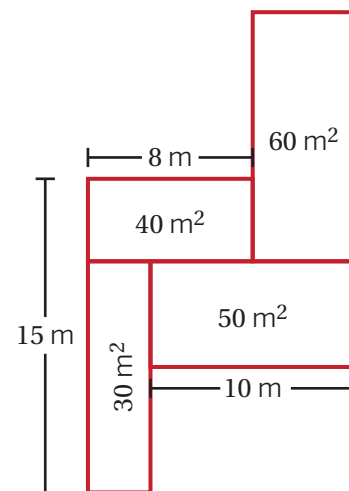
9. The distance from the tip of a slice of pizza to the crust is 6 inches. Diameter Radius Circumference

10. The length of the metal rim around a glass lens is 190 millimeters. Diameter Radius Circumference

Spiral Review

11. This shape is made up of four rectangles. Determine the total perimeter of the shape.

70 meters



Reflection

- Put a heart next to the problem you feel most confident about.
- Use this space to ask a question or share something you're proud of.

Lesson Summary

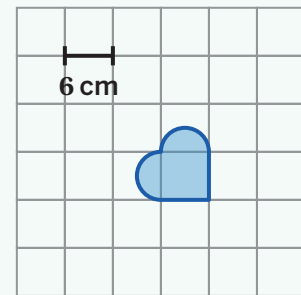
You can use what you know about the perimeter of squares and rectangles and the circumference of circles to find the perimeter of complex shapes.

- Determine what pieces it is made up of such as semicircles, quarter-circles, and straight pieces.
- Determine the length of each straight piece and the radius or diameter of each partial circle.
- Determine the perimeter or circumference of each piece.
- Add them together to get the total perimeter.

The perimeter of the heart shape is made up of 2 semicircles and 2 straight pieces. The size of one unit on the graph is 6 centimeters.

Each semicircle has a diameter of 6 centimeters. Together they make up one entire circle with a diameter of 6 centimeters and a circumference of $6 \cdot \pi = 6\pi$ centimeters. Each straight edge is 6 centimeters long.

The total perimeter is $6\pi + 12$ centimeters.



Things to Remember:

Lesson Practice

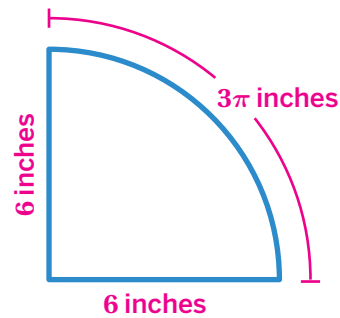
ACC6.5.14

Name: _____ Date: _____ Period: _____

Problems 1–2: Here is a quarter-circle. It was created using a circle with a 12-inch diameter. The circle was folded in half and then folded in half again.

1. Label the quarter-circle with any important measurements.

Sample shown.

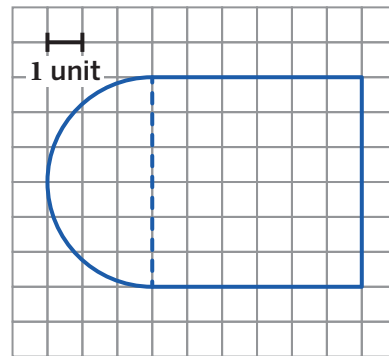


2. What is the perimeter of this quarter-circle?

$3\pi + 12 \approx 21.42$ inches

3. Here is a shape made of a semicircle and a square. What is the perimeter of this shape?

$3\pi + 18 \approx 27.42$ units



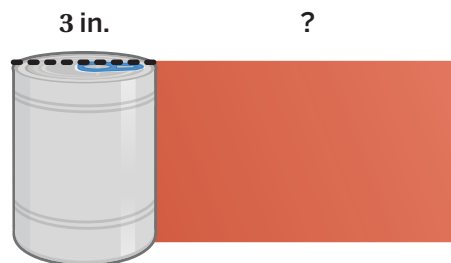
4. A soup can with a diameter of 3 inches has a wrapper that wraps exactly once around its outside. If the wrapper is rolled out, what is its approximate length? Show your thinking.

$$C = \pi d$$

$$C = \pi \cdot 3$$

$$C \approx 9.42$$

The wrapper is about 9.42 inches long.

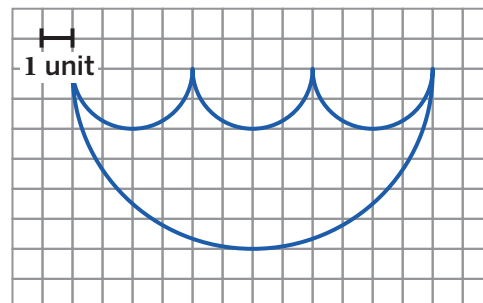


5. Determine the exact perimeter of this shape made from parts of circles.

12π units

Show or explain your thinking

Explanations vary. There are three semicircles with a radius of 2 and one semicircle with a radius of 6. $3 \cdot \frac{1}{2} \cdot 4\pi + \frac{1}{2} \cdot 12\pi = 12\pi$



Lesson Practice

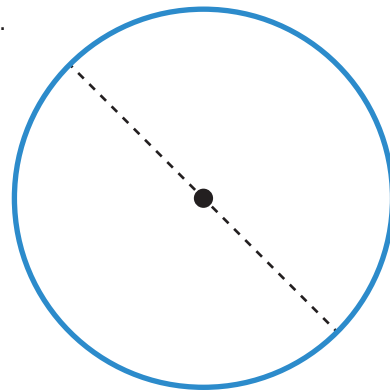
ACC6.5.14

Name: _____ Date: _____ Period: _____

Spiral Review

6. This circle has a diameter measurement of 5 centimeters. Which measurement is closest to the circumference of the circle in centimeters?

- A. 6.25 centimeters
- B. 7.85 centimeters
- C. 15.7 centimeters**
- D. 25 centimeters



Problems 7–9: Determine whether each measurement represents the radius, diameter, or circumference, and record it in the appropriate column of the table. Then determine the exact lengths of the other two measurements of the circle.

	Measurement	Radius	Diameter	Circumference
7.	The tires of a mining truck are 14 feet tall.	7 ft	14 ft	14π ft
8.	The fence around a circular pool is 76 feet long.	$\frac{38}{\pi}$ ft	$\frac{76}{\pi}$ ft	76 ft
9.	The center to the edge of a small plate measures 60 millimeters.	60 mm	120 mm	120π mm

Reflection

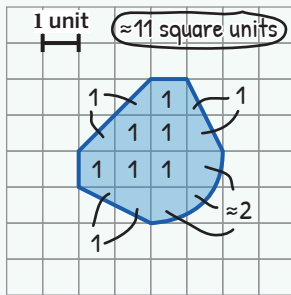
- Put a smiley face next to a problem you were stuck on and then figured out.
- Use this space to ask a question or share something you're proud of.

Lesson Summary

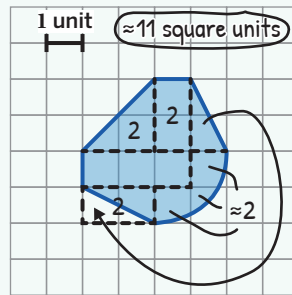
There are many ways you can find the *area* of a complex shape on a grid.

Here are some strategies you can use:

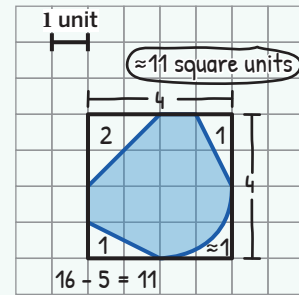
Counting Whole and Partial Squares



Decomposing and Rearranging



Enclosing and Subtracting



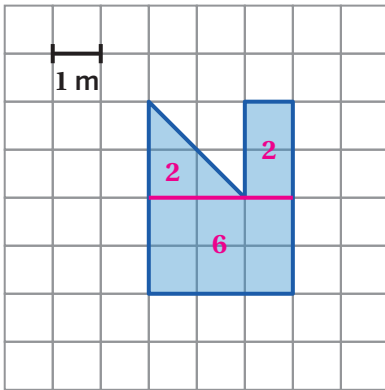
Things to Remember:

Lesson Practice

7.3.05

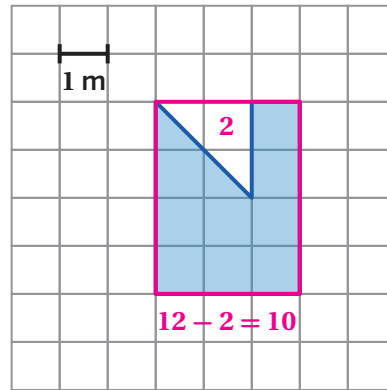
Name: _____ Date: _____ Period: _____

1. What is the area of this shape?
Show your thinking.



10 square meters. *Work varies.*

2. Show or describe another way to determine the area of this shape.

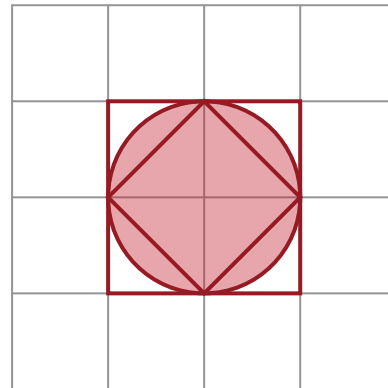


Responses vary.

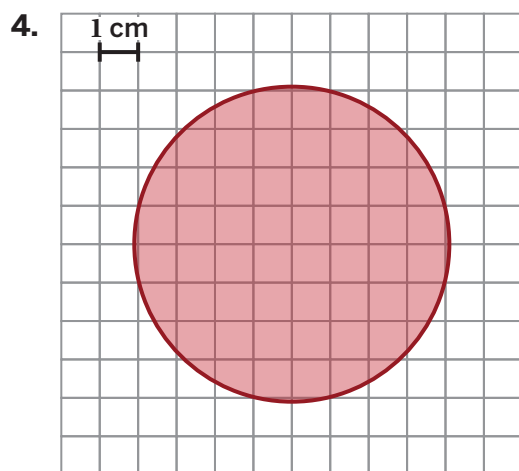
3. Here is a diagram of two squares and a circle.

Explain why the area of the circle is more than 2 square units but less than 4 square units.

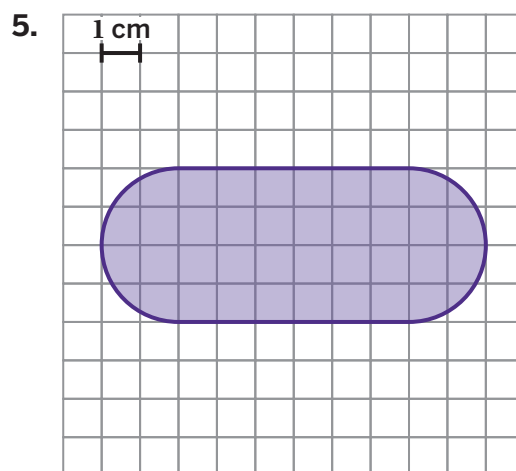
Responses vary. The unshaded square is larger than the circle, and its area is 4 square units. The shaded square is smaller than the circle, and its area is 2 square units. The area of the circle has to be between 2 and 4 square units.



Problems 4–5: Estimate the area of each shape.



Responses between 49 and 56 square centimeters are considered correct.



Responses between 32 and 38 square centimeters are considered correct.

Lesson Practice

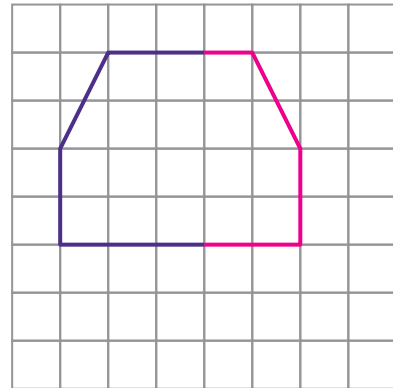
7.3.05

Name: _____ Date: _____ Period: _____

Problems 6–7: Basheera started drawing a polygon.

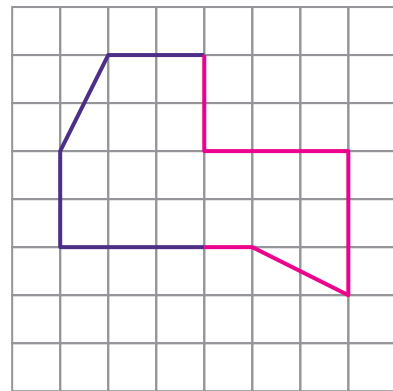
6. Complete Basheera's drawing so that the polygon has an area of 18 square units.

Responses vary.



7. Complete Basheera's drawing in a different way so that the polygon has an area of 18 square units.

Responses vary.



Spiral Review

8. Select *all* the expressions that are equivalent to 48.

☒ A. $3 \cdot 4^2$

☐ B. $6 + 6 \cdot 4$

☒ C. 50% of 96

☐ D. $(3 \cdot 4)^2$

☒ E. $4(6 + 6)$

9. What is the greatest common factor of 48 and 64?

16

10. What is the least common multiple of 4 and 6?

12

Reflection

- Circle a problem you want to talk to a classmate about.
- Use this space to ask a question or share something you're proud of.

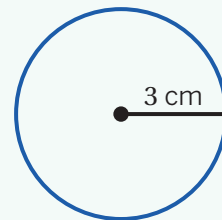
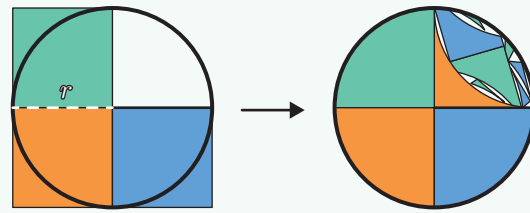
Lesson Summary

You can find the area of a circle if you know the length of its radius, r .

The *approximate* area of a circle is equal to the area of a little more than 3 radius squares. Each radius square has an area of r^2 .

The *exact* area of a circle is equal to the area of π radius squares. You can express this with the formula $A = \pi \cdot r^2$.

For example, to find the area of a circle with a radius of 3 centimeters, you can calculate 3^2 , then multiply the result by π . The area of the circle is 9π square centimeters.



Things to Remember:

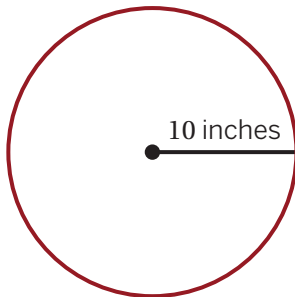
Lesson Practice

ACC6.5.16

Name: _____ Date: _____ Period: _____

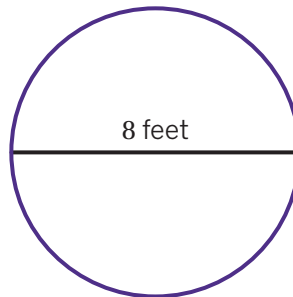
Problems 1–2: Determine the exact area of each circle.

1.



100π square inches

2.



16π square feet

Problems 3–4: The table shows the diameters of 4 different coins.

3. To determine how much metal is on one face of a coin, it is more useful to use the area rather than the circumference. Explain why this is the case.

Responses vary. The circumference would just tell you the distance around the edge of the coin, not the amount of metal needed to fill in the face of the coin. The area measures what is needed to cover the face of the coin, not just what is needed to go around it.

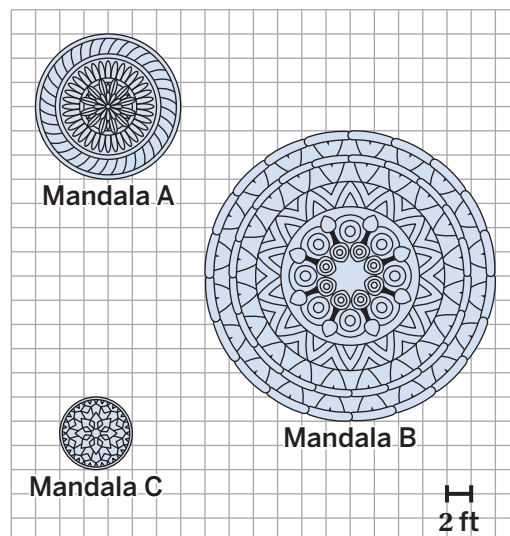
Coin	Diameter (cm)	Area (sq. cm)
Penny	1.9	0.9025π
Nickel	2.1	1.1025π
Dime	1.8	0.81π
Quarter	2.4	1.44π

4. Complete the table by calculating the exact area of each coin.

5. A mandala is a geometric figure that has spiritual relevance in many religions, including Hinduism and Buddhism. The word *mandala* is Sanskrit for *circle*.

Saanvi designed three mandalas. Determine the exact circumference and area for each mandala.

	Circumference (ft)	Area(sq. ft)
A	12π	36π
B	24π	144π
C	6π	9π



Lesson Practice

ACC6.5.16

Name: _____ Date: _____ Period: _____

Problems 6–7: The radius of Earth is approximately 6,400 kilometers. The equator is the circle around Earth that divides it into the northern and southern hemispheres.

6. Is the circumference of a circle or the area of a circle more useful for finding the length of the equator?

Circumference

7. What is the length of the equator?

$12800\pi \approx 40212$ kilometers

Spiral Review

Problems 8–10: This graph shows a proportional relationship between the number of hours a manufacturing factory is in operation and the number of kilowatt-hours (kWh) of electricity used. Use the graph to determine whether each statement is true or false.

8. Point F represents the number of kilowatt-hours of electricity used when the factory is in operation for 3 hours.

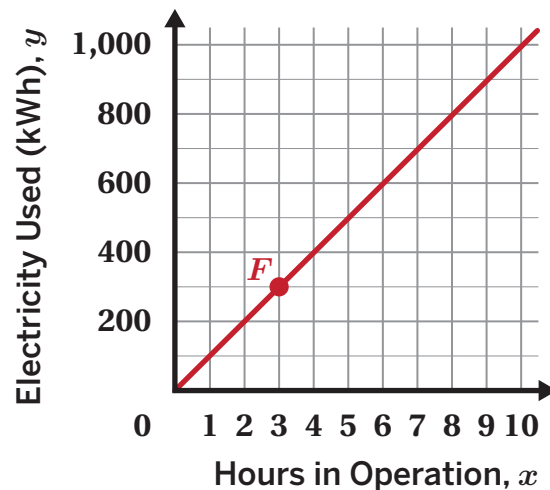
True

9. The factory uses 6 kilowatt-hours of electricity when it is in operation for 600 hours.

False

10. The factory uses 700 kilowatt-hours of electricity when it is in operation for 7 hours.

True



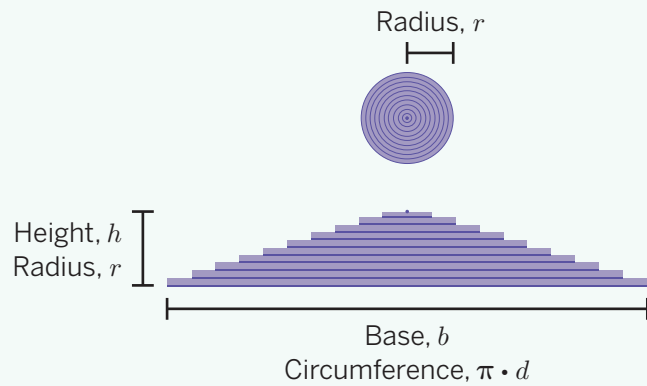
Reflection

1. Put a heart next to the problem you found most interesting.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

If you take apart a circle and rearrange it to resemble a triangle, the formula for the area of a circle can be related to the formula for the area of the triangle.

This helps us make sense of each part of the formula for the area of a circle.



- The radius of a circle is r and the circumference is $\pi \cdot d$, so you can substitute those values into the equation for the area of a triangle.
- You can replace d with $2 \cdot r$.
- $\frac{1}{2} \cdot 2 = 1$, which leaves $A = \pi \cdot r^2$.

$$A = \frac{1}{2} \cdot b \cdot h$$

$$A = \frac{1}{2} \cdot (\pi \cdot d) \cdot r$$

$$A = \frac{1}{2} \cdot \pi \cdot (2 \cdot r) \cdot r$$

$$A = \pi \cdot r \cdot r$$

$$A = \pi \cdot r^2$$

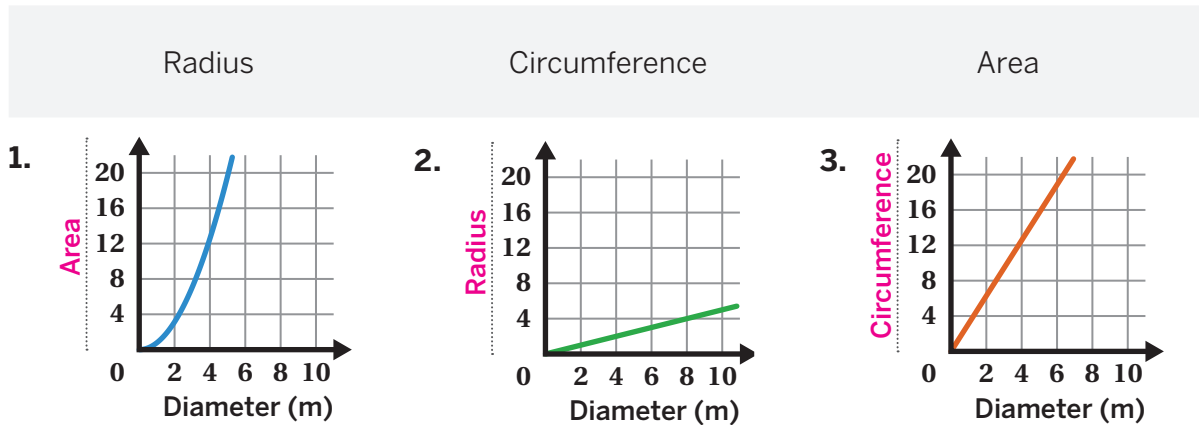
Things to Remember:

Lesson Practice

ACC6.5.17

Name: _____ Date: _____ Period: _____

Problems 1–3: Label each y -axis with a term from the word bank to describe the relationship within a circle that the graph represents.



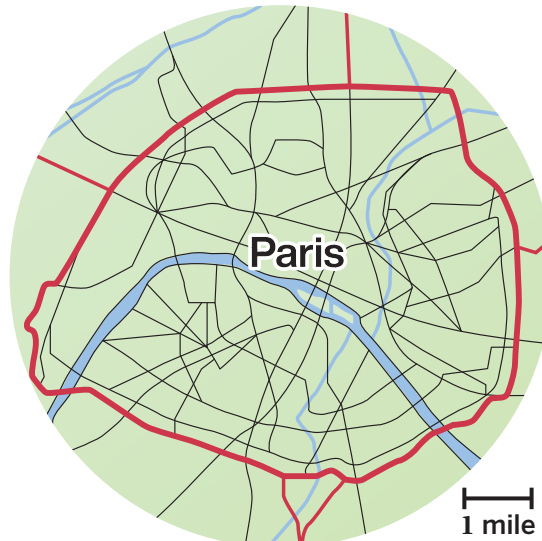
Problems 4–5: The city of Paris, France, is surrounded by an almost circular road called the Périphérique.

4. Use the map and the scale to estimate the length of the road. Explain your thinking.

Responses vary. If I estimate that the diameter is approximately 6.5 miles across, then the length of the road would be $6.5 \cdot \pi \approx 20.4$ miles.

5. Use the map and the scale to estimate the total area enclosed by the road. Explain your thinking.

Responses vary. If I estimate that the diameter is approximately 6.5 miles across, then the area enclosed by the road would be $3.25^2 \cdot \pi \approx 33.2$ square miles.



6. Select *all* the pairs of quantities that are proportional to each other. For the quantities that are proportional, write an equation that relates them. **Sample equations shown.**

- ☒ A. The radius and diameter measurements of a circle $d = 2r$
- ☒ B. The radius and circumference measurements of a circle $C = 2\pi r$
- ☐ C. The radius and area measurements of a circle
- ☒ D. The diameter and circumference measurements of a circle $C = \pi d$
- ☐ E. The diameter and area measurements of a circle

Lesson Practice

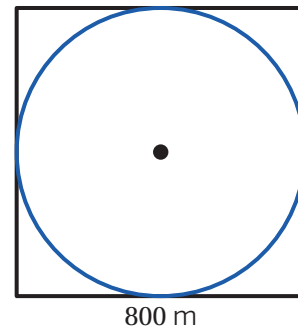
ACC6.5.17

Name: _____ Date: _____ Period: _____

Spiral Review

7. Determine the exact area of the circle. Explain your thinking.

160,000 π meters. Explanations vary. The circle meets the sides of the square without overlapping, so the length of the diameter of the circle is equal to the side length of the square, or 800 meters. The radius of the circle is 400 meters, so the area is $\pi \cdot r^2 = 160,000\pi$.

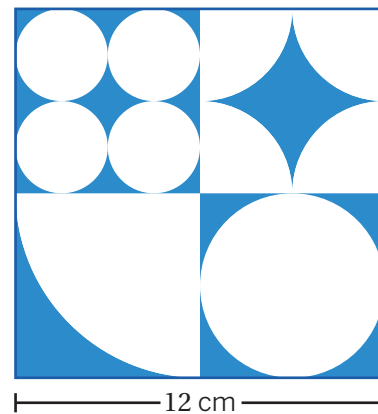


8. This 12-centimeter square contains squares, circles, and quarter-circles. Determine the exact area of the shaded region. Show or explain your thinking.

144 – 36 π square centimeters

Work varies.

144 – 4(2.25 π) – 9 π – 9 π – 9 π



9. A circle has a diameter of 13 units. What is the area of the circle to the nearest hundredth of a square unit?

A. 40.84

B. 132.73

C. 530.93

D. 2,123.72

Problems 10–11: Solve each equation. Show your thinking.

10. $x - 2\frac{2}{3} = 6\frac{2}{3}$

$x = 9\frac{1}{3}$

Work varies.

$x - 2\frac{2}{3} + 2\frac{2}{3} = 6\frac{2}{3} + 2\frac{2}{3}$

11. $\frac{1}{2}x = 12$

$x = 24$

Work varies.

$\frac{1}{2}x \div \frac{1}{2} = 12 \div \frac{1}{2}$

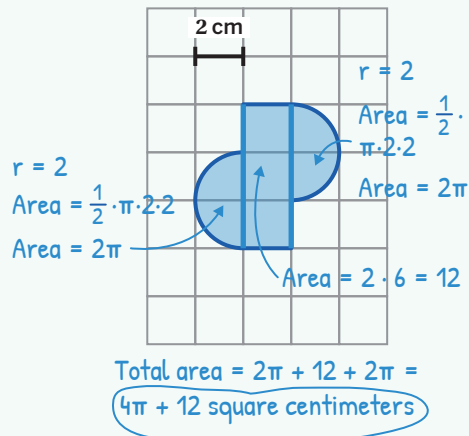
Reflection

- Put a question mark next to a problem you're feeling unsure of.
- Use this space to ask a question or share something you're proud of.

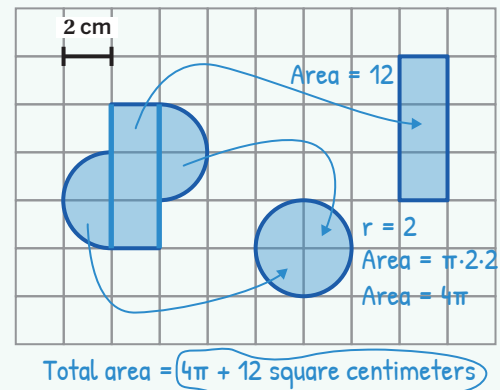
Lesson Summary

We can use different strategies and different ways of thinking to calculate the area of complex figures. Here are two ways we can determine the area of this shape:

Strategy 1



Strategy 2



The path to the solution may not be obvious at first, but by breaking the shape down into squares, circles, and parts of circles, we can figure things out!

Things to Remember:

Lesson Practice

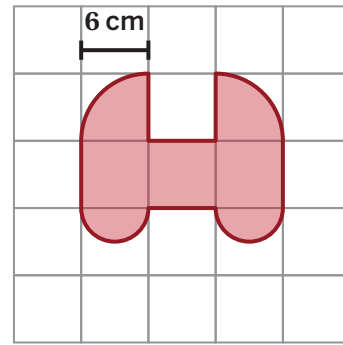
ACC6.5.18

Name: _____ Date: _____ Period: _____

1. Determine the area of this shape.

Responses vary as students may use π , 3.14, or $\frac{22}{7}$ in their calculations.

$$27\pi + 108 \approx 192.82 \text{ square centimeters}$$



Problems 2–4: Here is the architectural plan for a fenced-in garden. The shaded region represents an area for trees.

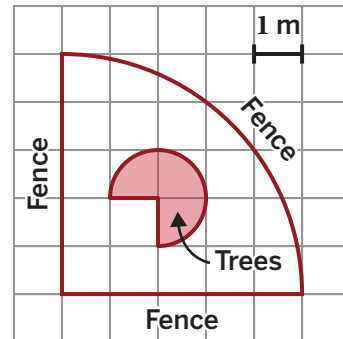
2. What is the area of the tree region?
0.75 π $\approx$ 2.36 square meters
3. The rest of the fenced-in garden is for flowers.
How much area will be covered in flowers?
Show or explain your thinking.

About 17.28 square meters

$$\text{Work varies. } 6.25\pi - 0.75\pi = 5.5\pi \approx 17.28$$

4. How long is the fence around the edge of the garden? (Note: The edge around the tree region is not a part of the fence.)

$$2.5\pi + 10 \approx 17.85 \text{ meters}$$



5. Each of these squares has a side length of 12 units. Compare the areas of the shaded regions in these three figures. Which figure has the largest shaded region? Show or explain your thinking.

Figure A

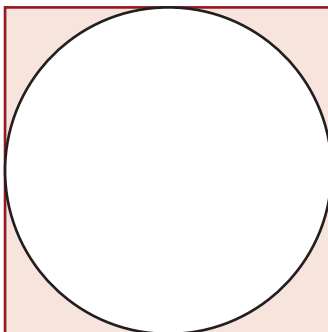


Figure B

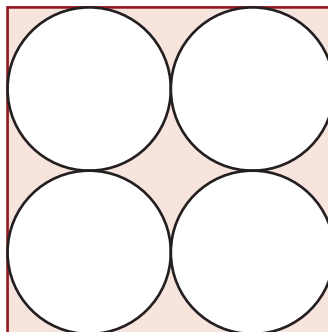
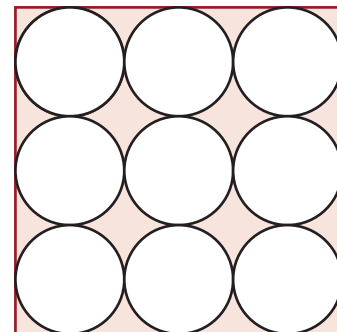


Figure C



The shaded region is the same size in each figure. Explanations vary.

In Figure A, the radius length of the circle is half the side length of the square, or 6 units.

$$A = \pi r^2$$

$$A = \pi \cdot 6^2$$

$$A = 36\pi$$

$$\text{Shaded region: } 144 - 36\pi$$

In Figure B, the radius length of each circle is one fourth the side length of the square, or 3 units.

$$A = \pi r^2$$

$$A = \pi \cdot 3^2$$

$$A = 9\pi$$

$$4 \text{ circles: } 4 \cdot 9\pi = 36\pi$$

$$\text{Shaded region: } 144 - 36\pi$$

In Figure C, the radius length of each circle is one sixth the side length of the square, or 2 units.

$$A = \pi r^2$$

$$A = \pi \cdot 2^2$$

$$A = 4\pi$$

$$9 \text{ circles: } 9 \cdot 4\pi = 36\pi$$

$$\text{Shaded region: } 144 - 36\pi$$

Lesson Practice

ACC6.5.18

Name: _____ Date: _____ Period: _____

6. A circle with a 12-inch long diameter is cut in half and then cut in half again. What is the area of the new shape? Show or explain your thinking.

$9\pi \approx 28.26$ square inches. *Explanations vary.*

$$A = \pi r^2$$

$$A = \pi \cdot 6^2$$

$$A = 36\pi$$

$$\frac{1}{4} \cdot 36\pi = 9\pi \approx 28.26$$

Spiral Review

7. A graffiti artist is spray painting a circular mural onto a wall. The mural will have a diameter that is 10 feet long. Each can of spray paint covers about 24 square feet. How many cans of spray paint will the artist need to create the mural? Explain your thinking.

4 cans. Explanations vary. If the diameter is 10 feet long, then the radius is 5 feet long. That means the area of the mural is $\pi \cdot 5^2 = \pi \cdot 25 \approx 79$ square feet. Because 1 can covers 24 square feet, the artist will need $79 \div 24$, or about 3.29 cans of paint. This means the artist will actually need to purchase 4 cans for the mural.

Problems 8–9: Determine each value.

8. What is the length of the radius of a circle with a diameter that measures 10 units?

5 units

9. What is the length of the diameter of a circle with a radius that measures 10 units?

20 units

10. Abena used wire fencing to form a border around a circular garden. If the radius of the circular garden was 10 yards long, what was the total length of the border, rounded to the nearest tenth of a yard?

A. 31.4 yards

B. 62.8 yards

C. 314.2 yards

D. 628.3 yards

Reflection

- Put a star next to a problem where you revised your thinking.
- Use this space to ask a question or share something you're proud of.

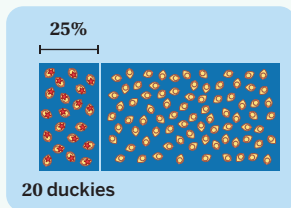
Lesson Summary

Percent means for every 100. It's represented by the percent symbol, %.

Each of the different ducky games in this lesson had a certain percentage of ducks with stars: 10%, 25%, 50%, or 75%. Fractions and tape diagrams can help us interpret these percentage problems.

Example Problem

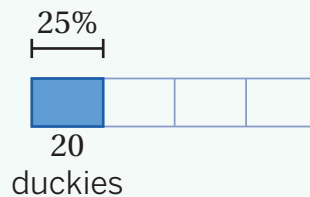
25% of the 80 duckies have stars.

**Using Fractions**

25% of something means $\frac{25}{100}$ or $\frac{1}{4}$.

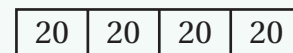
$\frac{1}{4}$ of 80 duckies is 20 duckies.

80 total duckies

**Using Tape Diagrams**

There are four 25s in 100, so the tape diagram can be split into 4 pieces. The total number of duckies can also be split into 4 parts, so there are 20 duckies in each section.

80 total duckies



Things to Remember:

Lesson Practice

ACC6.6.01

Name: _____ Date: _____ Period: _____

1. Here are 24 stars. Circle 25% of these stars.

Responses vary. 6 stars should be circled.



2. Shep made 40 muffins. 50% of the muffins are chocolate. How many muffins are chocolate?

20 muffins

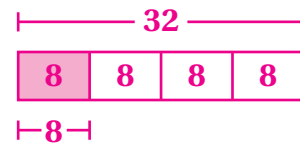
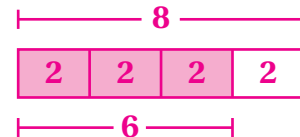
3. Which is greater? Show or explain your thinking.

A. 75% of 8

B. 25% of 32

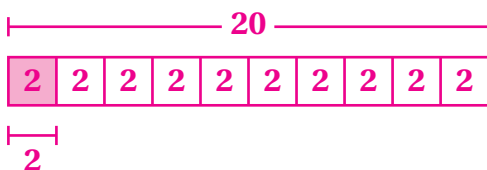
C. They are the same.

Explanations vary. Sample shown.

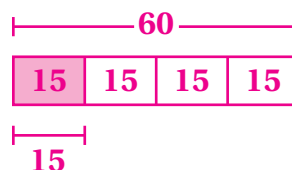


Problems 4–5: Complete each statement. Make a tape diagram if it helps with your thinking.

4. 10% of 20 is 2.



5. 25% of 60 is 15.



6. Explain how you could mentally calculate 10% of any number.

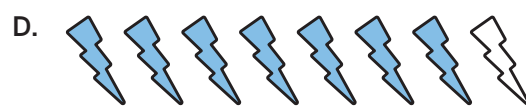
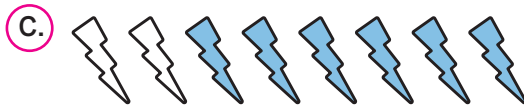
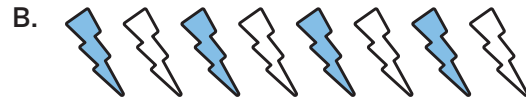
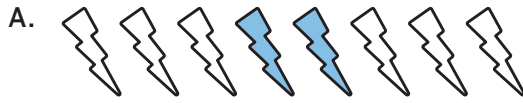
Responses vary. Divide the number by 10, or multiply the number by $\frac{1}{10}$.

Lesson Practice

ACC6.6.01

Name: _____ Date: _____ Period: _____

7. Which group shows 75% of the lightning bolts shaded?



Spiral Review

8. Abdel paid \$13 for 3 books. Jayden bought 12 books priced at the same rate. How much did Jayden pay for the 12 books? Explain your thinking.

\$52. Explanations vary. Jayden paid \$52 for the 12 books because he bought 4 times as many books as Abdel did at the same rate. $13 \cdot 4 = 52$

Money Paid (dollars)	Number of Books
13	3
52	12

Problems 9–11: Determine whether each product will be *less than*, *greater than*, or *equal to* 40.

9. $\left(\frac{6}{4}\right) \cdot 40$

Greater than

10. $\left(\frac{8}{8}\right) \cdot 40$

Equal to

11. $\left(\frac{1}{2}\right) \cdot 40$

Less than

Reflection

- Put a question mark next to a problem you're feeling unsure of.
- Use this space to ask a question or share something you're proud of.

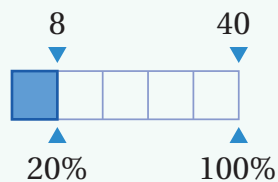
Lesson Summary

You can represent percentages using tape diagrams, double number lines, and tables. The strategies you've already used to solve ratio problems can help you think about and solve percentage problems, too!

Let's say a biker traveled 8 kilometers, which is 20% of their goal distance. What's their goal distance?

Here are three ways to represent and solve this percentage problem.

Tape Diagram

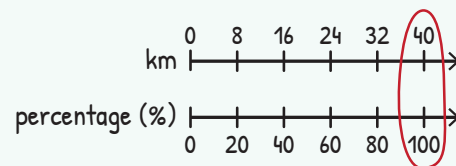


Table

km	%
8	20
40	100

Red arrows indicate a multiplier of 5 from 8 to 40 and from 20 to 100. The number 40 is circled in red.

Double Number Line



So the biker's goal distance is 40 kilometers.

Things to Remember:

Lesson Practice

ACC6.6.02

Name: _____ Date: _____ Period: _____

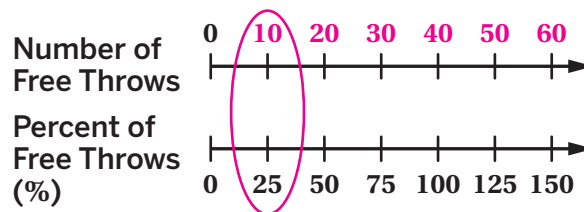
- What percent of each figure is shaded?
Record your answers in the table.

Figure A	30%
Figure B	40%
Figure C	75%



- Martina shot 40 free throws at basketball practice. 25% of her free throws went into the basket. How many of them went into the basket? Use the double number line if it helps with your thinking.

10 free throws



- On Tuesday, Parv made 12 cookies. On Wednesday, he made 150% as many cookies as he made on Tuesday. How many cookies did Parv make on Wednesday?

18 cookies

Problems 4–5: Leonardo works as a server in a restaurant. He gets tipped 20% of the cost of each order.

- What tip will he get if the food costs \$60?

\$12

- Leonardo got an \$18 tip. What was the cost of the food for this order?

\$90

- Nikhil says that to determine 20% of a number, you can divide the number by 5. For example, 20% of 60 is 12 because $60 \div 5 = 12$. Does Nikhil's method always work? Explain your thinking.

Yes, Nikhil's method always works. Explanations vary. 20% of a number means $\frac{20}{100}$ of that number, which is equivalent to $\frac{1}{5}$ of that number. Determining $\frac{1}{5}$ of any number is the same as dividing the number by 5.

Lesson Practice

ACC6.6.02

Name: _____ Date: _____ Period: _____

Spiral Review

Problems 7–8: Light travels about 180,000,000 kilometers in 10 minutes.

7. How many kilometers per minute is that?

18,000,000 kilometers per minute

8. How many kilometers per second is that?

300,000 kilometers per second

9. Match each expression with the tape diagram that represents it.

a.

30



..... **a** $\frac{3}{5} \cdot 30$

b.

30



..... **c** $\frac{1}{3} \cdot 5$

c.

5



..... **b** $\frac{5}{3} \cdot 30$

Reflection

1. Put a question mark next to a problem you're feeling unsure of.

2. Use this space to ask a question or share something you're proud of.

Lesson Summary

When solving percentage problems related to money, you can:

- Determine the value of 1% and multiply that by the percentage you're looking for.
- Determine how many cents per dollar a given percentage represents.

Let's say a pair of pants costs \$42. If the factory makes a *profit* of 14% on the price of a pair of pants, how many dollars of profit does the factory make from each sale?

Strategy 1

Cost (dollars)	Percentage (%)
42	100
$\frac{42}{100}$	1
$\frac{42}{100} \cdot 14$	14

$$\frac{42}{100} \cdot 14 = 5.88$$

The factory makes \$5.88 of profit from each sale.

Strategy 2

- 14% profit means 14 cents of each dollar is profit.

$$\frac{14}{100} = 0.14$$

- The price of the pants is \$42.

$$\frac{14}{100} \cdot 42 = 5.88$$

The factory makes \$5.88 of profit from each sale.

Things to Remember:

Lesson Practice

ACC6.6.04

Name: _____ Date: _____ Period: _____

Problems 1–4: Evaluate each expression.

1. 50% of 70
35

2. 10% of 70
7

3. 1% of 70
0.7

4. 2% of 70
1.4

5. A store is having a 30% off sale. The original price for a pair of headphones is \$150. How much would a customer save with this sale?
\$45

6. Order the following expressions from *least* to *greatest* value.

55% of 180	300% of 26	12% of 700
300% of 26	12% of 700	55% of 180
Least		Greatest

7. To find 40% of 75, Jamal calculates $\frac{2}{5} \cdot 75$. Does Jamal's calculation give the correct value for 40% of 75? Explain your thinking.

Yes. Explanations vary. Multiplying by $\frac{2}{5}$ is the same as finding 40% because $\frac{2}{5} = \frac{40}{100}$.

8. Emika has a monthly budget for her cell phone bill. Last month, she spent 120% of her budget, and the bill was \$60. What is Emika's monthly budget?
\$50

9. Kyrie spent 75 minutes practicing the piano over the weekend. Yasmine practiced the violin for 152% as much time as Kyrie practiced the piano. How long did Yasmine practice?
114 minutes

Lesson Practice

ACC6.6.04

Name: _____ Date: _____ Period: _____

10. Select *all* the expressions that could be used to calculate 45% of 60.

☐ A. $\frac{100}{45} \cdot 60$

☐ B. $\frac{60}{45} \cdot 100$

☒ C. $\frac{45}{100} \cdot 60$

☐ D. $\frac{100}{60} \cdot 45$

☐ E. $\frac{0.45}{100} \cdot 60$

☒ F. $\frac{60}{100} \cdot 45$

11. Fill in each blank using the digits 0 to 9 only once, so that the expression on the left is greater than the expression on the right. *Responses vary.*

Left

Right

% of 50

50% of

Spiral Review

12. Two stores sell identical sandwich rolls in different-sized packages. Store A sells a six-pack for \$5.28. Store B sells a four-pack for \$3.40. Which store offers the better price per roll?

A. Store A

☒ B. Store B

C. They are the same

Show or explain your thinking.

Explanations vary. Sandwich rolls cost \$0.88 each at Store A and \$0.85 each at Store B.

Reflection

1. Put a star next to a problem you want to understand better.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

You can use ratios to determine what percent one amount is compared to another amount. Let's say an adult giant panda weighs 90 kilograms and a giant panda cub weighs 36 kilograms. You can determine the cub's weight as a percent of the adult's weight using several strategies.

Double Number Line										
Ratio Tables	<table border="1"> <thead> <tr> <th>Mass (kg)</th> <th>Percent (%)</th> </tr> </thead> <tbody> <tr> <td>90</td> <td>100</td> </tr> <tr> <td>1</td> <td>$\frac{1}{90} \times 100$</td> </tr> <tr> <td>36</td> <td>$\frac{36}{90} \times 100$</td> </tr> </tbody> </table> Arrows indicate the relationships: from 90 to 1 is $\times \frac{1}{90}$, from 1 to 36 is $\times 36$; from 100 to $\frac{1}{90} \times 100$ is $\times \frac{1}{90}$, from $\frac{1}{90} \times 100$ to $\frac{36}{90} \times 100$ is $\times 36$.	Mass (kg)	Percent (%)	90	100	1	$\frac{1}{90} \times 100$	36	$\frac{36}{90} \times 100$	- Determine the unit rate (what percent matches 1 kilogram). - Use the unit rate to determine what percent 36 kilograms is of 90 kilograms.
Mass (kg)	Percent (%)									
90	100									
1	$\frac{1}{90} \times 100$									
36	$\frac{36}{90} \times 100$									
Expressions	$36 \div 90 \cdot 100 = \frac{36}{90} \cdot 100 = 40$ Evaluate $\frac{36}{90} \cdot 100$ to determine what percent the cub's weight, 36, is of the adult's weight, 90.									

Things to Remember:

Lesson Practice

ACC6.6.05

Name: _____ Date: _____ Period: _____

1. Select *all* the expressions that represent what percent 19 is of 20.

☒ A. $\frac{19}{20} \cdot 100$

☐ B. $\frac{19}{20} \div 100$

☐ C. $\frac{20}{19} \cdot 100$

☒ D. $19 \cdot \frac{100}{20}$

☐ E. $\frac{19}{100} \cdot 20$

2. At a hardware store, a tool set normally costs \$80. During a sale this week, the tool set costs \$12 less than normal. What percent of the original price can a customer save? Show or explain your thinking.

15%. Explanations vary. The savings is 15% of the normal price because $\frac{12}{80} \cdot 100 = 15$.

Problems 3–5: A 6th grade class did a weekend fitness challenge. Each student set a goal for 75 minutes of exercise.

3. Luca exercised for 54 minutes. What percent of the goal did Luca complete?

72%

4. Brianna completed 64% of her goal. How many minutes did she exercise for?

48 minutes

5. Jada exercised for 78 minutes. What percent of the goal did Jada complete?

104%

Lesson Practice

ACC6.6.05

Name: _____ Date: _____ Period: _____

Spiral Review

Problems 6–9: Determine each product.

6. $0.72 \cdot 15 = 10.8$

7. $0.72 \cdot 1.5 = 1.08$

8. $0.72 \cdot 0.15 = 0.108$

9. $72 \cdot 0.15 = 10.8$

Reflection

1. Put a heart next to the problem you feel most confident about.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

Working with real-world data and information can be interesting, but it presents challenges, like working with very large numbers or information presented in different forms.

Ratios, rates, and percentages can help you make sense of real-world situations and compare very large numbers.

Some of the benefits of ratios, rates, and percentages are:

- They allow you to compare quantities that are on different scales because they describe things in terms of multiplying and dividing instead of adding and subtracting.
- They bring everything to the same scale, most commonly with a reference point of either 1 or 100, which makes comparing numbers more straightforward.

For example, percentages can help us compare different-sized groups of people around the world to see what the distribution of people really looks like.

Things to Remember:

Lesson Practice

ACC6.6.07

Name: _____ Date: _____ Period: _____

Problems 1–3: The sale price of every item in a store is 85% of its original price. Complete the table to show the prices of each item.

	Item	Original Price (\$)	Sale Price (\$)
1.	Backpack	30.00	25.50
2.	Soccer ball	18.00	15.30
3.	Jacket	24.80	21.08

Problems 4–6: Last Sunday, an amusement park had 1,575 visitors.

4. 56% of the visitors were adults. Calculate the number of adults that visited the park.

882 adults

5. 16% of the visitors were teenagers. Calculate the number of teenagers that visited the park.

252 teenagers

6. 28% of the visitors were children ages 12 and under. Calculate the number of children ages 12 and under that visited the park.

441 children

7. A veterinarian examined 45 animals on Wednesday. Of the animals she examined, 20% were cats. How many cats did the veterinarian examine on Wednesday?

A. 18

B. 9

C. 5

D. 14

Lesson Practice

ACC6.6.07

Name: _____ Date: _____ Period: _____

Spiral Review

Problems 8–10: Fill in each blank to complete the sentence.

8. 5% of 70 is 3.5.

9. 25% of 24 is 6.

10. 12% of 700 is 84.

Reflection

1. Put a question mark next to a problem you were feeling stuck on.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

Tape diagrams and tables can help us make sense of problems involving percent increase and percent decrease.

The terms *percent increase* and *percent decrease* describe an increase or decrease of a quantity as a percentage of the starting amount.

One method to solve these types of problems is to start with the original amount and then add or subtract the amount that matches the percent of increase or decrease.

Things to Remember:

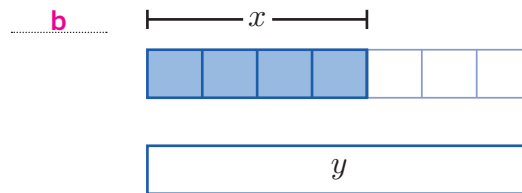
Lesson Practice

ACC6.6.08

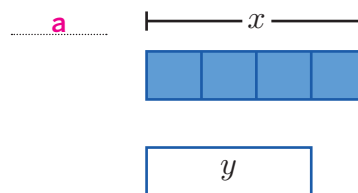
Name: _____ Date: _____ Period: _____

1. Match each situation with a diagram.

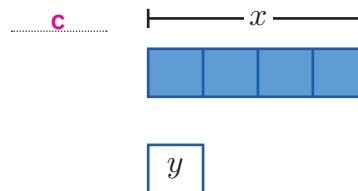
- a. Hoang drinks x ounces of juice. Nekeisha drinks $\frac{1}{4}$ less than that.



- b. Hoang runs x miles. Nekeisha runs $\frac{3}{4}$ more than that.



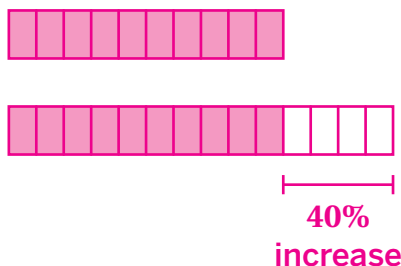
- c. Hoang buys x pounds of almonds. Nekeisha buys $\frac{1}{4}$ of that.



Problems 2–3: Draw a diagram to represent each situation.

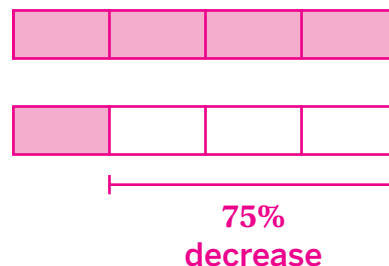
2. The amount of flour that the bakery used this month increased 40% compared to last month.

Responses vary.



3. The amount of milk that the bakery used this month decreased 75% compared to last month.

Responses vary.



4. At the beginning of the month, there were 80 ounces of peanut butter in the pantry. Since then, a family has eaten 30% of it. Which expression represents the ounces of peanut butter left in the pantry?

A. $0.7 \cdot 80$

B. $0.3 \cdot 80$

C. $8 - 0.30$

D. $(1 + 0.3) \cdot 80$

Lesson Practice

ACC6.6.08

Name: _____ Date: _____ Period: _____

Problems 5–7: Fill in the blanks to describe each increase or decrease as a percentage of the original amount.

5. This year, there was 40% more snow than last year.

The amount of snow this year is 140% of the amount last year.

6. This year, there were 25% fewer sunny days than last year.

The number of sunny days this year is 75% of the number of sunny days last year.

7. A restaurant adds a 93% markup to the price of the ingredients to set the menu price.

The menu price is 193% of the price of ingredients.

Spiral Review

Problems 8–9: A store sells strawberries for \$1.38 per pound.

8. Write an equation relating the cost, c , and the pounds of strawberries, p .

$c = 1.38p$ (or equivalent)

9. One strawberry order costs \$8.97. How many pounds did the person order?

6.5 pounds

Problems 10–11: Solve each equation.

10. $x \cdot \frac{7}{3} = 1$

$x = \frac{3}{7}$ (or equivalent)

11. $1 \div \frac{11}{2} = x$

$x = \frac{2}{11}$ (or equivalent)

Reflection

1. Put a star next to a problem you're still wondering about.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

We can use equations to help us make sense of situations involving percent increase or percent decrease.

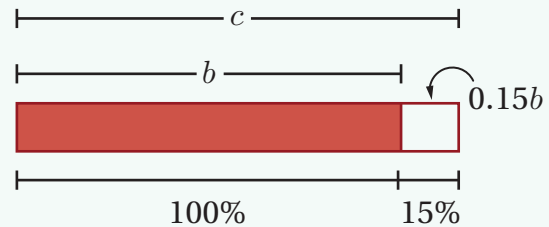
For example, c is 15% more than b .

Three equations can be written to model the relationship between b and c :

$$c = b + 0.15b$$

$$c = (1 + 0.15)b$$

$$c = 1.15b$$



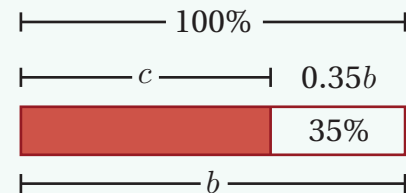
In this example, c is 35% less than b .

Three equations can be written to model the relationship between b and c :

$$c = b - 0.35b$$

$$c = (1 - 0.35)b$$

$$c = 0.65b$$



Things to Remember:

Lesson Practice

ACC6.6.09

Name: _____ Date: _____ Period: _____

1. Draw a diagram that represents this situation:

The number of people in a town has increased by 50% in the past decade.

Responses vary.



2. Write a story that matches the diagram.

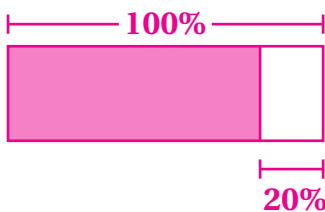
Responses vary. The number of pets in my family increased by 25%.



3. Draw a diagram that represents this situation:

The amount of paper that the copy shop used this month decreased by 20% compared to what they used last month.

Responses vary.



4. A new video game costs \$60 on the day it gets released. The price of a new game generally drops by about 12% within a few weeks. What will the price of this new game be a few weeks after release? Show or explain your thinking.

\$52.80. Explanations vary.

$100\% - 12\% = 88\%$

Let c represent the original cost of the game and s represent the price.

Then $0.88 \cdot c = s$. If $c = 60$, then $0.88 \cdot 60 = s$, so $52.80 = s$.

5. Jada and Braylen both wrote an equation to represent an increase of 3%, where x represents the amount before the increase and y represents the new amount.

Jada wrote the equation $y = 1x + 0.03x$. Braylen wrote the equation $y = 1.03x$.

Whose equation is correct? Circle one.

Jada

Braylen

Both

Neither

Explain your thinking.

Explanations vary. If I substitute a number in for either equation and solve for y , I get a value that is 3% more than the value I started with.

Lesson Practice

ACC6.6.09

Name: _____ Date: _____ Period: _____

Problems 6–7: A sneaker store raised its prices 15% compared to last year.

6. If x is the price before the increase and y is the price after the increase, which equations are correct? Select *all* that apply.

☒ A. $y = 1.15x$

☐ B. $y = x + 0.15$

☒ C. $y = x + 0.15x$

☐ D. $y = 15 + x$

☒ E. $y = (1 + 0.15)x$

7. A pair of designer sneakers was \$120 last year. What is the price this year?

\$138

8. Two stores each advertise a discount on the same type of sweatshirt. At both stores, the original price of the sweatshirt is \$25.

- Store A discounts the price of the sweatshirt by 20%.
- Store B discounts the price of the sweatshirt by 15%.

How much less is the discounted price of the sweatshirt at Store A than the discounted price of the sweatshirt at Store B?

A. \$0.75

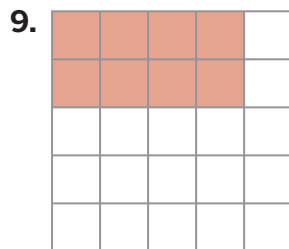
B. \$1.00

C. \$1.25

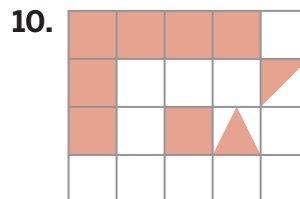
D. \$1.75

Spiral Review

Problems 9–10: Determine the percentage of each grid that is shaded.



32%



40%

Reflection

1. Put a smiley face next to a problem you were stuck on and then figured out.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

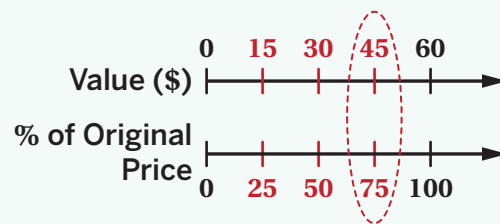
A double number line is a helpful tool for understanding percentage problems.

When using double number lines, it helps to first identify which value aligns with which percentage. It may be helpful to think of the values as a new amount and an original, or old, amount.

Once the known values and percentages are aligned, filling in more values on both number lines can help you solve the problem. For percentage problems, the 0 on each number line should be aligned.

Example:

A furniture store offers 25% off every piece of furniture to make room in the warehouse. If a chair normally sells for \$60, what is its sale price?



Things to Remember:

Lesson Practice

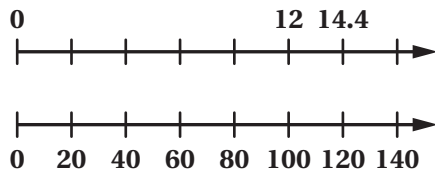
ACC6.6.10

Name: _____ Date: _____ Period: _____

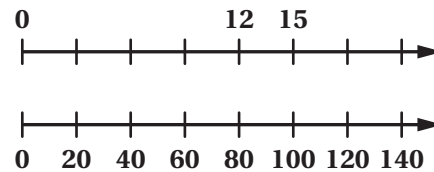
- Abdullah and Kanna are working on this problem:

A juice box has 20% more juice in its new packaging. The original packaging held 12 fluid ounces. How many fluid ounces of juice does the new packaging hold?

Abdullah's Double Number Line



Kanna's Double Number Line



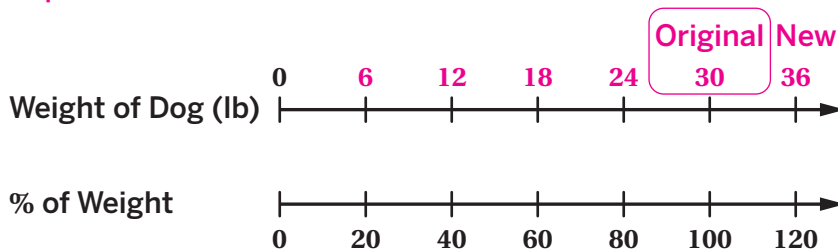
Whose double number line represents the problem? Explain your thinking.

Abdullah's. Explanations vary. The number 12 should be aligned with 100% because that is the original amount of fluid ounces.

Problems 2–3: Read each scenario and then answer the question. Complete the double number line if it helps with your thinking.

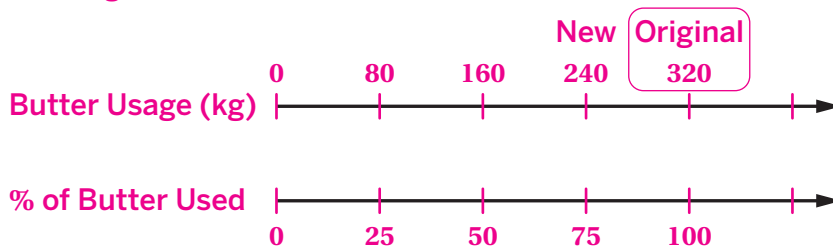
- A dog weighs 20% more than it did three months ago. It weighs 36 pounds now. How much did the dog weigh three months ago?

30 pounds



- A bakery used 25% less butter this month than last month. The bakery used 240 kilograms of butter this month. How much did it use last month?

320 kilograms



Lesson Practice

ACC6.6.10

Name: _____ Date: _____ Period: _____

Problems 4–5: Next week, the price of oranges at the farmer’s market will decrease by 20%.

4. Select *all* the equations that represent the relationship between the price of oranges this week, x , and the price of oranges next week, y .

☐ A. $y = \frac{1}{5}x$

☒ B. $y = \frac{4}{5}x$

☒ C. $y = x - \frac{1}{5}x$

☐ D. $y = 0.2x$

☒ E. $y = 0.8x$

5. The price of oranges this week is \$4.50. What will the price be next week?

\$3.60

6. The number of fish in a lake decreased by 15% between last year and this year. This year, there are 51 fish in the lake. What was the population last year?

60 fish

7. Without determining the actual percent increase, explain why it is not reasonable to say that a price increase from \$4.00 to \$5.00 for a bag of clementines represents a 125% increase.

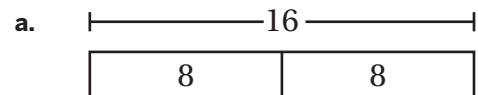
Responses vary. A 125% increase would mean the new price is 225% of the original price, so it more than doubles. \$5.00 is not more than twice \$4.00.

Spiral Review

Problems 8–10: Match each situation to the diagram that best represents it.

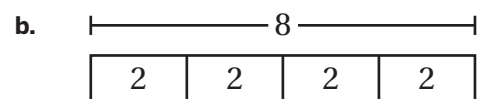
8. I earn \$8 an hour for mowing the lawn.

How much will I earn if I work for 2 hours? **a.**

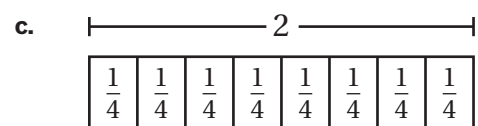


9. I divided 2 cakes equally between 8 people.

How much cake does each person get? **c.**



10. I folded 8 socks into pairs. How many pairs do I have? **d.**



Reflection

- Put a heart next to the problem you’re most proud of.
- Use this space to ask a question or share something you’re proud of.

Lesson Summary

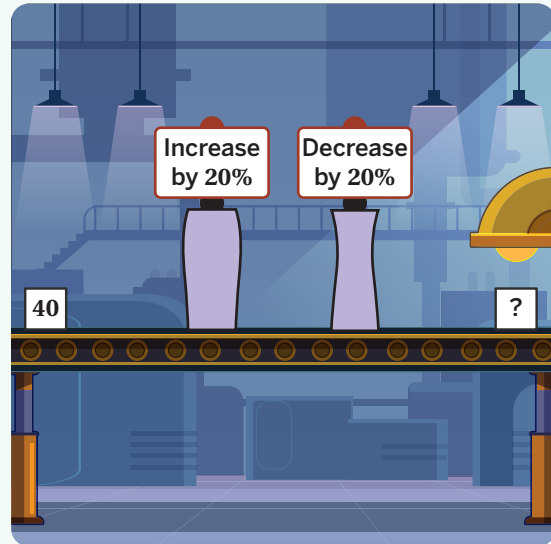
Percent machines take an input value and increase or decrease that value by a percentage to produce an output.

Increasing and then decreasing by the same percentage will not produce the original input value.

For example:

- An input value of 40 is increased by 20%.
 $40 \cdot 1.2 = 48$ or $40 \cdot 0.2 + 40 = 48$
- The new value is decreased by 20%.
 $48 \cdot 0.8 = 38.4$ or $48 - (48 \cdot 0.2) = 38.4$

It may surprise you to discover the final result is *not* the original input value of 40! This is because the input value represents the whole when you determine 20% of 40. But in the second calculation, the input value changes. You are now determining 20% of 48 and then subtracting that value from 48.

**Things to Remember:**

Lesson Practice

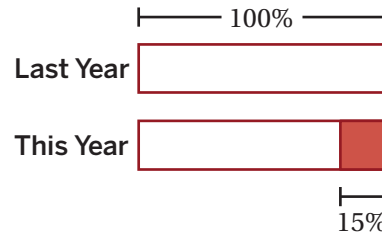
ACC6.6.11

Name: _____ Date: _____ Period: _____

1. Match each situation with a diagram that represents it.

- a. The amount of apples this year decreased by 15% compared to last year's amount.

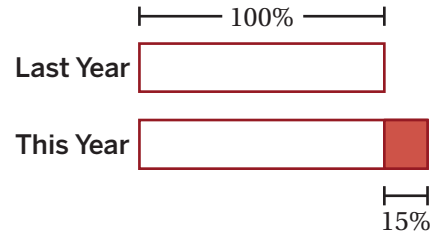
a, c



- b. The amount of cherries this year increased by 15% compared to last year's amount.

- c. The amount of pears this year is 85% of last year's amount.

b, d



- d. The amount of oranges this year is 115% of last year's amount.

2. Mar's aunt bought a trading card many years ago. The value of the card increased by 80%. Its value is now \$270. What was the value of the card when Mar's aunt bought it?

\$150

Problems 3–5: A small town had a population of 4,000 in 1990.

3. By 2000, the population increased to 5,000. What was the percent change from 1990 to 2000? Show or explain your thinking.

25%. *Explanations vary.* $\frac{5000}{4000} \cdot 100 = 125$ and $125\% - 100\% = 25\%$

4. In 2010, the population decreased back to 4,000. What was the percent change from 2000 to 2010? Show or explain your thinking.

20%. *Explanations vary.* $\frac{4000}{5000} \cdot 100 = 80$ and $100\% - 80\% = 20\%$

5. Explain why the percent change in Problem 3 is not equal to the percent change in Problem 4.

Responses vary. Although in each decade the population changed by 1,000, the original population in each situation is different.

Lesson Practice

ACC6.6.11

Name: _____ Date: _____ Period: _____

6. A restaurant pays all of its servers by the hour. Tyler earned \$136 for 8 hours of work last week and got a 5% raise this week. What is Tyler's new hourly wage?

\$17.85 per hour

Spiral Review

Problems 7–8: Write an equation to represent each situation.

7. A worker is paid the minimum wage in their state, which is \$10.00 per hour. What is their total earnings, t , for h hours worked?

$t = 10h$ (or equivalent)

8. What is the value of y if it represents 40% of a given value x ?

$y = 0.40x$ (or equivalent)

9. Eliza walked 12 miles. Then she walked $\frac{1}{4}$ of that distance. How many miles did she walk altogether? Select *all* that apply.

☐ A. $12 + \frac{1}{4}$

☐ B. $12 \cdot \frac{1}{4}$

☒ C. $12 + 12 \cdot \frac{1}{4}$

☒ D. $12\left(1 + \frac{1}{4}\right)$

☐ E. $12 \cdot \frac{3}{4}$

☒ F. $12 \cdot \frac{5}{4}$

Reflection

1. Circle the problem you're most interested in knowing more about.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

The listed price and the total a customer ends up paying are often two different quantities. Tax, tip, and discounts are some of the reasons why the listed price and final price are different.

These changes are often calculated as percentages of the listed price.

For example, a tip is an amount of money that a person gives someone who provides a service, such as restaurant servers, hairdressers, and delivery drivers. If a person plans to leave a 20% tip, then the total cost with tip will be 120% of the bill.

Things to Remember:

Lesson Practice

ACC6.6.13

Name: _____ Date: _____ Period: _____

1. Wessam orders a meal that costs \$15. There is no tax on the meal, and Wessam decides to tip 20%. Select the expression that represents the total cost, in dollars, including the tip.

- A. $15 + 0.20$ B. $15 + 1.20$
C. $15 \cdot 0.20$ **D. $15 \cdot 1.20$**

2. In a city in Ohio, the sales tax rate is 7.25%. Complete the table.

Item	Price Before Tax (\$)	Sales Tax (\$)	Price Including Tax (\$)
Pillow	8.00	0.58	8.58
Blanket	24.00	1.74	25.74
Trash can	16.00	1.16	17.16

Problems 3–4: A family eats at a restaurant. The bill is \$42. There is no tax on the meal, and the family leaves a tip and spends \$49.77 total.

3. How much money does the family tip?
\$7.77
4. How much is the tip as a percent of the bill?
18.5%

Problems 5–7: A music store buys instruments and then sells them for 30% more than they paid.

5. If the store buys a guitar for \$45, what will the store sell it for?
\$58.50
6. If the price tag on a trumpet says \$104, how much did the store pay for it?
\$80
7. During a 20% off sale, the store offers a clarinet for \$93.60. How much did the store pay for the clarinet?
\$90

Lesson Practice

ACC6.6.13

Name: _____ Date: _____ Period: _____

8. A clothing store is having a sale: 25% off shirts and 30% off jeans.

- Crow bought a shirt with an original price of \$24.
- Crow bought a pair of jeans with an original price of \$32.

Determine how much money Crow saved.

\$15.60

Spiral Review

9. In 1969, the annual global mean temperature was 14°C . In 2019, the annual global mean temperature was 14.8°C . Two different news websites reported this change.

Website A

“In the last 50 years, the annual global mean temperature has increased by almost 6%.”

Website B

“There has been an increase in the annual global mean temperature of approximately 5% during the last half century.”

Which website more accurately reported the change? Explain your thinking.

Website A. Explanations vary. $\frac{14.8}{14} \cdot 100 \approx 105.71$, and $105.71\% - 100\% = 5.7\%$. Because 5.7% rounds up to 6%, Website A is more accurate.

Reflection

1. Put a question mark next to a problem you're feeling unsure of.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

There are two different federal minimum wages — one for workers who receive tips and one for workers who don't.

For tipped workers, such as restaurant servers, pay depends not only on their hourly wage and the number of hours they work, but also the number of tables served, the average bill at those tables, and the average percent tip.

Let's say a restaurant server earns \$2.13 per hour and works 30 hours per week. They serve about 40 tables in a week, where the typical bill is \$75 and the tip is 18%. Here's an equation that shows how much this server earns in a typical week:

$$2.13 \cdot 30 + 40 \cdot 75 \cdot 0.18 = \$603.90$$

We can compare this amount to other ways of paying servers, such as a simple rate of \$15 per hour ($15 \cdot 30 = \450 in a week), with no tips. Switching to this way would mean about a 25% decrease in pay for the restaurant server in our example because

$$\frac{603.90 - 450}{603.90} \approx 0.2548.$$

Things to Remember:

Lesson Practice

ACC6.6.14

Name: _____ Date: _____ Period: _____

Problems 1–2: A customer leaves a 15% tip on a \$20 meal.

1. Select the expression that represents the value of the tip.

- A. $15 \cdot 20$ B. $20 + 1.5 \cdot 20$
 C. $1.15 \cdot 20$ **D. $\frac{15}{100} \cdot 20$**

2. Select the expression that represents the *total* bill.

- A. $15 \cdot 20$ B. $20 + 1.5 \cdot 20$
C. $1.15 \cdot 20$ D. $\frac{15}{100} \cdot 20$

Problems 3–4: Mauricio is a server at a restaurant. In an average 8-hour work day, he serves 10 tables, with an average bill of \$45 per table. He typically receives a 20% tip on each bill and earns \$7.25 per hour.

3. How much money does Mauricio earn in a typical day?

\$148

4. Let's say the typical tip increased to 23% of the bill. By what percent would Mauricio's earnings increase? Show or explain your thinking.

About 9%. Explanations vary. Mauricio's earnings would increase from \$148 to $8 \cdot 7.25 + 10 \cdot 45 \cdot 0.23 = \161.50 . This is about a 9% increase because $\frac{161.50 - 148}{148} \approx 0.0912$.

Problems 5–7: Here is some information about three parks. Complete each sentence.

5. Golden Gate Park is about **21** % larger than Central Park.
6. Longview Lake Park is about **134** % larger than Golden Gate Park.
7. If the size of Central Park increased by 200%, would it be larger than Longview Lake Park? Explain your thinking.

Yes. Explanations vary. Increasing by 200% is the same as tripling. $843 \cdot 3$ is greater than 2,381.

Park	Area (acres)
Central Park (New York City)	843
Golden Gate Park (San Francisco)	1,017
Longview Lake Park (Kansas City)	2,381

Lesson Practice

ACC6.6.14

Name: _____ Date: _____ Period: _____

Spiral Review

8. Match each situation with an equation.

- | | |
|--|------------------------------|
| a. Tay sleeps for x hours. Omar sleeps for 20% less than that. | <u> b </u> $y = 0.6x$ |
| b. Tay practices piano for x hours. Omar practices for 40% less than that. | <u> d </u> $y = 0.75x$ |
| c. Tay drinks x ounces of juice. Omar drinks 130% more than that. | <u> a </u> $y = 0.8x$ |
| d. Tay spends x dollars. Omar spends 25% less than that. | <u> c </u> $y = 2.3x$ |
| e. Tay completes x puzzles. Omar completes 60% more than that. | <u> f </u> $y = 1.4x$ |
| f. Tay eats x grams of almonds. Omar eats 40% more than that. | <u> e </u> $y = 1.6x$ |

9. The radius of circle S is half the radius of circle L . The radius of circle L is 10 millimeters. Which measurement is closest to the area of circle S ?

- | | |
|------------------------------------|-----------------------------|
| <u>A.</u> 78.54 square millimeters | B. 31.42 square millimeters |
| C. 314.16 square millimeters | D. 15.71 square millimeters |

Problems 10–12: Solve each equation.

10. $\frac{2}{3}x = \frac{8}{15}$
 $x = \frac{4}{5}$

11. $1.8 + x = 7.2$
 $x = 5.4$

12. $5\frac{4}{5} = 3\frac{2}{3} + x$
 $2\frac{2}{15} = x$

Reflection

1. Put a star next to a problem where you revised your thinking.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

Percent error describes the difference between a desired value and the actual value, expressed as a percent of the desired value.

To determine the percent error, the amount of the error is compared to the desired value. You can use this formula:

$$\text{Percent error} = \frac{(\text{difference between actual value and the desired value})}{\text{desired value}} \cdot 100$$

For example, a milk carton is supposed to contain 16 fluid ounces, but it only contains 15 fluid ounces.

- The error is 1 fluid ounce.
- The percent error is 6.25% because $\frac{1}{16} \cdot 100 = 6.25$.

In some situations, there is no clear “desired” value. In those cases, the denominator is the value that has no error. Here are some examples:

- For a thermometer that reads 73°, if the real temperature is 70°, the percent error for that thermometer’s reading is $\frac{3}{70}$, or 4.3%.
- For an estimate of 800 jelly beans in a jar, if the jar actually has 947 jelly beans, the percent error of the estimate is $\frac{147}{947}$, or 15.5%.

Things to Remember:

Lesson Practice

ACC6.6.15

Name: _____ Date: _____ Period: _____

1. A baker needs 500 grams of flour for a batter, but accidentally only puts 300 grams of flour into the mixing bowl. What is the percent error?
40%
2. Any can of seltzer should have 12 fluid ounces of liquid to be appropriately carbonated. One can on the assembly line contains 13.5 fluid ounces. What is the percent error?
12.5%
3. A radar gun measured the speed of a baseball at 93 miles per hour. If the baseball was actually going 90 miles per hour, what was the percent error in this measurement?
About 3.3%

Problems 4–5: In Burlington, all city departments get an annual budget. They are allowed to spend 1.5% over their budget. If they spend more than that, the mayor conducts a review.

4. The Department of Health has a budget of \$90,000. What is the largest acceptable amount for their spending? Show or explain your thinking.
\$91,350. Explanations vary.
 $90000 \cdot 0.015 = 1350$
 $90000 + 1350 = 91350$
5. The Department of Parks and Recreation has a budget of \$30,000, and they spent \$31,000. Will the mayor conduct a review of their spending? Show or explain your thinking.
Yes. Explanations vary.
 $31000 - 30000 = 1000$
 $1000 \div 30000 = 0.0333\ldots$
 $0.033 \cdot 100 = 3.3$
3.3% is more than 1.5%.

Lesson Practice

ACC6.6.15

Name: _____ Date: _____ Period: _____

6. Hoang is buying 5 concert tickets that cost \$95.00 each. The concert venue gives Hoang a coupon with a 10% discount. A sales tax of 8% is applied after the discount. How much did Hoang pay after the discount and sales tax?

A. \$92.34 B. \$427.50 C. \$461.70 D. \$513.00

7. Amal bikes x kilometers. Kimora bikes $\frac{3}{10}$ less than that. Using y for Kimora's distance, write an equation that describes the relationship between the two quantities.

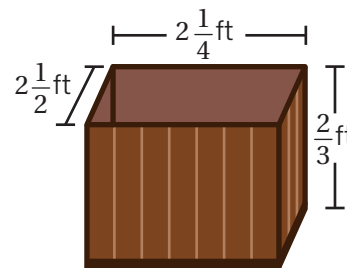
$y = \frac{7}{10}x$ (or equivalent)

Spiral Review

Problems 8–10: Raven is building a new planter for her class garden.

8. The base of her planter is $2\frac{1}{2}$ feet by $2\frac{1}{4}$ feet. What is its area?

$5\frac{5}{8}$ square feet



9. If the planter is $\frac{2}{3}$ feet high, what volume of soil does she need to fill it?

$3\frac{3}{4}$ cubic feet

10. Each bag holds $\frac{3}{4}$ cubic feet of soil. How many bags of soil does Raven need?

5 bags of soil

Reflection

1. Put a question mark next to a response you'd like to compare with a classmate's.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

Situations involving percent increase and decrease are everywhere in our society.

For example, news articles often contain facts and statistics about pollution, such as:

In 2019, the U.S. generated 72.8 million tons of plastic waste.

This was 55% more waste than in 2000.

Information like this can be used to generate interesting questions. When writing these questions, it's important to be precise with language. We could ask: *How much waste was there in 2000?* But a more precise question might be: *How many tons of plastic waste did the U.S. generate in 2000?*

We can use strategies from this unit, such as equations, double number lines, tables, and tape diagrams, to answer these kinds of questions.

Things to Remember:

Lesson Practice

7.4.10

Name: _____ Date: _____ Period: _____

Problems 1–2: A city has a 5% sales tax.

1. A toothbrush costs \$3.40 before tax. How much does it cost including tax?

\$3.57

2. A book costs \$32.55 after tax. How much did it cost before tax?

\$31

Problems 3–5: Oliver went to the store and purchased these items.

3. Oliver set a monthly grocery budget for himself.
This grocery list costs 6% of the budget.
What is his monthly budget? **\$300**

4. Complete each sentence:

Milk made up about **20** % of Oliver's spending on this trip.

This was about **1.2** % of his grocery budget for the month.

Milk (1 gal)	\$3.61
Beef (1 lb)	\$7.30
Apples (1 lb)	\$2.39
Bananas (1 lb)	\$0.99
Oranges (1 lb)	\$1.96
Potatoes (1 lb)	\$1.75
Total	\$18.00

5. Use the given information to write another sentence about Oliver's spending. Include a percentage.

Responses vary. Beef made up about 40% of Oliver's spending on this trip.

Lesson Practice

7.4.10

Name: _____ Date: _____ Period: _____

6. A greeting card costs \$6 before tax. A customer has a coupon for a 10% discount. Then a 5% sales tax is added. How much will the customer pay for the greeting card?

\$5.67

7. The price of gold is often reported per ounce. At the end of 2005, gold was \$513 per ounce. At the end of 2015, it was \$1,060 per ounce. By what percent did the price increase?

About 107%

8. A grocery store allows customers to use multiple coupons when checking out. Suppose you have a \$5 off coupon and a 10% off coupon. The register will calculate the new price after each coupon is used. Does the order you use the coupons make a difference? Explain your thinking.

Yes. Explanations vary. I will test what will happen if I buy an item that costs \$10 and check both possible orders for the coupons.

Using \$5 off first:

$$10 - 5 = 5$$

$$5 \cdot 0.9 = 4.50$$

The item will cost \$4.50.

Using 10% off first:

$$10 \cdot 0.9 = 9$$

$$9 - 5 = 4$$

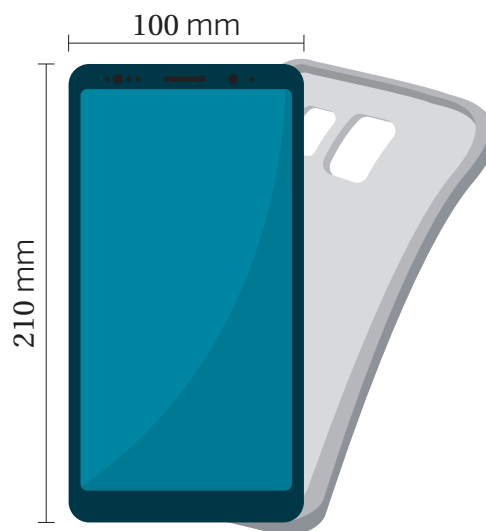
The item will cost \$4.00.

When buying an item that costs \$10, you save an extra \$0.50 by using the 10% off coupon first.

Spiral Review

9. Here are the dimensions of a phone case designed by the Soft Shield company. The size of a case can be up to 1% off and still fit a phone. Determine the largest and smallest width and height of the case that will fit a phone.

The phone case's width can be as small as 99 millimeters and as wide as 101 millimeters. Its height can be as small as 207.9 millimeters and as large as 212.1 millimeters.



10. What is the decimal equivalent of $\frac{13}{100}$?

0.13

Reflection

1. Put a star next to a problem that looked more difficult than it really was.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

When a problem involves a proportional relationship, determining the constant of proportionality can be helpful.

You can see this relationship between the columns of this table. The heights are multiplied by $2\frac{1}{2}$ to get the widths.

When you multiply one quantity in a proportional relationship by a value, the other quantity will change by the same factor whether or not the values are whole numbers.

You can see this relationship between the rows of the table. When the height is multiplied by $1\frac{3}{4}$, the width is multiplied by the same number.

Height (in.)	Width (in.)
2	5
$3\frac{1}{2}$	$8\frac{3}{4}$

Height (in.)	Width (in.)
2	5
$3\frac{1}{2}$	$8\frac{3}{4}$

Things to Remember:

Lesson Practice

ACC6.6.16

Name: _____ Date: _____ Period: _____

Problems 1–3: A snail is moving away from a rock at a constant rate. This table shows the distance the snail is from the rock at certain times.

Distance (in.)	Time (min)
0	0
1	$1\frac{1}{3}$

1. How many minutes does it take for the snail to reach a distance of 9 inches from the rock?

12 minutes

2. How far will the snail be from the rock after 9 minutes?

$6\frac{3}{4}$ inches

3. Select *all* the equations that represent the relationship between the distance in inches, d , and time in minutes, t .

☐ A. $d = \frac{4}{3}t$

☒ B. $d = \frac{3}{4}t$

☒ C. $t = \frac{4}{3}d$

☐ D. $t = \frac{3}{4}d$

Problems 4–5: At a deli counter, a customer buys:

- $1\frac{3}{4}$ pounds of ham for \$14.50.
- $2\frac{1}{2}$ pounds of turkey for \$26.25.
- $\frac{3}{8}$ pounds of roast beef for \$5.50.

4. Which deli meat is the least expensive per pound? Show or explain your thinking.

Ham. Explanations vary.

Ham: $14.50 \div 1\frac{3}{4} = 8\frac{2}{7} \approx 8.29$

Turkey: $26.25 \div 2\frac{1}{2} = 10.50$

Roast beef: $5.50 \div \frac{3}{8} \approx 14.67$

5. Which deli meat is the most expensive per pound?

Roast beef

Lesson Practice

ACC6.6.16

Name: _____ Date: _____ Period: _____

6. To make a shade of paint called Jasper Green, mix 4 quarts of green paint with $\frac{2}{3}$ of a cup of black paint.

How much green paint should be mixed with 4 cups of black paint to make Jasper Green?

24 quarts

Spiral Review

Problems 7–8: This table shows a relationship between x and y .

x	y
1	8
2	16
3	24

7. What is a constant of proportionality in the relationship?

8 or $\frac{1}{8}$

8. Write an equation to represent the relationship.

$y = 8x$ (or equivalent)

9. Angel checks out 12 library books, and Shikoba checks out $\frac{1}{3}$ less than that. How many books does Shikoba check out?

8 books

10. Which is greater? Circle one.

40% of 12

12% of 40

They are the same

Show or explain your thinking.

Explanations vary. $0.4 \cdot 12 = 0.12 \cdot 40$

11. 12 is 40% of what number? Show or explain your thinking.

30. Work varies. $12 \div 0.40 = 30$

Reflection

- Circle the problem you feel least confident about.
- Use this space to ask a question or share something you're proud of.

Lesson Summary

Proportional relationships may involve fractional amounts. You can solve problems involving fractions by using the same strategies you use to solve problems with whole numbers.

- To determine the constant of proportionality within a recipe, divide the amount of an ingredient by the total number of servings.
- Constants of proportionality can help to compare proportional relationships involving fractional quantities.

Here is a recipe for banana bread. To find the amount of sugar per serving, divide $\frac{3}{4}$ cups of sugar by 6 servings. This gives you $\frac{3}{4} \div 6$, or $\frac{1}{8}$ cups of sugar per serving.

Banana Bread Recipe

Number of servings: 6

- 2 lb of bananas
- $\frac{1}{2}$ cups of butter
- $\frac{3}{4}$ cups of sugar
- $2\frac{1}{2}$ cups of flour
- 1 tsp of baking soda

Things to Remember:

Lesson Practice

ACC6.6.17

Name: _____ Date: _____ Period: _____

1. A recipe calls for $\frac{1}{2}$ cups of sugar and 1 cup of flour. Complete the table to show how much sugar and flour is needed for different batches of the recipe.

Sugar (cups)	Flour (cups)
$\frac{1}{2}$	1
$\frac{3}{4}$	$1\frac{1}{2}$
$\frac{7}{8}$	$1\frac{3}{4}$
1	2
$1\frac{1}{4}$	$2\frac{1}{2}$

Problems 2–4: A punch recipe calls for $1\frac{1}{2}$ quarts of sparkling water and $\frac{3}{4}$ quarts of grape juice.

2. How much sparkling water would you need to mix with 9 quarts of grape juice?

18 quarts

3. How much grape juice would you need to mix with $3\frac{3}{4}$ quarts of sparkling water?

$\frac{15}{8}$ quarts (or equivalent). Note: The ratio $1\frac{1}{2}$ quarts of sparkling water to $\frac{3}{4}$ quarts of grape juice is equivalent to a 2 : 1 ratio.

4. How much of each ingredient would you need to make 75 quarts of punch?

Sparkling water: **50 quarts**

Grape juice: **25 quarts**

Problems 5–6: To make a specific color of green paint, a painter mixes $\frac{1}{2}$ of a gallon of blue paint with $\frac{4}{5}$ of a gallon of yellow paint.

5. How many gallons of yellow paint are needed to mix with 3 gallons of blue paint?

$4\frac{4}{5}$ gallons (or equivalent)

6. How many gallons of each color are needed to make 26 total gallons of this green paint?

Blue paint: **10 gallons**

Yellow paint: **16 gallons**

Lesson Practice

ACC6.6.17

Name: _____ Date: _____ Period: _____

Spiral Review

7. Complete the table to represent $y = \frac{2}{3}x$.

x	y
12	8
24	16

8. Select *all* the ratios that are equivalent to 4 : 5.

☒ A. 2 : 2.5

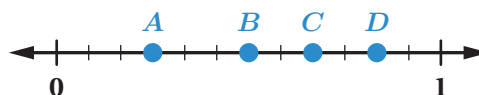
☐ B. 3 : 4

☒ C. 3 : 3.75

☒ D. 8 : 10

☐ E. 14 : 27.5

9. Here are four points plotted on a number line. Which point best represents $66\frac{2}{3}\%$ of the distance between 0 and 1?



A. Point A

B. Point B

☒ C. Point C

D. Point D

10. What is 40% of 160?

64

11. What is 160% of 40?

64

Reflection

- Put a heart next to the problem you found most interesting.
- Use this space to ask a question or share something you're proud of.

Lesson Summary

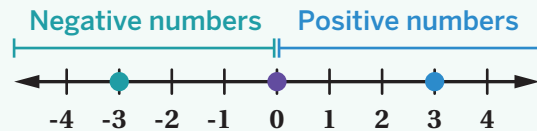
Positive numbers are numbers that are greater than 0. **Negative numbers** are numbers that are less than 0. Zero is neither positive nor negative.

You can extend a number line to the right of 0 to show positive numbers, and you can extend a number line to the left of 0 to show negative numbers.

For example:

The number 3 is 3 units to the right of 0 on the number line.

The number -3 is 3 units to the left of 0 on the number line.



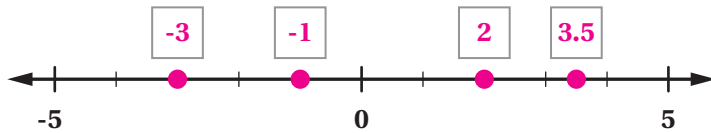
Things to Remember:

Lesson Practice

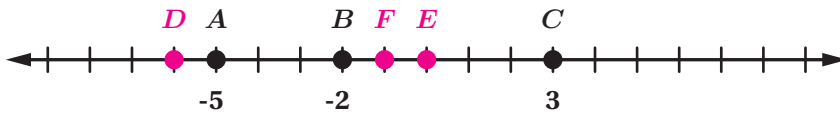
ACC6.7.01

Name: _____ Date: _____ Period: _____

1. Fill in the blanks on the number line.



Problems 2–6: Here is a number line.



2. Describe where you would plot -100 on the number line.

Responses vary. The number -100 would be really far to the left of 0 on the number line.

3. Point *D* is 1 unit to the left of point *A*. Plot point *D*.

Response shown on the number line.

4. Point *E* is at 0. Plot point *E*.

Response shown on the number line.

5. List both locations that are 4 units away from point *B*.

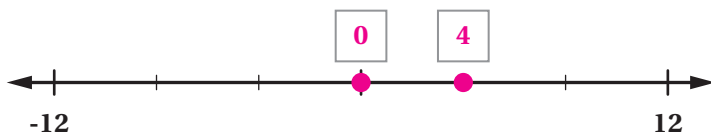
-6 and 2

6. Point *F* is the same distance from point *A* and point *C*. Plot point *F*.

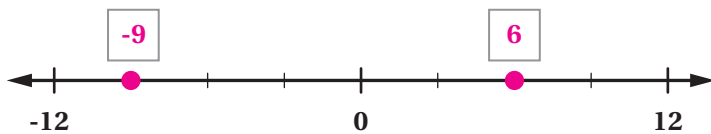
Response shown on the number line.

Problems 7–9: Fill in the blanks on the number lines.

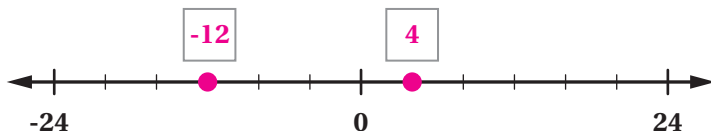
7.



8.



9.



10. Which pair of numbers are on opposite sides of zero on the number line?

A. 0 and 5

B. 2 and -5

C. -2 and -5

D. 2 and 5

Lesson Practice

ACC6.7.01

Name: _____ Date: _____ Period: _____

Spiral Review

11. A rectangle has an area of 24 square centimeters. If one side is $2\frac{2}{5}$ centimeters long, how long is the other side?

$2\frac{2}{5}$ cm

24 sq. cm

Explain your thinking.

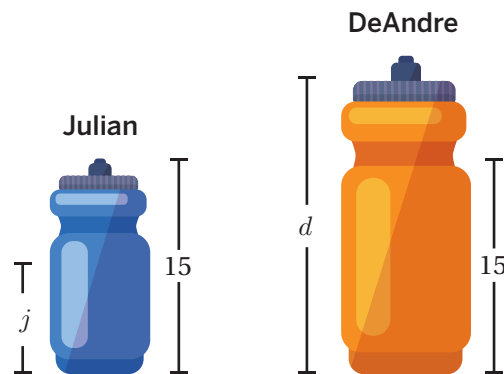
10 centimeters. Work varies. $24 \div 2\frac{2}{5} = 24 \div \frac{12}{5} = 24 \cdot \frac{5}{12} = 10$

Problems 12–14: Two friends at soccer practice are drinking from their water bottles. Julian drinks $\frac{3}{5}$ of his 15-ounce bottle. DeAndre drinks 15 ounces of water, which is $\frac{3}{5}$ of his bottle.

12. How are these two situations alike? How are they different?

Responses vary.

- Both situations are about a person drinking from a water bottle.
- Both situations use the number 15 and the fraction $\frac{3}{5}$.
- In Julian's situation, the whole bottle is 15 ounces. In DeAndre's situation, the 15 ounces is how much water he drinks.
- DeAndre's water bottle is bigger than Julian's water bottle.



13. Write an equation to represent the amount of water Julian drinks, j .

$\frac{3}{5} \cdot 15 = j$ (or equivalent)

14. Write an equation to represent the total amount of water in DeAndre's bottle, d .

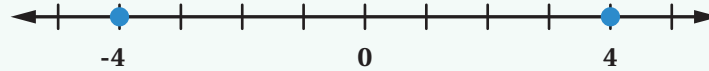
$\frac{3}{5} \cdot d = 15$ (or equivalent)

Reflection

1. Put a smiley face next to a problem you were stuck on and then figured out.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

Two numbers are **opposites** if they are the same distance from 0 on different sides of the number line. For example, -4 and 4 are opposites because they are both 4 units away from 0.



Every number has an opposite, including fractions and decimals. 0 is its own opposite. The opposite of the opposite of a number is the number itself. For example, $-(-2) = 2$.

All positive and negative whole numbers and 0 are a group of numbers called *integers*. All positive and negative numbers that can be written as fractions, including whole numbers, are called *rational numbers*.

2 and -2 are both integers and rational numbers.

8.3, -8.3, $\frac{3}{2}$, and $-\frac{3}{2}$ are rational numbers, but *are not* integers.

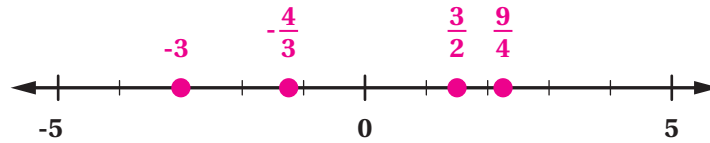
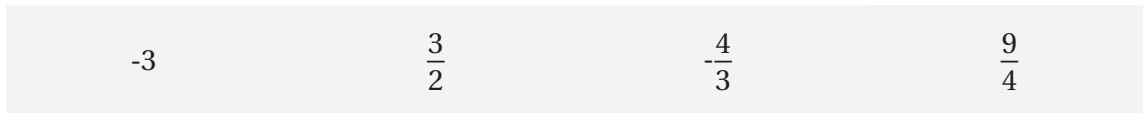
Things to Remember:

Lesson Practice

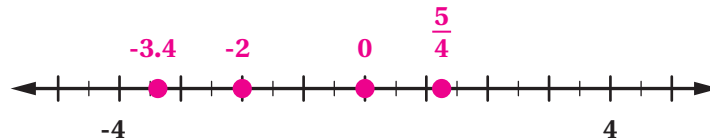
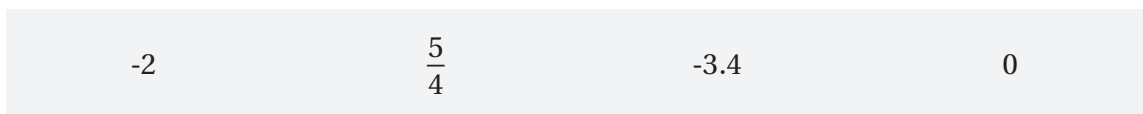
ACC6.7.02

Name: _____ Date: _____ Period: _____

- Plot each number in its approximate location on the number line.



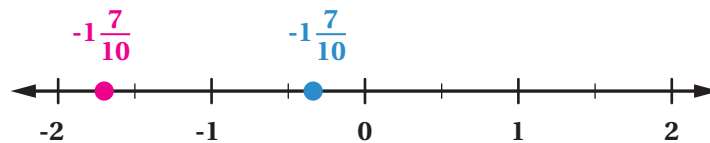
- Plot each number in its approximate location on the number line.



Problems 3–6: Complete each statement below.

- The opposite of -2 is 2.
- The opposite of $\frac{5}{4}$ is $-\frac{5}{4}$.
- The opposite of -3.4 is 3.4.
- The opposite of 0 is 0.

Problems 7–8: Rebecca incorrectly plotted the point $-1\frac{7}{10}$ on the number line.



- What question could you ask to help Rebecca understand her mistake?

Questions vary. Is $-1\frac{7}{10}$ closer to -1 or -2?

- Plot $-1\frac{7}{10}$ in the correct location on the number line.

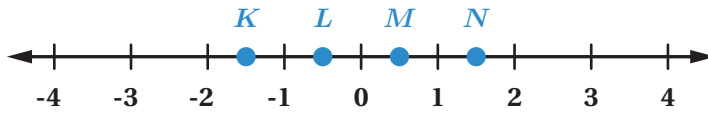
Response shown on the number line.

Lesson Practice

ACC6.7.02

Name: _____ Date: _____ Period: _____

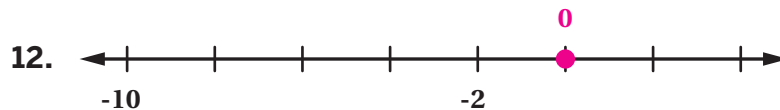
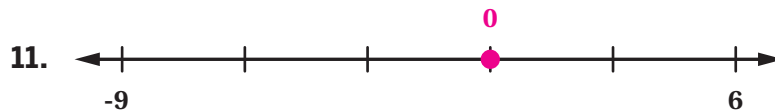
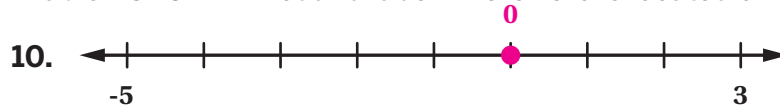
9. Points K , L , M , and N are plotted on the number line.



What point represents the location of -0.5 ?

- A. Point K **B. Point L** C. Point M D. Point N

Problems 10–12: Plot and label where zero is located on each number line.



Spiral Review

Problems 13–16: Solve each equation.

13. $3x = 6$
 $x = 2$

14. $\frac{1}{3}x = 6$
 $x = 18$

15. $\frac{1}{3} = 6x$
 $x = \frac{1}{18}$ (or equivalent)

16. $\frac{1}{3} = \frac{1}{6}x$
 $x = 2$ (or equivalent)

Reflection

1. Circle a problem you want to talk to a classmate about.
2. Use this space to ask a question or to share something you're proud of.

Lesson Summary

You can use a vertical number line to represent positive and negative numbers. On a vertical number line, points above 0 are positive and points below 0 are negative.

When talking about elevation, 0 feet represents sea level. This means that a positive elevation is above sea level and a negative elevation is below sea level.

When talking about temperature, 0°C means the temperature is freezing. If the temperature in Mt. Olympus is -10°C , that means it has a temperature of 10°C below 0°C , or below freezing.

If the temperature in Cotopaxi is -3°C , you can write $-10 < -3$, which means that it is colder in Mt. Olympus than it is in Cotopaxi.

Things to Remember:

Lesson Practice

ACC6.7.03

Name: _____ Date: _____ Period: _____

Problems 1–3: Determine if each statement is true or false.

Statement	True	False
1. An elevation of 35 feet is the same as an elevation of -35 feet.		✓
2. A city that has an elevation of -17 meters is closer to sea level than a city that has an elevation of -40 meters.	✓	
3. A temperature of -4°F is the same as a temperature of 4°F below zero.	✓	

Problems 4–7: Here is a table that shows some elevations in several cities around the world.

Location	Elevation (ft)
San Juan, Puerto Rico	-4
New Orleans, Louisiana	-7
Amsterdam, Netherlands	-2
Jakarta, Indonesia	3
Taipei, Taiwan	5
Tunis, Tunisia	0

4. A city, not listed in the table, has a higher elevation than San Juan, Puerto Rico.

Select *all* the numbers that could represent the city's elevation in feet.

- ☒ A. 0 ☒ B. 4
☒ C. -2 ☐ D. -10
☐ E. -7

5. Plot all of the elevations in the table on the number line.

Responses shown on the number line.

6. Mio says: *I know that 2 is less than 4, so -2 must be less than -4. This means Amsterdam has a lower elevation than San Juan.* Is Mio correct? Explain your thinking.

No. Explanations vary. Mio is incorrect because -4 is located below -2 on the number line.

7. Nasir says: *3 is less than -7 because 3 is closer to 0 on the number line.* Is Nasir correct? Explain your thinking.

No. Explanations vary. Nasir is incorrect because 3 is positive and -7 is negative. Positive numbers are always greater than negative numbers, no matter how close they are to 0.



Lesson Practice

ACC6.7.03

Name: _____ Date: _____ Period: _____

Problems 8–10: Here is a table that shows some of the lowest temperatures recorded in five U.S. locations.

Location	Temperature (°F)
Boca, CA	-45
Coventry, CT	-32
Tallahassee, FL	-2
Mt. Carroll, IL	-38
CCC Fire Camp F-16, GA	-17

8. Which of these locations had the lowest record temperature?

Boca, CA

9. Which location had a lower record temperature: Tallahassee, FL, or CCC Fire Camp F-16, GA? Write a statement using $<$ or $>$ to compare the recorded temperatures.

CCC Fire Camp F-16, GA. $-17 < -2$ or $-2 > -17$

10. Which location had a lower record temperature: Coventry, CT, or Mt. Carroll, IL? Write a statement using $<$ or $>$ to compare the recorded temperatures.

Mt. Carroll, IL. $-32 > -38$ or $-38 < -32$

Spiral Review

Problems 11–14: Determine the value of each expression.

11. $2^3 \cdot 4 = 32$

12. $\frac{2^3}{4} = 2$

13. $2^4 - 4 = 12$

14. $2^3 + 4^3 = 72$

Reflection

- Put a heart next to a problem you feel most confident about.
- Use this space to ask a question or to share something you're proud of.

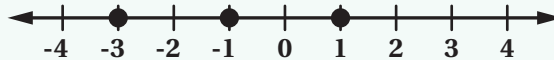
Lesson Summary

You can use a number line to compare numbers with different **signs** (like -4 and 3) or numbers with the same sign (like -4 and -3).

The order of numbers from least to greatest is the same order as they appear on the number line from left to right. This means that negative numbers farther from 0 are less than negative numbers that are closer to 0.

For example, let's say you want to compare -3 and -1. On a number line, -1 is to the right of -3. This means that -1 is greater than -3, or $-3 < -1$. This also makes sense because -1 is closer to 0 than -3 is.

A number line can also help you order numbers from least to greatest. 1 is greater than -1 and -3 because it is the farthest to the right on the number line.



Things to Remember:

Lesson Practice

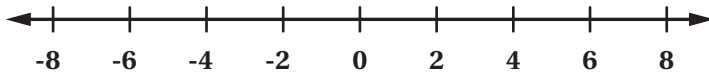
ACC6.7.04

Name: _____ Date: _____ Period: _____

1. Complete each statement with a number that makes it true.

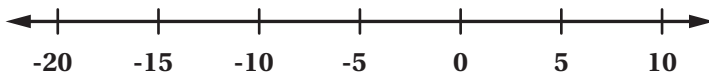
Use the number line if it helps with your thinking. *Responses vary.*

a $\square < 5$ b $\square < -5$ c $-5 < \square$ d $-5 > \square$



2. Complete each statement with the symbol $<$, $>$, or $=$. Use the number line if it helps with your thinking.

a $-5 < 2$ b $5 > -5$ c $-12 > -15$ d $-12.5 < -12$



3. Circle whether each statement is *true* or *false*.

Statement A: -8.4 is to the right of -8.7 on the number line.

True False

Statement B: -2.4 is greater than -2.3.

True False

Statement C: $-\frac{11}{12} < -\frac{7}{12}$

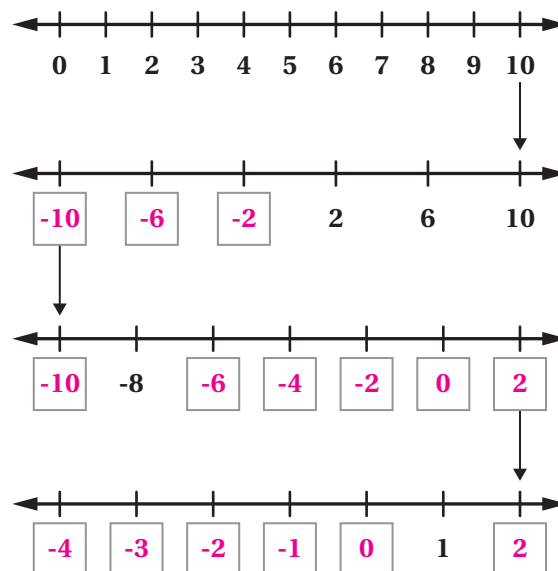
True False

Choose one statement and explain your thinking.

Responses and explanations vary. Statement B is false because -2.4 is farther left on the number line, so $-2.4 < -2.3$.

4. Here is a number line maze.

Each number line has a different scale but at least one matching number, which is labeled with an arrow. Determine the missing values in this number line maze.



Lesson Practice

ACC6.7.04

Name: _____ Date: _____ Period: _____

5. Here are five numbers: $-\frac{2}{5}$, -1 , $\frac{4}{3}$, 1 , $-\frac{3}{2}$.

These numbers are plotted on a horizontal number line. Which statement about the locations of the numbers is true?

- A. $-\frac{2}{5}$ is the farthest to the left, and 1 is farthest to the right.
 B. -1 is the farthest to the left, and 1 is farthest to the right.
 C. $\frac{4}{3}$ is the farthest to the left, and $-\frac{3}{2}$ is farthest to the right.
 D. $-\frac{3}{2}$ is the farthest to the left, and $\frac{4}{3}$ is farthest to the right.

Spiral Review

6. Solve each equation. Write each solution as a fraction and as a decimal.

Equation	Solution (Fraction)	Solution (Decimal)
$2x = 3$	$x = \frac{3}{2}$ (or equivalent)	$x = 1.5$
$5y = 3$	$y = \frac{3}{5}$ (or equivalent)	$y = 0.6$
$0.3z = 0.09$	$z = \frac{3}{10}$ (or equivalent)	$z = 0.3$

Problems 7–10: Each lap around a track is 400 meters. How many meters will someone run in:

7. 2 laps?
800 meters
8. 5 laps?
2,000 meters
9. x laps?
 $400x$ meters

10. Sol ran 7,600 meters. How many laps is that? Explain your thinking.

19 laps. *Explanations vary. If Sol ran 8,000 meters, that would be 20 laps. Because Sol ran 400 meters less than 8,000 meters, Sol ran $20 - 1 = 19$ laps.*

Reflection

- Put a star next to one problem you are still wondering about.
- Use this space to ask a question or share something you're proud of.

Lesson Summary

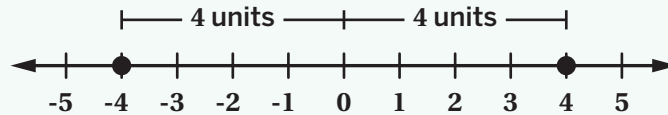
The **absolute value** of a number is a way to describe its distance from 0. For example:

The absolute value of -4 is 4,
because it is 4 units away from 0.

$$|-4| = 4$$

The absolute value of 4 is also 4,
because it is 4 units away from 0.

$$|4| = 4$$



The distance from 0 to itself is 0, so $|0| = 0$.

Absolute values are helpful when you are interested in the size of a difference or measurement but its direction is not important.

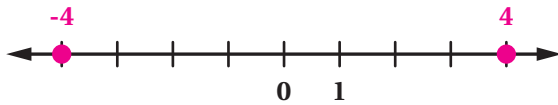
Things to Remember:

Lesson Practice

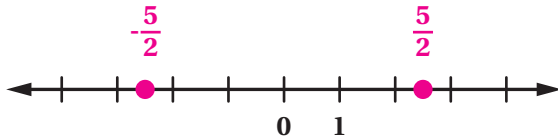
ACC6.7.05

Name: _____ Date: _____ Period: _____

1. Plot and label all numbers that are 4 units away from 0.



2. Plot and label all numbers with an absolute value of $\frac{5}{2}$.



3. Ivory claims that a number and its opposite will always have the same absolute value. Is Ivory correct? Explain your thinking.

Yes. Explanations vary. Opposite numbers are the same distance away from 0. Because absolute value is another way of saying how far away a number is from 0, opposites have the same absolute value.

Problems 4–11: Complete each number sentence with the symbol $<$, $>$, or $=$.

4. -3.2 $<$ 1.5

5. $|-3.2|$ $>$ $|1.5|$

6. 2 $>$ -1.5

7. $|2|$ $>$ $|-1.5|$

8. $\frac{3}{2}$ $>$ -1.5

9. $|\frac{3}{2}|$ $=$ $|-1.5|$

10. $|-2.7|$ $<$ $|-4.5|$

11. $|-2.7|$ $>$ -4.5

12. Which list of absolute value expressions is ordered from *least* to *greatest* value?

A. $|1|, |\frac{9}{4}|, |-5|, |-7|$

B. $|-7|, |-5|, |1|, |\frac{9}{4}|$

C. $|-7|, |-5|, |\frac{9}{4}|, |1|$

D. $|\frac{9}{4}|, |1|, |-5|, |-7|$

Lesson Practice

ACC6.7.05

Name: _____ Date: _____ Period: _____

- 13.** Make each statement true by using each number at most once.

Responses vary. $|-3| = 3$, $|-2| > 1$, and $-1 < |2|$

$$|\square| = \square$$

$$|\square| > \square$$

$$\square < |\square|$$

-3

-2

-1

0

1

2

3

Spiral Review

Problems 14–17: Determine the value of each quotient.

14. $24 \div 15 = 1.6$
(or equivalent)

15. $0.24 \div 0.15 = 1.6$
(or equivalent)

16. $0.24 \div 0.015 = 16$

17. $0.024 \div 0.015 = 1.6$
(or equivalent)

Reflection

1. Circle the problem you feel most confident about.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

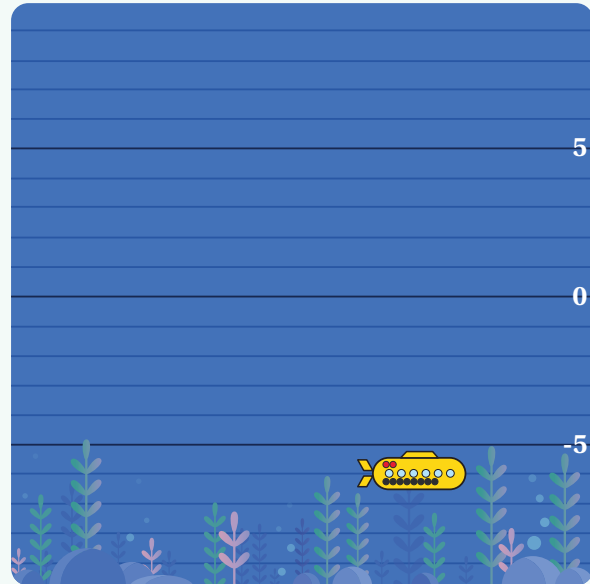
Using models, such as floats and anchors on a vertical number line, can be useful when representing addition and subtraction of positive and negative numbers.

For example, imagine a submarine whose position is at -6 units. The submarine will move from its position as 3 floats are added and 2 anchors are removed.

- 3 floats being added represents moving up 3 units or adding 3.
- 2 anchors being removed represents moving up 2 units because $-(-2) = 2$.

The submarine's new position would be $-6 + 3 + 2 = -1$ unit.

To move the submarine to 0 units from -1 unit, 1 float can be added. -1 and 1 are an *opposite pair*, which means they sum to 0.



Things to Remember:

Lesson Practice

ACC6.7.06

Name: _____ Date: _____ Period: _____

Problems 1–5: One moment in December, it was -8°C in Harbin, China, and -2°C in Beijing, China. Use the number line if it helps with your thinking.

1. Which city was colder?

Harbin

2. At the same moment, it was 7 degrees warmer in Shanghai than it was in Beijing. What was the temperature in Shanghai?

5°C

3. How many degrees warmer was it in Shanghai than in Harbin?

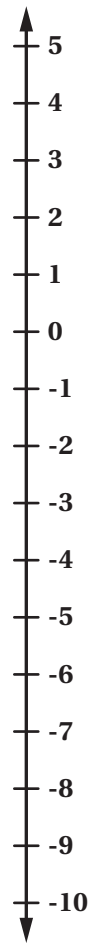
13 degrees

4. Later in the day, Beijing got 5 degrees colder. What was Beijing's new temperature?

-7°C

5. Later in the day, Harbin's temperature was 1°C . By how much did Harbin's temperature change? Explain your thinking.

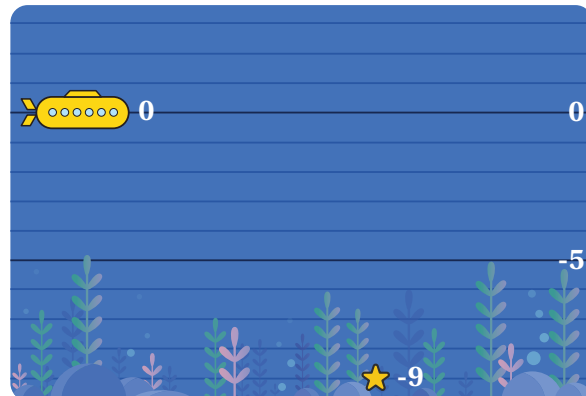
9 degrees. *Explanations vary. The distance from -8 to 1 on a number line is 9 units.*



Problems 6–7: This submarine has 0 floats and 0 anchors. The submarine can hold up to 10 floats and 10 anchors.

6. List *all* the combinations of floats and anchors that could collect the star at -9 units.

- 9 anchors
- 10 anchors and 1 float



7. How do you know there are no other combinations?

Responses vary. It takes 9 anchors to get down to -9. Every other anchor needs to be balanced with a float. Since the submarine only has space for 10 anchors, you can only add 1 float.

Lesson Practice

ACC6.7.06

Name: _____ Date: _____ Period: _____

Spiral Review

8. Fill in each blank using the symbols $>$, $<$, or $=$.

3	$>$	-3
12	$<$	24
-12	$>$	-24
7	$<$	7.2
-7	$>$	-7.2

Problems 9–11: A color of green paint is made by mixing 2 cups of yellow paint with 3.5 cups of blue paint.

9. Complete the table to show amounts of yellow and blue paint that will make the same color of green but in a smaller amount.

Responses vary. Sample shown in table.

Yellow Paint (cups)	Blue Paint (cups)
1	1.75

10. Complete the table to show amounts of yellow and blue paint that will make the same color of green but in a larger amount.

Responses vary. Sample shown in table.

Yellow Paint (cups)	Blue Paint (cups)
4	7

11. Will a mixture that is 3 cups of yellow and 4.5 cups of blue be more blue, more yellow, or the same color of green as the original mixture? Explain your thinking.

More yellow. Explanations vary. For a mixture with 3 cups of yellow paint, 5.25 cups of blue paint are needed to get the same color of green as the original mixture. 4.5 cups of blue paint is less than 5.25 cups, which would make the green look more yellow.

Reflection

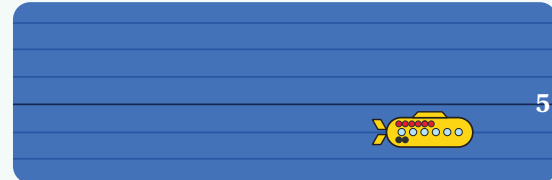
- Put a question mark next to a response you'd like to compare with a classmate's.
- Use this space to ask a question or share something you're proud of.

Lesson Summary

Different combinations of floats and anchors can give you the same result. Here are some examples:

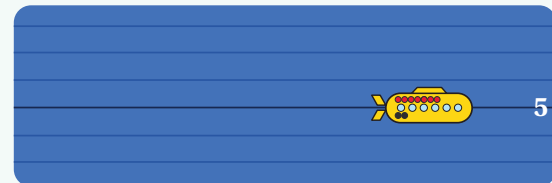
- If a submarine starts at 4 units, adding 2 floats or removing 2 anchors will both result in the submarine moving up to 6 units. So adding a positive number is the same as subtracting a negative number.

Adding Floats	Removing Anchors
$4 + 2 = 6$	$4 - (-2) = 6$



- If a submarine starts at 5 units, removing 1 float or adding 1 anchor will both result in the submarine moving down to 4 units. So subtracting a number is the same as adding its opposite.

Removing Floats	Adding Anchors
$5 - 1 = 4$	$5 + (-1) = 4$



When you add two values that are opposites, the sum is always 0. These numbers are also called *additive inverses* of each other.

Things to Remember:

Lesson Practice

ACC6.7.07

Name: _____ Date: _____ Period: _____

Problems 1–4: Determine the value of each expression.

1. $5 + (-3) = 2$

2. $-5 + 3 = -2$

3. $-5 - 3 = -8$

4. $-5 - (-3) = -2$

5. The temperature was 13°F and then dropped 5 degrees. What was the final temperature?

8°F

6. The temperature was -13°F and then dropped 5 degrees. What was the final temperature?

-18°F

7. The temperature was -13°F and then rose to 5°F . What was the *change* in temperature? Explain your thinking.

18 degrees. Explanations vary. If I use a vertical number line and plot a point at -13 and go up to 5, I first count 13 steps up to reach 0 and then have to go up 5 more steps to reach 5.

8. A swimmer was 8 feet underwater. Then he swam 3 feet deeper. Hikari wrote the expression $-8 - 3$. Charlie wrote the expression $-8 + (-3)$. Explain why both Hikari and Charlie are correct.

Responses vary. 8 feet underwater can be represented by -8. Swimming 3 feet deeper can be represented by -3. Whether I start at -8 and subtract 3 or add -3, I will still move down 3 feet to a position of -11 feet.

Lesson Practice

ACC6.7.07

Name: _____ Date: _____ Period: _____

Problems 9–10: The table shows eight expressions.

9. Determine the value of each expression.

Expression	Value
$1 + 2 - 3$	0
$1 + 2 - 3 + 4$	4
$1 + 2 - 3 + 4 - 5$	-1
$1 + 2 - 3 + 4 - 5 + 6$	5
$1 + 2 - 3 + 4 - 5 + 6 - 7$	-2
$1 + 2 - 3 + 4 - 5 + 6 - 7 + 8$	6
$1 + 2 - 3 + 4 - 5 + 6 - 7 + 8 - 9$	-3
$1 + 2 - 3 + 4 - 5 + 6 - 7 + 8 - 9 + 10$	7

10. What is the value of the next expression? The 10th expression? The 20th expression?

9th expression: -4

10th expression: 8

20th expression: 13

Spiral Review

Problems 11–13: Complete each statement with a value that makes the statement true.

11. _____ < 13

Responses less than 13 are considered correct.

12. _____ < -0.1

Responses less than -0.1 are considered correct.

13. _____ > -2

Responses greater than -2 are considered correct.

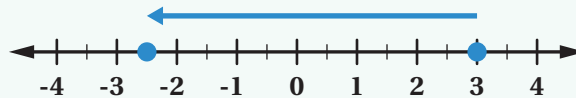
Reflection

1. Put a star next to a problem you're still wondering about.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

When you add positive and negative decimals and fractions, it might help to use a number line and think of each equation as representing $\text{start} + \text{change} = \text{end}$.

For example, in the equation $3 + (-5.5) = x$, 3 represents the starting location, -5.5 represents the change (moving 5.5 units to the left), and x represents the end location.



Things to Remember:

Lesson Practice

ACC6.7.08

Name: _____ Date: _____ Period: _____

Problems 1–3: Determine the value of the variable that makes each equation true.

1. $40 + a = 30$

$a = -10$

2. $-3.5 + c = 4.5$

$c = 8$

3. $d + 2.4 = 0.9$

$d = -1.5$

Problems 4–5: Evaluate each expression.

4. $2 - 3 = -1$

5. $-2 - 3 = -5$

6. Select the equation that is represented by this number line.



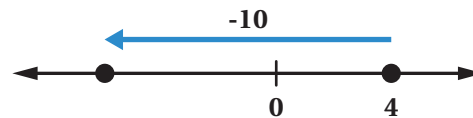
☒ A. $-1 + x = 4$

B. $4 + x = -1$

C. $-1 + 4 = x$

D. $4 + (-1) = x$

7. Select the equation that is represented by this number line.



A. $4 + x = -10$

B. $-10 + x = 4$

☒ C. $4 + (-10) = x$

D. $x + (-10) = 4$

Lesson Practice

ACC6.7.08

Name: _____ Date: _____ Period: _____

Spiral Review

8. Last week, it rained g inches. This week, the amount of rain decreased by 5%. Which expressions represent the amount of rain that fell this week? Select *all* that apply.

- ☐ A. $g - 0.05$ ☒ B. $g - 0.05g$ ☒ C. $0.95g$
☐ D. $0.05g$ ☒ E. $(1 - 0.05)g$

9. Here is an equation representing the relationship between the volume measured in cups, c , and the same volume measured in ounces, z : $c = \frac{1}{8}z$

Does this equation represent a proportional relationship? Circle one.

☒ Yes ☐ No

Explain your thinking.

Explanations vary. The equation is in the form $y = kx$ and the constant of proportionality is $\frac{1}{8}$.

10. Here is an equation representing the length, l , and width, w , for a rectangle whose area is 60 square units: $l = \frac{60}{w}$

Does this equation represent a proportional relationship? Circle one.

☐ Yes ☒ No

Explain your thinking.

Explanations vary. The ratio of w to l will not remain constant. When w is 1, l is 60, but when w is 2, l is 30.

11. Tiam bought a train ticket online. The original price of the train ticket was \$83.00. Tiam used a coupon to receive a 20% discount. A sales tax of 9% was applied after the discount. How much did Tiam end up paying?

- A. \$18.09 B. \$56.44 ☒ C. \$72.38 D. \$75.00

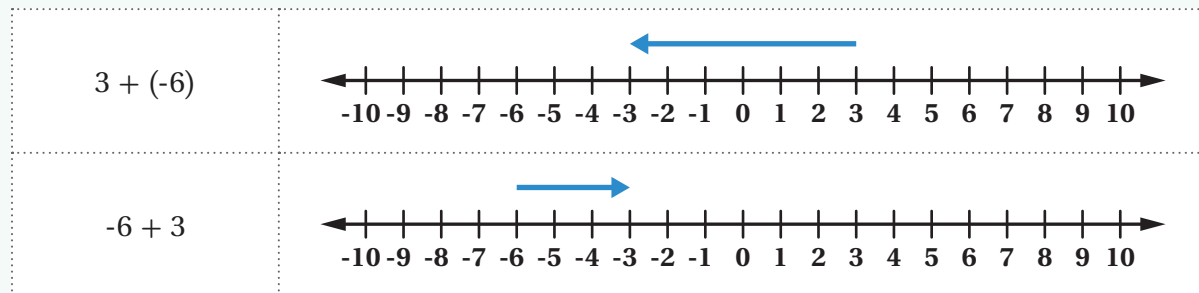
Reflection

1. Put a heart next to the problem you feel most confident about.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

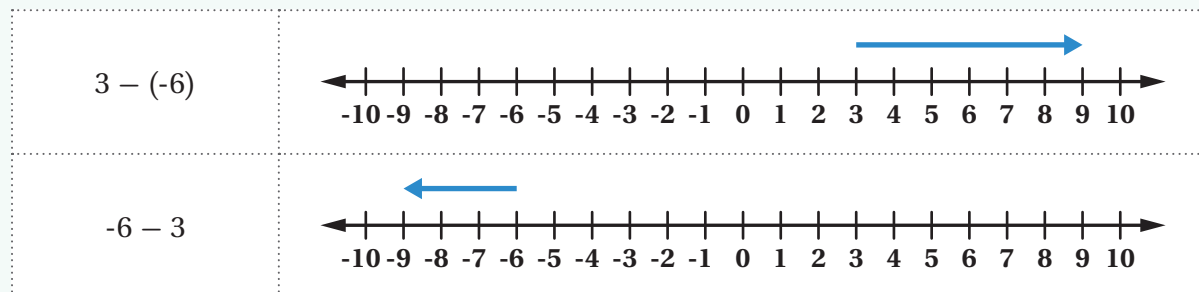
When determining the sum of terms in an expression, the order of the values does not change the final result. This is an example of the *commutative property*.

To represent addition on a number line, start at one of the values and use the other value for the direction and distance of the change.



When determining the difference, the order of the values in an expression *does* affect the final result.

To represent subtraction on a number line, start at the first value and then use the second value to determine the direction and distance of the change. If subtracting a positive number, move to the left; if subtracting a negative number, move to the right.



Things to Remember:

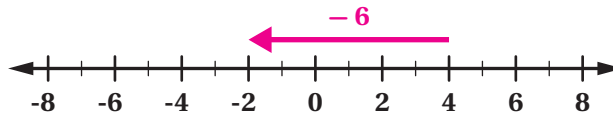
Lesson Practice

ACC6.7.09

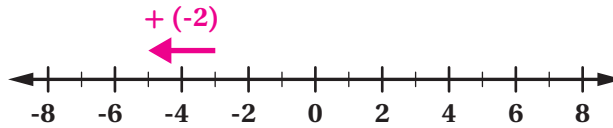
Name: _____ Date: _____ Period: _____

Problems 1–3: Use the number lines to determine the value of each expression.

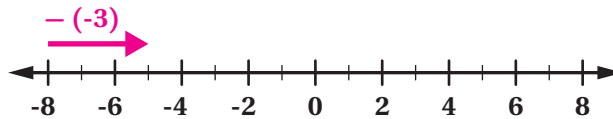
1. $4 - 6 = -2$



2. $-3 + (-2) = -5$



3. $-8 - (-3) = -5$



Problems 4–9: Complete the tables and answer the follow-up questions.

4.

Expression	Value
$2 + (-3)$	-1
$-8 + 4$	-4
$-2 + (-7)$	-9
$-3.6 + (-2.6)$	-6.2

5.

Expression	Value
$-3 + 2$	-1
$4 + (-8)$	-4
$-7 + (-2)$	-9
$-2.6 + (-3.6)$	-6.2

6. Use your work from the previous tables to describe any patterns you notice.

Responses vary. The values of the sums are the same and order doesn't matter for addition.

7.

Expression	Value
$3 - 2$	1
$5 - (-9)$	14
$-6 - (-3)$	-3
$-1.5 - (-4.7)$	3.2

8.

Expression	Value
$2 - 3$	-1
$-9 - 5$	-14
$-3 - (-6)$	3
$-4.7 - (-1.5)$	-3.2

9. Use your work from the previous tables to describe any patterns you notice.

Responses vary. If the order of the numbers being subtracted changes, the value changes to its opposite.

Spiral Review

10. According to the U.S.D.A., hamburger patties should be cooked until the internal temperature is 160°F , for safety. Valeria takes out a frozen patty to cook. Its current temperature is -7°F . How much does its temperature need to increase to reach 160°F ?

167 degrees

Problems 11–13: Bettie's Boutique is having a 20% off sale.

11. Complete the table to determine the sale price for each item.

Responses shown in table.

Item	Original Price	Sale Price
Hat	\$15	\$12
Tie	\$25	\$20
Scarf	\$35	\$28

12. Explain why the relationship between original price and sale price is proportional.

Responses vary. The relationship is proportional because the sale price is a percentage of the original price.

13. Select *all* the equations that represent the relationship between the original price, x , and the sale price, y .

☒ A. $y = 0.8x$

☐ B. $y = x - 0.2$

☒ C. $y = x - 0.2x$

☐ D. $y = 0.2x$

☒ E. $y = (1 - 0.2)x$

Reflection

- Circle a problem you want to talk to a classmate about.
- Use this space to ask a question or share something you're proud of.

Lesson Summary

When adding and subtracting **integers**, fractions, and decimals, there are multiple paths to the same value. Here are some strategies for adding and subtracting positive and negative numbers:

- Imagine the problem as floats and anchors or think about it on a number line. For example, $(-3) + (-4)$ is like starting with (-3) and adding 4 anchors or moving 4 to the left on a number line.
- Rewrite subtraction as addition.
For example, $-3 - 4$ can be rewritten as $-3 + (-4)$, which is -7 .
- Combine numbers that add or subtract to make 0.
For example, when adding -5 and 6 , you can break 6 into $5 + 1$. Using properties of operations, we can add $-5 + 5 + 1$ in pieces. The $-5 + 5$ portion of the expression adds to 0 and $0 + 1 = 1$, so the final value is 1 .

Things to Remember:

Lesson Practice

ACC6.7.10

Name: _____ Date: _____ Period: _____

Problems 1–2: Order the expressions from *least* to *greatest*.

1.

$-5 + (-4)$	$4 + 5$	$-4 + 5$	$4 + (-5)$
$-5 + (-4)$	$4 + (-5)$	$-4 + 5$	$4 + 5$
Least			Greatest

2.

$-5 - (-4)$	$-4 - (-5)$	$-4 - 5$	$4 - (-5)$
$-4 - 5$	$-5 - (-4)$	$-4 - (-5)$	$4 - (-5)$
Least			Greatest

3. Is the solution to $-2.7 + x = -3.5$ positive or negative? Explain your thinking.

Negative. Explanations vary. On a number line, -3.5 is to the left of -2.7. If x were positive, $-2.7 + x$ would be to the right of -2.7. So x must be negative.

Problems 4–7: Determine the value of the variable that makes each equation true.

4. $33 + a = -33.8$
 $a = -66.8$

5. $-9 - b = 3.5$
 $b = -12.5$

6. $c = \left(-\frac{3}{4}\right) + \frac{3}{2}$
 $c = \frac{3}{4}$ (or equivalent)

7. $d + 0.7 = -4$
 $d = -4.7$

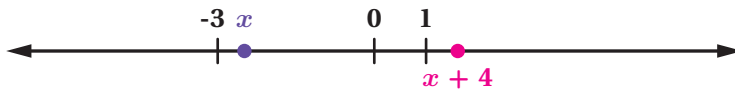
Lesson Practice

ACC6.7.10

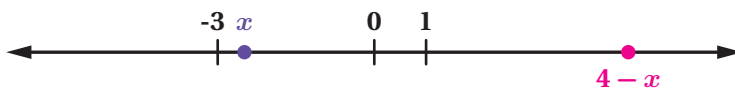
Name: _____ Date: _____ Period: _____

Spiral Review

8. Plot the approximate location of $x + 4$ on the number line.



9. Plot the approximate location of $4 - x$ on the number line.



10. Aditi's Attic is having a \$5 off sale. Is the relationship between original price and sale price proportional? Explain your thinking.

No. Explanations vary. Each price has been reduced by the same amount. If it was a proportional relationship, the difference between the original price and sale price would be smaller for lower prices and larger for higher prices.

Original Price	Sale Price
\$15	\$10
\$25	\$20
\$35	\$30

11. Select the expression that is equivalent to $2\frac{3}{4} - \left(-\frac{7}{8}\right)$.

A. $2\frac{3}{4} - \frac{7}{8}$

B. $2\frac{3}{4} + \frac{7}{8}$

C. $2\frac{3}{4} + \left(-\frac{7}{8}\right)$

D. $-2\frac{3}{4} + \left(-\frac{7}{8}\right)$

Reflection

- Put a smiley face next to the problem you learned from most.
- Use this space to ask a question or share something you're proud of.

Lesson Summary

We can use models, such as floats and anchors, to make sense of multiplying integers. Here are some examples for a submarine that starts at 0 units.

Action	Representation	Submarine's Direction	Final Value
Adding 2 groups of 3 floats	$2 \cdot 3$	Up	6
Removing 2 groups of 3 floats	$-2 \cdot 3$	Down	-6
Adding 2 groups of 4 anchors	$2 \cdot (-4)$	Down	-8
Removing 2 groups of 4 anchors	$-2 \cdot (-4)$	Up	8

Things to Remember:

Lesson Practice

ACC6.7.11

Name: _____ Date: _____ Period: _____

Problems 1–3: Determine the value of the variable that makes each equation true.

1. $3 \cdot a = 12$

$a = 4$

2. $-3 \cdot 4 = b$

$b = -12$

3. $-3 \cdot c = 12$

$c = -4$

Problems 4–6: A weather station on top of a mountain reports that the temperature is currently 0°C and has been decreasing at a constant rate of 3 degrees per hour.

4. What will the temperature be in 5 hours?

-15°C

5. What was the temperature 1 hour ago?

3°C

6. What was the temperature 3 hours ago?

9°C

Problems 7–9: For each equation, check the box to show whether it is true or false. If the equation is false, change one value of the equation to make it true, and write the revised equation in the last column of the table.

	Equation	True	False	Revised Equation
7.	$3 \cdot (-6) = -18$	☒	☐	
8.	$5 \cdot (-2) = 10$	☐	☒	Responses vary. $5 \cdot (-2) = -10$
9.	$(-4) \cdot (-1) = -4$	☐	☒	Responses vary. $(-4) \cdot (-1) = 4$

Lesson Practice

ACC6.7.11

Name: _____ Date: _____ Period: _____

Spiral Review

Problems 10–13: The sales tax rate in Tyler’s state is 6.6%.

- 10.** Write an equation to represent how the cost of an item changes after sales tax is added. Let x represent an original cost and y represent the cost after sales tax.

$$y = 1.066x$$

- 11.** Tyler orders a meal that costs \$15. What is the after-tax amount?

\$15.99

- 12.** If Tyler leaves an 18% tip on the after-tax amount, how much will he end up paying?

\$18.87

- 13.** If Tyler pays with a \$20 bill and gets no change, what percent tip did he leave?

About 25% or 26.73%

Reflection

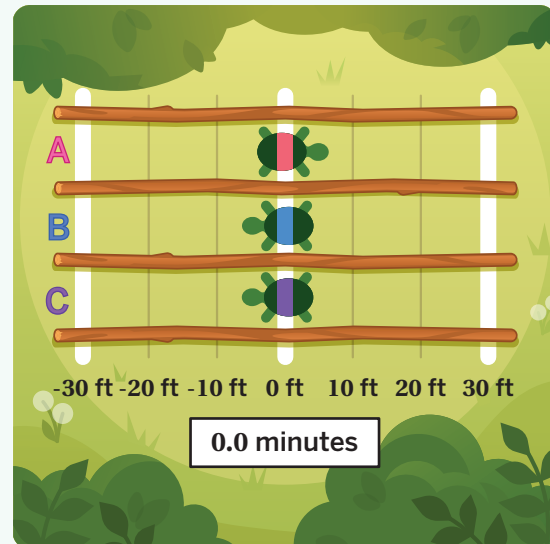
1. Put a question mark next to a problem you’re feeling unsure of.
2. Use this space to ask a question or share something you’re proud of.

Lesson Summary

Multiplying positive and negative numbers can help represent position, rate, and time. The position of an object is equal to the walking rate multiplied by the time or $\text{rate} \cdot \text{time} = \text{change in position}$.

Here are three turtles. They are all together at 0 feet.

- Turtle A walks to the right 6 feet per minute. 3 minutes ago, Turtle A was at -18 feet because $6 \cdot (-3) = -18$.
- Turtle B walks to the left 5 feet per minute. 6 minutes ago, Turtle B was at 30 feet because $(-5) \cdot (-6) = 30$.
- Turtle C walks to the left 3 feet per minute. In 4 minutes, Turtle C will be at -12 feet because $(-3) \cdot 4 = -12$.



Things to Remember:

Lesson Practice

ACC6.7.12

Name: _____ Date: _____ Period: _____

Problems 1–4: Determine the value of each expression.

1. $5(-3) = -15$

2. $-5 \cdot 3 = -15$

3. $(-5)(-3) = 15$

4. $-5 \cdot (-0.3) = 1.5$

Problems 5–8: Anton the Ant can travel up and down an ant nest. Currently, his crawling rate is -7 centimeters per second, which means he's crawling down the ant nest. Anton passes ground level at position 0.

5. Which equation represents Anton's position 8 seconds *after* he passes ground level?

A. $-7 \cdot 8 = 56$

B. $-7 \cdot 8 = -56$

C. $-7 \cdot -8 = 56$

D. $-7 \cdot -8 = -56$

6. Which equation represents Anton's position 8 seconds *before* he passes ground level?

A. $-7 \cdot 8 = 56$

B. $-7 \cdot 8 = -56$

C. $-7 \cdot -8 = 56$

D. $-7 \cdot -8 = -56$

7. Write an ant story that could be represented by the equation $5 \cdot (-6) = -30$.

Responses vary. An ant is crawling up an ant nest at a rate of 5 centimeters per second. The ant passes ground level. The expression shows the position of the ant 6 seconds before it passes ground level.

8. Anton has friends in nearby nests named Anto and Anty. For each description, write an equation in the table that represents it. Assume each ant starts at position 0.

Description	Equation
Anto crawls for 28 seconds at -2 centimeters per second.	$-2 \cdot 28 = -56$
Anty crawls for 3 seconds at -8 centimeters per second.	$-8 \cdot 3 = -24$

Lesson Practice

ACC6.7.12

Name: _____ Date: _____ Period: _____

Spiral Review

Problems 9–10: A tank of water breaks and begins draining at a constant rate.

9. Complete the table.

Time (min)	Water in the Tank (L)
0	770
1	756
2	742
...	...
5	700

10. Is there a proportional relationship between time and the amount of water left in the tank? Explain your thinking.

No. Explanations vary. The water in the tank at the beginning is 770 liters. For it to be a proportional relationship, it would need to begin at 0 liters.

11. Nuka buys a pair of shoes. The shoes cost \$40. The store is having a 20% off sale on everything in the store. How much money does Nuka save from the sale?

\$8.00

Reflection

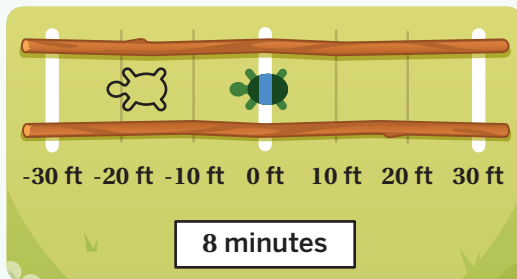
1. Circle the problem you feel most confident about.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

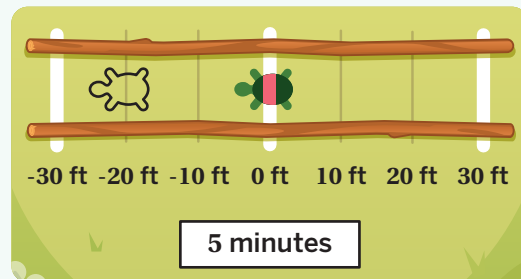
You can use your knowledge of position, rate, and time to multiply and divide positive and negative numbers.

Here are two examples.

Turtle A walks -2 feet per minute. If it starts at 0 feet, how long will it take to walk to -16 feet? Dividing position by walking rate tells us the time, so $\frac{-16}{-2} = 8$ minutes.



Turtle B takes 5 minutes to walk 20 feet to the left, or -20 feet. This means that each minute, Turtle B walks -4 feet. Dividing distance by time tells us the walking rate, so $\frac{-20}{5} = -4$.



Thinking about an expression in terms of position, rate, and time can help you determine whether the value is negative or positive.

Things to Remember:

Lesson Practice

ACC6.7.13

Name: _____ Date: _____ Period: _____

1. Select *all* the expressions that have a negative value.

☒ A. $\frac{-15}{12}$

☒ B. $\frac{15}{-12}$

☐ C. $\frac{-15}{-12}$

☒ D. $-\frac{15}{12}$

☐ E. $\frac{15}{12}$

Problems 2–3: Determine the value of x that makes each equation true.

2. $-3x = 6.3$

$x = -2.1$

3. $\frac{x}{-1.2} = -0.3$

$x = 0.36$

Problems 4–5: A machine that drills holes for wells drilled at a constant rate to a depth of -72 feet in one day (24 hours).

4. How much did the depth change each hour? Make sure to show whether the change is positive or negative.

-3 feet per hour

5. What was the depth after 15 hours?

-45 feet

6. Select *all* the values that are equivalent to $-\frac{12}{7}$.

☒ A. $-\frac{12}{-7}$

☐ B. $-5\frac{1}{7}$

☒ C. $-1\frac{5}{7}$

☐ D. $1\frac{5}{7}$

☐ E. $\frac{-12}{-7}$

Lesson Practice

ACC6.7.13

Name: _____ Date: _____ Period: _____

Spiral Review

7. The equation $30 + (-30) = 0$ is an example of two numbers whose sum is 0.
Write a different equation with two numbers whose sum is 0.

Responses vary. $44 + (-44) = 0$

8. Write an equation with three numbers whose sum is 0.

Responses vary. $33 + (-22) + (-11) = 0$

9. Write an equation with *four* numbers, none of which are opposites, whose sum is 0. One example of opposites is -30 and 30.

Responses vary. $44 + (-33) + (-22) + 11 = 0$

10. Write an equation using *four* numbers that multiply to 0.

Responses vary. $33 \cdot 22 \cdot 11 \cdot 0 = 0$

Reflection

1. Put a star next to a problem you could explain to a classmate.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

You can use properties of operations, order of operations, and multiplication and division of integers as strategies to solve integer puzzles.

For example, if we want to make this inequality true, it may be helpful to think about the signs of the sum inside the parentheses and the number outside of the parentheses.

Make the inequality true.

$$\text{.....} (\text{.....} + \text{.....}) > 0$$

$-8(-4 + 1) > 0$ is true because $-8(-3)$ is positive.

$-8(4 + 1) > 0$ is false because $-8(5)$ is negative.

Things to Remember:

Lesson Practice

ACC6.7.15

Name: _____ Date: _____ Period: _____

Problems 1–4: Determine the value of the variable that makes each equation true.

1. $-22 + a = -5$

$a = 17$

2. $-22 - 5 = b$

$b = -27$

3. $-5c = -22$

$c = 4.4$

4. $\frac{d}{-5} = 22$

$d = -110$

5. Which expression has the greater value?

A. $(-22) - (-5)$

B. $(-5) - (-22)$

C. They have the same value

Explain your thinking.

Explanations vary. $(-5) - (-22)$ equals 17 because $(-5) - (-22) = (-5) + 22$, whereas $(-22) - (-5)$ equals -17 because $(-22) - (-5) = (-22) + 5$. Because $-17 < 17$, $(-5) - (-22)$ has a greater value than $(-22) - (-5)$.

Problems 6–7: Let $x = -2$, $y = 4$, and $z = 2$.

6. Order these expressions from *least* to *greatest*.

$x - z$	$x - 2y$	$x \cdot y$	xyz
xyz	$x - 2y$	$x \cdot y$	$x - z$

Least

Greatest

7. Would your order be different if the value of x was 2 instead? Explain your thinking.

Yes. Explanations vary. Because 2 is positive, the signs of the expressions xyz and $x \cdot y$ would be the opposite of what they were when $x = -2$, which would change the order of the values.

8. For the expressions $\frac{a}{b}$ and $a + b$, choose values for a and b so that $\frac{a}{b}$ is positive and $a + b$ is negative.

Responses vary. $a = -2$, $b = -1$

Lesson Practice

ACC6.7.15

Name: _____ Date: _____ Period: _____

Spiral Review

Problems 9–11: Lucia held a penny underwater in a pool and let it go. The penny traveled downward at a rate of 0.8 feet per second. Use the number line if it helps with your thinking.

9. How many seconds did it take for the penny to move -4.4 feet (4.4 feet downward)?

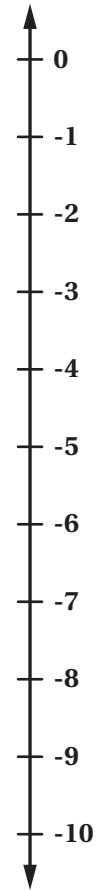
5.5 seconds

10. 3 seconds after letting go, the penny was at position -3.4 feet. What position was the penny let go from?

-1 foot

11. What was the position of the penny after it sank for 10 seconds? Explain your thinking.

-9 feet. Explanations vary. The penny started at -1 and then fell for 10 seconds, so it went $-0.8 \cdot 10 = -8$ more feet. So after 10 seconds, it's at $-1 + (-8) = -9$ feet.



Reflection

1. Circle the problem you enjoyed doing the most.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

Adding and subtracting positive and negative numbers can help you solve problems involving real-world situations.

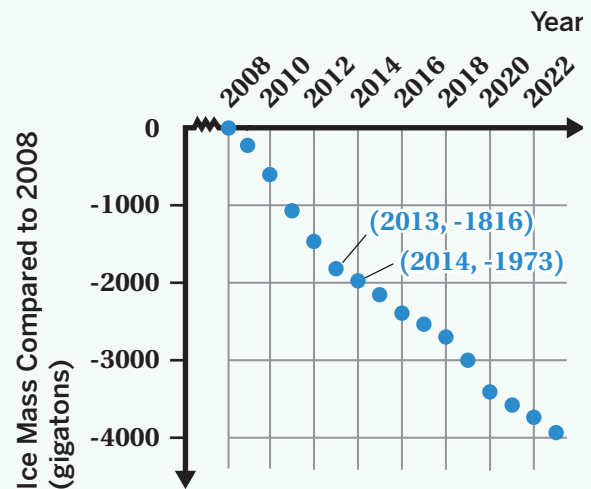
You can tell from the graph that the ice mass in Greenland is decreasing each year. Subtraction lets you determine how much it decreased between any two years.

For example, to determine the change in the ice mass from 2013 to 2014, you need to find the difference between -1,973 and -1,816.

Since -1,973 is less than -1,816, the difference will be negative.

$$-1973 - (-1816) = -1973 + 1816 = -157$$

The change in ice mass from 2013 to 2014 is -157 gigatons. You can use this information to determine if the ice mass is continuing to change at the same rate in future years, or if the ice mass is changing at a faster or slower rate.



Things to Remember:

Lesson Practice

ACC6.7.16

Name: _____ Date: _____ Period: _____

Problems 1–3: Determine the value of the variable that makes each equation true.

1. $3 \cdot (-2.5) = x$

$x = -7.5$

2. $-3y = 33$

$y = -11$

3. $-3 - \frac{1}{2} = z$

$z = -3.5$ (or equivalent)

4. This table shows the transactions, in dollars, in a checking account for the month of January.

Determine the total of the transactions for the month. Show or explain your thinking.

\$252.20. Explanations vary.

$-38.50 + (-265.00) = -303.50$

$126.30 + 429.40 = 555.70$

$555.70 + (-303.50) = 252.20$

January
-\$38.50
\$126.30
\$429.40
-\$265.00

Problems 5–7: On January 22, 1943, the town of Spearfish, South Dakota, set the record for the world's fastest temperature change.

- At 7:30 AM, the temperature was -4°F .
 - By 7:32 AM, the temperature was 45°F .
 - By 9:00 AM, the temperature was 54°F .
 - By 9:27 AM, the temperature was -4°F .
5. What was the temperature change from 7:30 AM to 7:32 AM? Make sure to show whether the change was positive or negative.

49 degrees

6. What was the temperature change from 9:00 AM to 9:27 AM? Make sure to show whether the change was positive or negative.

-58 degrees

7. Years later, the town of Bristol had a big temperature drop. Between 6:02 PM and 6:10 PM, the temperature went from 9°F to -11°F . How does this temperature change compare to Spearfish's big drop that started at 9 AM?

Responses vary. Spearfish's drop was bigger as measured by degrees: -58 compared to -20. But Bristol had a bigger change in terms of degrees per minute, -2.5 compared to about -2.15.

Lesson Practice

ACC6.7.16

Name: _____ Date: _____ Period: _____

Spiral Review

8. Anushka thinks her freezer is broken, so she tracks the temperature every 2 hours. The table shows the temperature at different times of the day. At what time was her freezer's temperature closest to freezing level (0°C)?

- A. 9 AM B. 11 AM
C. 1 PM D. 3 PM

Time	Temperature ($^{\circ}\text{C}$)
9 AM	0.4
11 AM	-0.3
1 PM	0.1
3 PM	-0.7

Problems 9–10: These are equation puzzles. Fill in the blanks so that each row and column makes a true equation. *Responses vary.*

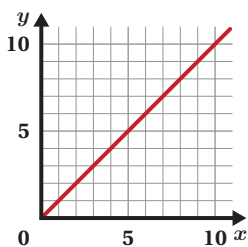
9. Use numbers to complete the puzzle. 10. Use the symbols $+$, $-$, $\times$, and $\div$.

4	$\div$	-4	$\times$	0	=	0
$\times$		+		+		
-1	$\times$	2	-	-2	=	0
+		-		+		
4	$\div$	-2	+	2	=	0
=		=		=		
0		0		0		

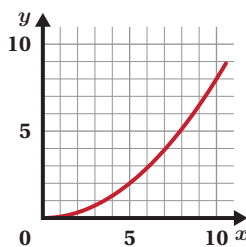
-4	$\times$	-1	-	3	=	1
+		-		$\div$		
1	$\times$	-3	$\div$	-3	=	1
+		-		-		
4	-	1	+	-2	=	1
=		=		=		
1		1		1		

11. Select *all* the graphs that don't represent a proportional relationship.

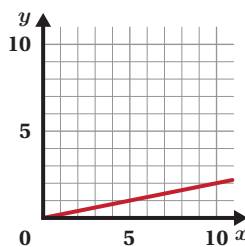
☐ A.



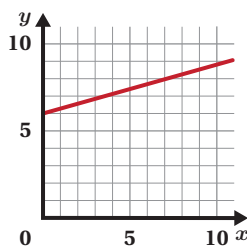
☒ B.



☐ C.



☒ D.



Reflection

- Circle the problem you feel least confident about.
- Use this space to ask a question or share something you're proud of.

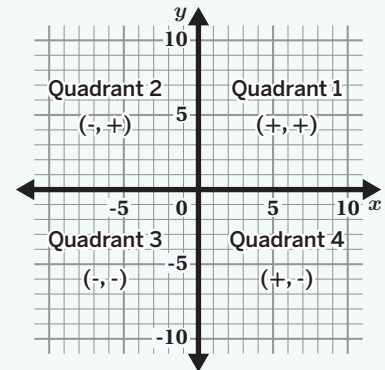
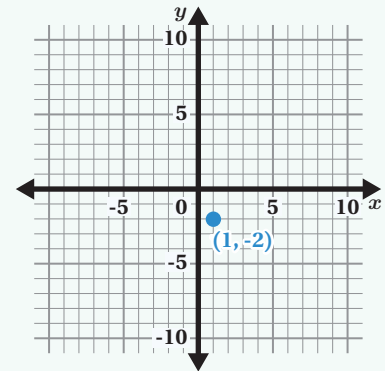
Lesson Summary

You can include positive and negative numbers along the x - and y -axes, just like on a number line. The x - and y -axes cross at the *origin*, or the point $(0, 0)$.

Ordered pairs are written (x, y) , where the x -value is the horizontal location (left and right) and the y -value is the vertical location (up and down). For example, the point $(1, -2)$ is 1 unit to the right and 2 units down from the origin.

The four regions of the *coordinate plane* are called **quadrants**. They are numbered 1–4 starting with the top right quadrant and going in a circle counter-clockwise.

The image shows each quadrant, along with the sign of the x - and y -values in that quadrant.



Things to Remember:

Lesson Practice

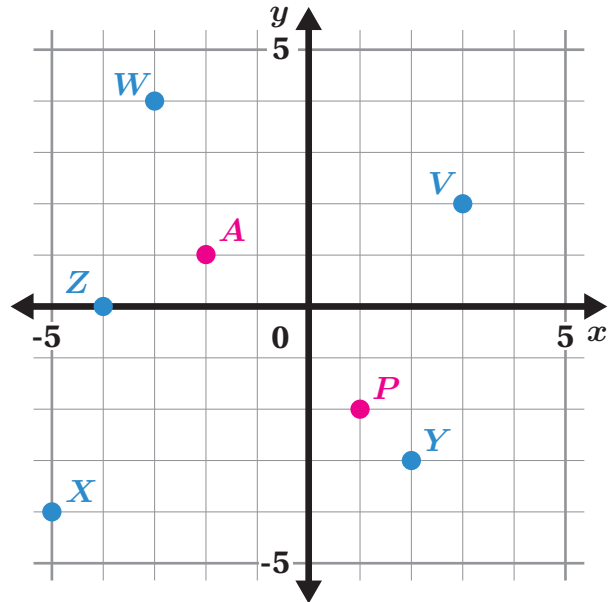
ACC6.7.17

Name: _____ Date: _____ Period: _____

Problems 1–6: Here is a coordinate plane.

- Write the coordinates of each point in the table.

Point	Coordinates
V	(3, 2)
W	(-3, 4)
X	(-5, -4)
Y	(2, -3)
Z	(-4, 0)



- Plot and label point *A* at $(-2, 1)$.
- Plot and label point *P* at $(1, -2)$.
- Plot two points that are each 2 units away from point *V*. Label each point with its coordinates.

Responses vary.

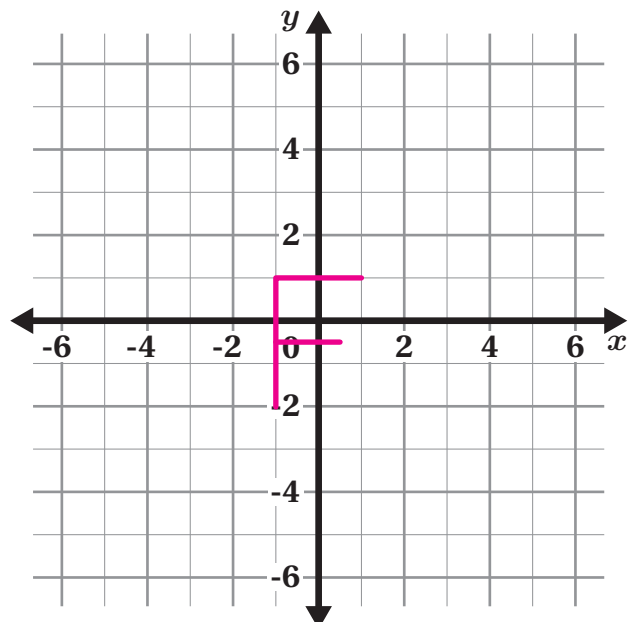
- Point *Q* is more than 3 units directly to the left of point *Y*. Write at least one thing you know about the coordinates of point *Q*.

Responses vary.

- Both coordinates are negative.
- The *y*-coordinate will be -3.

- Draw the first letter of your name on the coordinate plane so that at least part of your drawing is in each quadrant.

Responses vary. Sample shown on the graph.



Lesson Practice

ACC6.7.17

Name: _____ Date: _____ Period: _____

Spiral Review

Problems 7–8: A city has a 5% sales tax.

7. A toothbrush costs \$3.40 before tax. How much does it cost including tax?

\$3.57

8. A book costs \$32.55 after tax. How much did it cost before tax?

\$31

Problems 9–12: Solve each equation.

9. $3a = 12$

$a = 4$

10. $b + 3.3 = 8.9$

$b = 5.6$

11. $1 = \frac{1}{4}c$

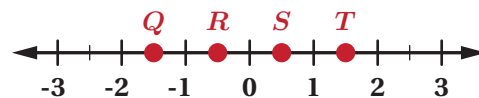
$c = 4$

12. $2d = 6.4$

$d = 3.2$

13. Here is a number line with points Q , R , S , and T .
Which point represents $-1\frac{1}{2}$?

Point Q



Reflection

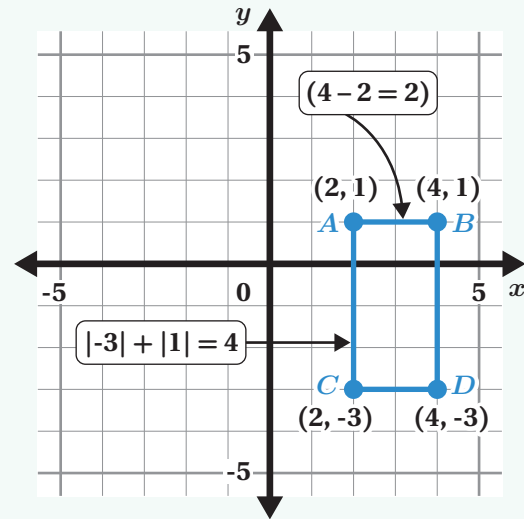
1. Put a smiley face next to a problem you were stuck on and then figured out.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

You can plot points on the coordinate plane to create *polygons*. When the vertices of a polygon are horizontally or vertically aligned on the graph, you can count the number of units between them to determine the length of that side.

You can also calculate the side lengths using the coordinates of each vertex. Here are two calculation strategies:

- If the coordinates are in the same quadrant, like points A and B , find the length by subtracting the coordinates that are different.
- If the coordinates are in different quadrants, like points A and C , use the absolute value to determine the distance each point is from the axis between them.



Things to Remember:

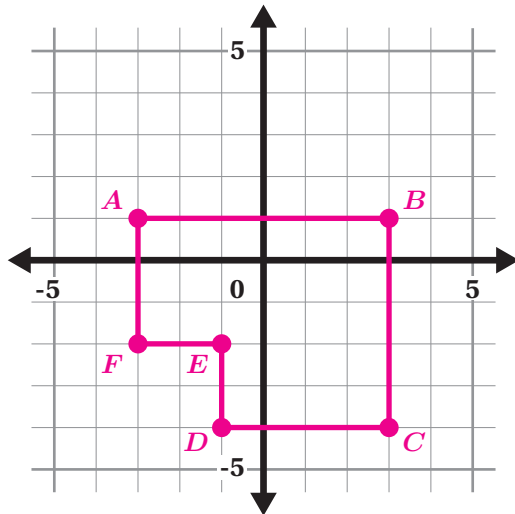
Lesson Practice

ACC6.7.19

Name: _____ Date: _____ Period: _____

Problems 1–4: Here is a set of coordinates.

- Plot each point.



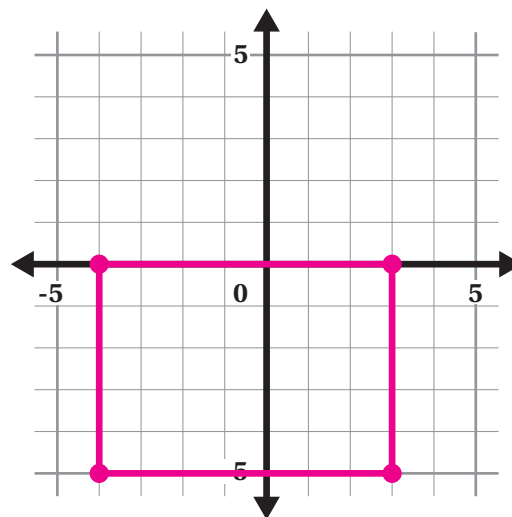
Point	Coordinates
<i>A</i>	$(-3, 1)$
<i>B</i>	$(3, 1)$
<i>C</i>	$(3, -4)$
<i>D</i>	$(-1, -4)$
<i>E</i>	$(-1, -2)$
<i>F</i>	$(-3, -2)$

- Connect the points to create polygon $ABCDEF$.
- Determine the length of the segment between point A and point B .
6 units

- Determine the perimeter of polygon $ABCDEF$.
22 units

Problems 5–7: Three points of a rectangle are $(3, 0)$, $(3, -5)$, and $(-4, -5)$.

- What are the coordinates of the missing point?
 $(-4, 0)$
- Sketch the rectangle and calculate its perimeter.
24 units
- Calculate the area of the rectangle.
35 square units



Lesson Practice

ACC6.7.19

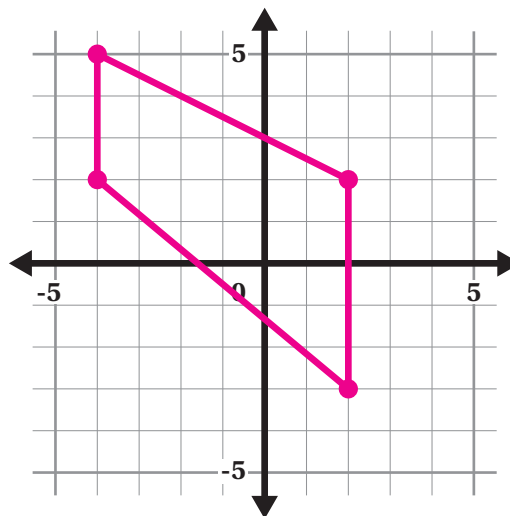
Name: _____ Date: _____ Period: _____

8. Gabriela drew a quadrilateral with these descriptions:

- It has exactly one pair of parallel sides.
- Two of the vertices are $(-4, 5)$ and $(2, -3)$.
- At least one side has a length of 5 units.

Create a quadrilateral that Gabriela could have made.

Polygons vary.



Spiral Review

9. Is $\frac{12}{5} \div \frac{3}{5}$ greater than, less than, or equal to 1?

Greater than 1

10. What is the value of $\frac{9}{5} \div \frac{3}{5}$?

3

11. At top speed, an elephant can run 25 miles per hour and a giraffe can run 16 miles in $\frac{1}{2}$ hour. Which animal runs faster?

Giraffe

Reflection

1. Put a heart next to a problem you understand well.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

A **statistical question** is a question that needs more than one piece of data to answer it. **Numerical data** are numbers, quantities, or measurements that can be meaningfully compared.

Here is an example:

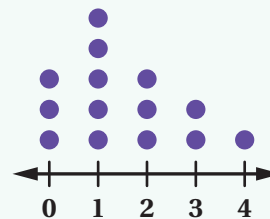
- “Which classroom in your school has the most books?” is a statistical question because you need to know the number of books in *each* classroom to answer it.
- “How many books are in your classroom?” is not a statistical question because you only need to know the number of books in *one* classroom to answer it.

You can organize data that answers a statistical question into a list or a **dot plot**.

List

0, 0, 0, 1, 1, 1, 1, 1, 2, 2, 2, 3, 3, 4

Lists allow you to see all of the data.
Lists can be used for both numerical
and categorical data.

Dot Plot

Dot plots are a visual representation
of numerical data and allow you to
compare multiple data sets.

Things to Remember:

Lesson Practice

ACC6.8.01

Name: _____ Date: _____ Period: _____

Problems 1–5: Five sixth-grade students at a school were each asked the following survey questions:

Question A: What grade are you in?

Question B: How many books did you read in the last year?

Question C: How many inches are in 1 foot?

Question D: How many dogs and cats do you have?

Their answers are shown in the table. Write the letter of the question that could have produced each line of data.

	Question	Diya	Chey	Tiana	Shep	Callen
1.	D	0	1	1	3	0
2.	C	12	12	12	12	12
3.	A	6	6	6	6	6
4.	B	11	5	18	20	9

5. How are Questions A and C different from the other questions?

Responses vary. Questions A and C have only one correct answer. The other questions have a lot of different possible answers.

Problems 6–7: Zee asked 10 students how many minutes it takes them to get to school each morning.

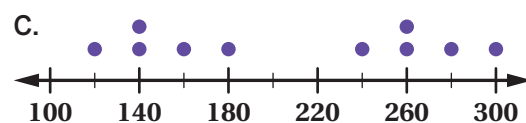
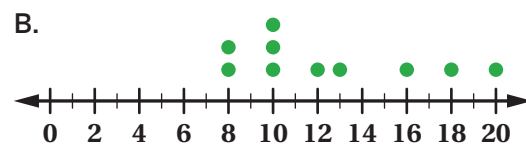
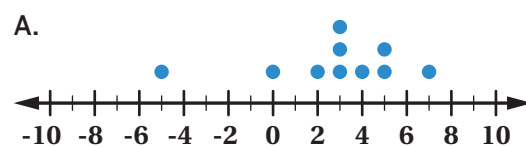
6. Which dot plot could represent the data that Zee collected?

B

7. Which dot plot could not represent Zee's data? Explain your thinking.

Responses vary.

- Dot Plot A cannot represent Zee's data because it would not make sense for someone to take -5 minutes to get to school.
- Dot Plot C would not make sense because 300 minutes is 5 hours, which is probably too long for someone to get to school.



Lesson Practice

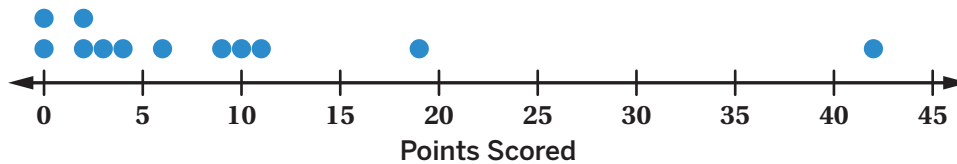
ACC6.8.01

Name: _____ Date: _____ Period: _____

8. Which question is a statistical question?

- A. How old are you?
- B. How old are the students in our school?**
- C. How old is your teacher?
- D. What is the difference between your age and your teacher's age?

9. This dot plot represents the points scored by 12 players on a basketball team during an important game.



Write a story about the basketball game.

Responses vary.

Spiral Review

10. Order these values from *least* to *greatest*.

$ -17 $	$ -18 $	-18	$ 19 $	20
-18	-17	-18	19	20
Least				Greatest

Problems 11–13: Determine each quotient.

11. $45052 \div 28 = \mathbf{1609}$ 12. $6052 \div 17 = \mathbf{356}$ 13. $60.52 \div 1.7 = \mathbf{35.6}$

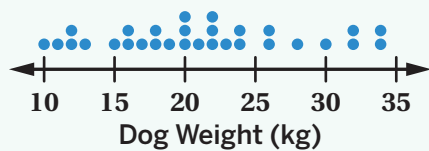
Reflection

- Put a star next to your favorite problem.
- Use this space to ask a question or share something you're proud of.

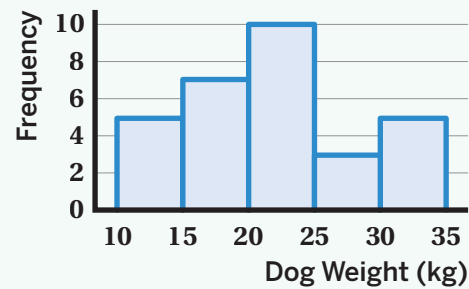
Lesson Summary

You can use dot plots and **histograms** to visualize numerical data. Here is an example of a data set of the weights of 30 dogs presented in a dot plot and in a histogram.

Dot Plot



Histogram



In a histogram, data values are grouped into **bins** that cover a range of values, and each bin has the same width. The height of each bar represents the total number of values in that range, including the left boundary (least value) but excluding the right boundary (greatest value). For example, the height of the tallest bar, from 20 to 25, represents weights of 20 kilograms up to (but not including) 25 kilograms.

Things to Remember:

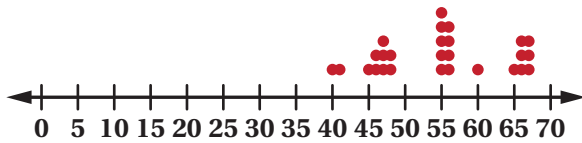
Lesson Practice

ACC6.8.03

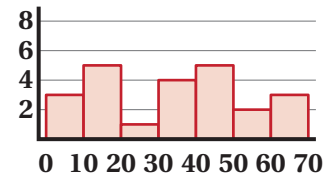
Name: _____ Date: _____ Period: _____

1. Match each histogram with the dot plot that represents the same data set.

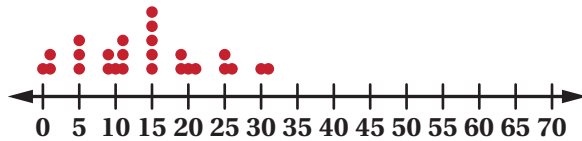
a.



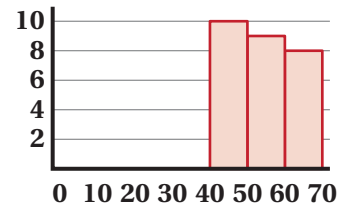
c.



b.



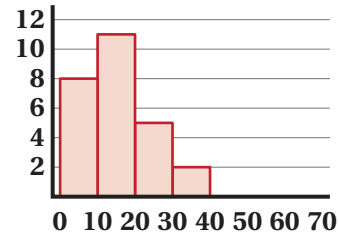
a.



c.

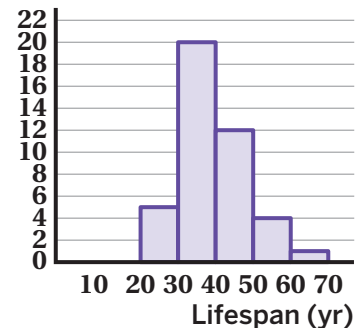


b.



2. Here is a histogram showing the lifespans of 42 chimpanzees that lived in the wild. How many chimpanzees lived at least 50 years and less than 70 years?

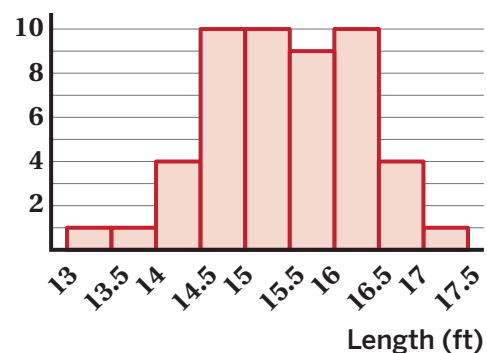
5 chimpanzees



Problems 3–4: A marine biologist is studying a group of sharks. She made a histogram of the lengths of a group of adult sharks.

3. Select *all* of the false statements.

- ☒ A. A total of 9 sharks were measured.
- ☐ B. Two of the sharks are less than 14 feet long.
- ☐ C. A typical shark is about 14.5 to 16.5 feet long.
- ☒ D. The longest shark measured was 10 feet long.
- ☒ E. The smallest shark measured was 11 feet long.



4. Explain your thinking for one of the false statements.

Explanations vary.

Lesson Practice

ACC6.8.03

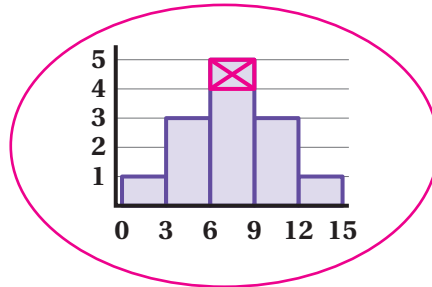
Name: _____ Date: _____ Period: _____

Problems 5–6: Here are three representations of the same data set.

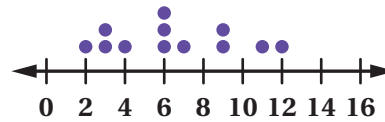
List

2, 3, 3, 4, 6, 6, 6,
7, 9, 9, 11, 12

Histogram



Dot Plot



- Circle the representation that doesn't match the other two.
- Revise your circled representation to make it match the other two representations.

Spiral Review

- Order these expressions from *least* to *greatest*.

$-15.6 \div (-3)$

$-5.2 \cdot (-3)$

$-5.2 - (-3)$

$-5.2 + (-3)$

$15.6 \div (-3)$

$-5.2 + (-3)$

$15.6 \div (-3)$

$-5.2 - (-3)$

$-15.6 \div (-3)$

$-5.2 \cdot (-3)$

Least

Greatest

Problems 8–11: Determine the value of each expression.

8. $3.727 + 1.384 = 5.111$

9. $3.727 - 1.384 = 2.343$

10. $5.01 \cdot 4.8 = 24.048$

11. $5.01 \div 4.8 = 1.04375$

Reflection

- Put a question mark next to a response you'd like to compare with a classmate's.
- Use this space to ask a question or share something you're proud of.

Lesson Summary

A **statistic** is a single number that measures something about a data set. One way to measure the center of a data set is by determining the **mean**, or average, of all the data values. You can think of the mean as “an equal share.”

For example, suppose this data set represents how many liters of water are in 5 bottles: 1, 4, 2, 3, 0. To calculate the mean, you first add up all of the values to determine the total (10 liters), then divide that sum by the number of values (5 bottles). This example can be represented by the expression $(1 + 4 + 2 + 3 + 0) \div 5$, or $10 \div 5$. So, the mean amount of water in the 5 bottles is 2 liters (per bottle). The mean is a whole number in this example, but it is possible for the mean to be a decimal number.

Things to Remember:

Lesson Practice

ACC6.8.04

Name: _____ Date: _____ Period: _____

1. A preschool teacher plans to reorganize these 4 boxes of playing blocks so that each box contains an equal number of blocks. How could the teacher determine the number of blocks to put in each box?

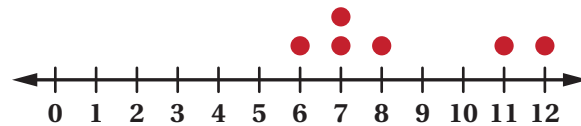
Box 1	Box 2	Box 3	Box 4
32 blocks	18 blocks	41 blocks	9 blocks

Responses vary. The teacher could add all of the blocks and then divide that number by 4.

2. 3 classes worked together to raise money for their classroom libraries. They agreed to share the money equally. The first class raised \$25.50, the second class raised \$49.75, and the third class raised \$37.25. What is each class's equal share? Show or explain your thinking.

\$37.50. Explanations vary. $\frac{25.50 + 49.75 + 37.25}{3} = \frac{112.50}{3}$ or **\$37.50**

3. Kimaya guesses that 11 is the mean of this data set. Without calculating, determine if Kimaya's guess is correct. Explain your thinking.



No. Explanations vary. Most of the numbers are less than 11 and the only number greater than 11 is 12. So the mean is probably less than 11.

Problems 4–5: For 12 days, Mai recorded how many minutes long her bus rides to school were. Here are the times she recorded.

Time on the Bus (min)											
9	12	6	9	10	7	6	12	9	8	10	10

4. Determine the mean for Mai's data. Show or explain your thinking.

9 minutes. Explanations vary. $\frac{9 + 12 + 6 + 9 + 10 + 7 + 6 + 12 + 9 + 8 + 10 + 10}{12} = \frac{108}{12}$ or **9**

5. What does the mean tell us about Mai's trip to school?

Responses vary. The mean tells us that it takes Mai 9 minutes to get to school on most days.

6. In English class, Hanjun's teacher gives 4 quizzes, each worth 5 points. After 3 quizzes, Hanjun has scores of 4, 3, and 4. How many points does Hanjun need to get on the last quiz to have an average score of 4? Show or explain your thinking.

5 points. Explanations vary. Because 3 is one less than 4, Hanjun has to score $4 + 1 = 5$ points on the final quiz.

$$\begin{aligned} \frac{4 + 3 + 4 + x}{4} &= 4 \\ 4 \cdot \left(\frac{4 + 3 + 4 + x}{4} \right) &= 4 \cdot 4 \\ 4 + 3 + 4 + x &= 16 \\ 11 + x &= 16 \\ x &= 5 \end{aligned}$$

Lesson Practice

ACC6.8.04

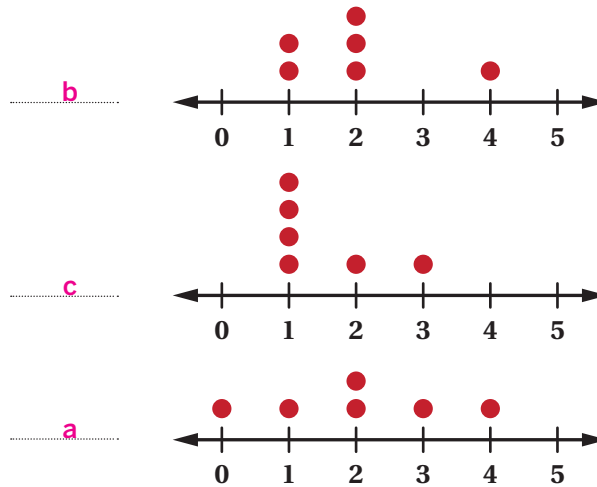
Name: _____ Date: _____ Period: _____

7. Select a dot plot that matches each statement. You can only select each dot plot once.

a. The mean is 2.

b. Only one value is greater than the mean.

c. Two values are greater than the mean.



8. In a round of mini golf, Angel records the number of putts on each hole.

Number of Putts								
2	3	1	4	5	2	3	4	3

What is the mean number of putts per hole? Show or explain your thinking.

3 putts per hole. Explanations vary. $\frac{2+3+1+4+5+2+3+4+3}{9} = \frac{27}{9}$ or 3

Spiral Review

Problems 9–11: Evaluate the expression $4x^3$ for each value of x .

9. 1
4

10. 2
32

11. $\frac{1}{2}$
 $\frac{1}{2}$

12. Is 4.5 a solution to the equation $1.5 + x = 6$? Circle one.

Yes

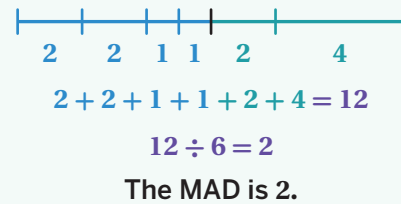
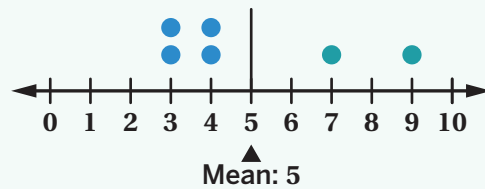
No

Reflection

- Put a smiley face next to a problem you were stuck on and then figured out.
- Use the space to ask a question or share something you're proud of.

Lesson Summary

You can describe how spread out the values in a data set are with a single number, the **mean absolute deviation (MAD)**. The MAD is calculated by determining the mean of the absolute deviations (i.e., the average of the distances between each data value and the mean).



The mean absolute deviation is an example of a measure of spread. A measure of spread is a way to measure the consistency of the values in a data set. The smaller the value of the MAD, the less spread out the data points are around the mean, and the more consistent the data is. The larger the MAD, the more spread out the data points are around the mean, and the less consistent the data is.

Things to Remember:

Lesson Practice

ACC6.8.05

Name: _____ Date: _____ Period: _____

Problems 1–2: This table shows the amount of time it takes 6 students to get to school. Their mean travel time is 22 minutes.

Time (minutes)	10	10	18	20	30	44
Absolute Deviation	12	12	4	2	8	22

1. Complete the table by calculating the absolute deviation of each value from the mean.

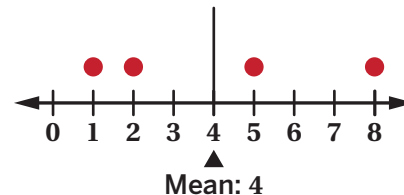
2. Calculate the mean absolute deviation (MAD) of this data set.

10 minutes

3. Calculate the MAD of this data set.

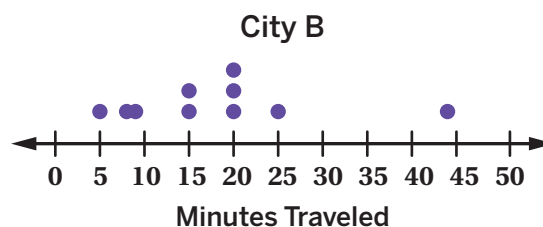
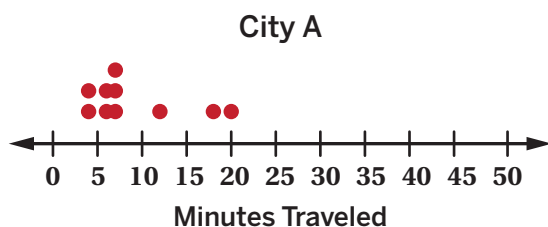
Use the table if it helps with your thinking.

2.5



Data Point	1	2	5	8
Absolute Deviation	3	2	1	4

Problems 4–5: These dot plots show the travel times for 10 students from two cities.



4. The MADs have been calculated for you. Match each MAD to the correct city.

MAD	City
4.4	A
7.8	B

5. Without calculating, explain how you matched the MADs to the data sets.

Explanations vary. I knew the larger MAD had to go with City B because the data is more spread out than City A.

Lesson Practice

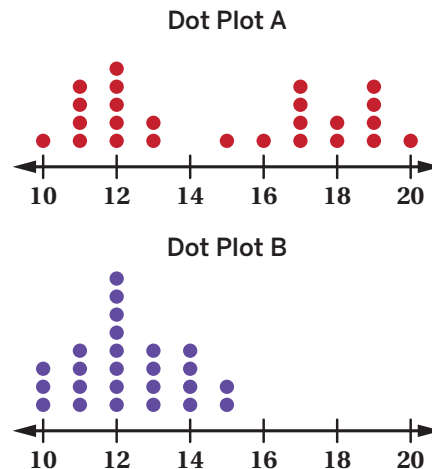
ACC6.8.05

Name: _____ Date: _____ Period: _____

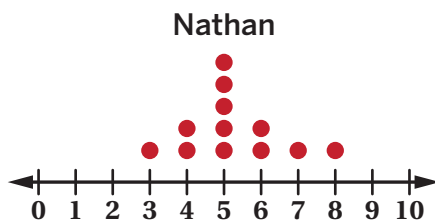
6. Here are two dot plots showing the recorded speeds of two manatees.

Which of these statements is true?

- A. The data for Dot Plot B is more spread out than the data for Dot Plot A.
- B. The manatee recorded in Dot Plot A swims at a more consistent speed.
- C. The MAD of Dot Plot A is greater than the MAD of Dot Plot B.**
- D. Both dot plots have approximately the same mean.



Problems 7–8: Nathan and Esi recorded the number of baskets they each made out of 10 attempts. They each collected 12 data points. Nathan's mean number of baskets was 5.25, and his MAD was 1. Esi's mean number of baskets was 5.



Esi	7	8	9	0	1	2	1	3	2	10	9	8
Distance From 5	2	3	4	5	4	3	4	2	3	5	4	3

7. Calculate Esi's MAD. Use the table if it helps with your thinking.

3.5

8. Which player is more consistent?

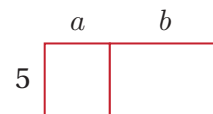
Nathan

Spiral Review

9. Select *all* the expressions that are equivalent to $2(6 + 3x)$.

- ☐ A. $8 + 5x$
- ☐ B. $12x + 6$
- ☒ C. $12 + 6x$
- ☒ D. $6(2 + x)$
- ☐ E. $2(9x)$

10. Select *all* the expressions that represent the total area of the rectangle.



- ☒ A. $5(a + b)$
- ☐ B. $5 + ab$
- ☒ C. $5a + 5b$
- ☐ D. $2(5 + a + b)$
- ☐ E. $5ab$

Reflection

- Put a heart next to a problem that you understand well.
- Use the space to ask a question or share something you're proud of.

Lesson Summary

You can describe a data set using another measure of center called the **median**. The median is the “middle” value in a data set when the values are listed in order from least to greatest (or greatest to least). Half of the data values are less than or equal to the median, and half of the data values are greater than or equal to the median.

To determine the median from an ordered representation of the data, you can repeat a process of eliminating the pairs of least and greatest values.

Here are some examples.

~~0~~ ~~1~~ ~~1~~ **2** ~~2~~ ~~4~~ ~~5~~

**Odd Number
of Values**

Once all pairs have been eliminated, only one value remains in the middle, making it the median.

Median: 2

~~0~~ ~~1~~ ~~1~~ **1** **2** ~~2~~ ~~4~~ ~~5~~

**Even Number
of Values**

Once all pairs have been eliminated, two values remain.

Their average is the median.

$$(1 + 2) \div 2 = 1.5$$

Median: 1.5

Things to Remember:

Lesson Practice

ACC6.8.06

Name: _____ Date: _____ Period: _____

Problems 1–2: Here is a data set.

1	6	7	6	2	9	3
---	---	---	---	---	---	---

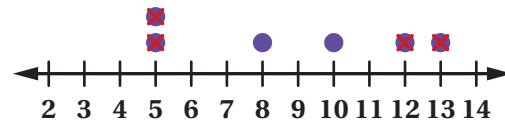
1. Create a dot plot for this data.



2. Determine the median of this data.

6

Problems 3–4: Kayla wants to determine the median of the data in this dot plot. She starts solving the problem but isn't sure what to do next.



3. What could you say to help Kayla determine the median?

Responses vary. Because there are an even number of points, I would tell Kayla to calculate the average of the two points in the middle: 8 and 10.

4. What is the median of this data?

9

5. The table shows Prisha's scores after attempting the first level of a video game 10 times. What is her median score?

130	150	120	170	130
120	160	160	190	140

145

6. Pilar recorded the number of points she scored in her last 7 basketball games. She says that her median score was 8 points. Is she correct? Explain your thinking.

No. Explanations vary. 8 is not the middle value of Pilar's data set. It's just the middle of the list. To determine the middle value of these numbers, Pilar should start by rearranging them from smallest to largest.

Pilar
13, 20, 9, 8, 11, 11, 15

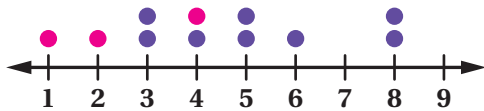
Lesson Practice

ACC6.8.06

Name: _____ Date: _____ Period: _____

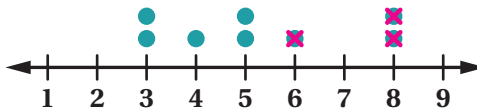
7. Add points to this dot plot to make the median 4.

Responses vary.



8. Remove points from this dot plot to make the median 4.

Responses vary.

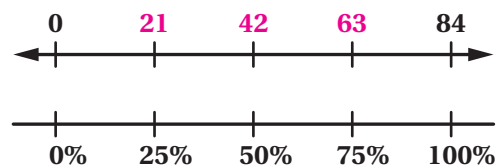


Spiral Review

9. Determine 25%, 50%, and 75% of 84.

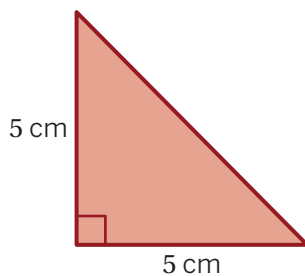
Use the double number line if it helps with your thinking.

Sample shown on diagram.



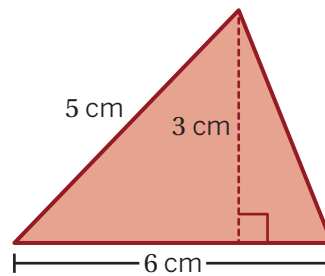
Problems 10–12: Calculate the area of each triangle.

10.



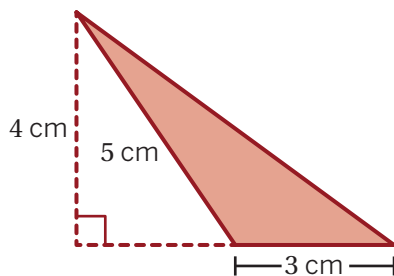
12.5 (or equivalent) square centimeters

11.



9 square centimeters

12.



6 square centimeters

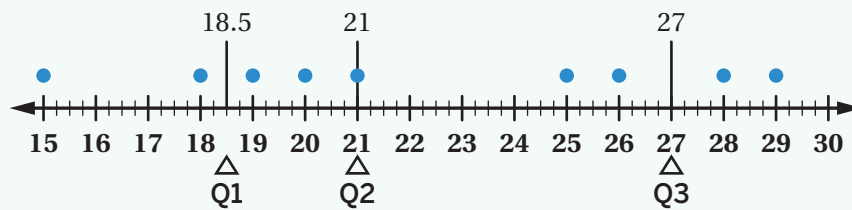
Reflection

- Put a star next to a problem you could explain to a classmate.
- Use this space to ask a question or share something you're proud of.

Lesson Summary

You can describe the middle half of a data set by dividing it into four equal sections called quartiles.

You can determine the value of the quartiles by splitting the entire data set in half and then splitting the halves again. The middle half is all the data points that are between $Q1$ and $Q3$. Representations such as dot plots are helpful for identifying quartiles to describe data sets.



The *first quartile* ($Q1$) is the median of the lower half of the data set.

The *second quartile* ($Q2$) is the median of the entire data set.

The *third quartile* ($Q3$) is the median of the upper half of the data set.

Things to Remember:

Lesson Practice

ACC6.8.07

Name: _____ Date: _____ Period: _____

Problems 1–2: Here are the ages of 20 people at a family reunion, ordered from youngest to oldest.

3, 8, 9, 10, 11, 11, 12, 18, 18, 28, 30, 35, 37, 40, 53, 54, 58, 65, 70, 72

1. The value of Quartile 2 (Q2) is 29. What does that tell us about the people at the family reunion?

Responses vary. It means that about half of the people at the family reunion are younger than 29 years old and about half are older than 29 years old.

2. Determine the value of Q1 and Q3.

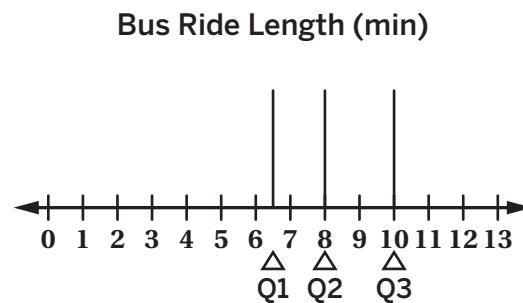
Quartile 1 (Q1): **11**

Quartile 3 (Q3): **53.5**

Problems 3–5: Haru recorded how long his bus ride to school took for 16 days. Here are the values of the quartiles of his data.

3. About how many rides would you expect to be less than 6.5 minutes long?

About 4 bus rides



4. About how many rides would you expect to be less than 10 minutes long?

About 12 bus rides

5. About what percent of the rides would you expect to be between 6.5 minutes and 10 minutes long?

About 50% of the bus rides

6. The heights, in inches, of Javier's classmates are 60, 68, 56, 60, 62, 58, 55, 67, 59, 61, 62, 64, 63, 63, 59, 62, 66, and 61. Determine the values of Q1, Q2 (median), and Q3.

	Height (in.)
Q1	59
Q2 (Median)	61.5
Q3	63

Lesson Practice

ACC6.8.07

Name: _____ Date: _____ Period: _____

7. Makayla and Axel both try to determine the quartiles for this dot plot.

Makayla says:

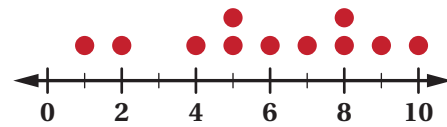
Q1: 4.5 Q2: 6 Q3: 8

Axel says:

Q1: 4 Q2: 6 Q3: 8

Who is correct? Explain your thinking.

Axel. Explanations vary. If the data set has an odd number of points, the median should not be included when determining Q1 and Q3.



Spiral Review

Problems 8–10: Calculate each percentage.

8. 25% of 40

10

9. 25% of 120

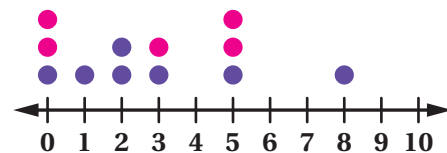
30

10. 25% of 90

22.5

11. Add 5 points to this dot plot without changing the mean absolute deviation (MAD).

Response shown on graph.



Reflection

1. Circle the problem that was the most challenging for you.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

You can create a **box plot** to visualize a data set. While a box plot shows the same data as a dot plot, it gives us new information about the data. Rather than showing every data point, a box plot separates the data into quartiles.

We can use box plots to describe the spread of the data in two ways.

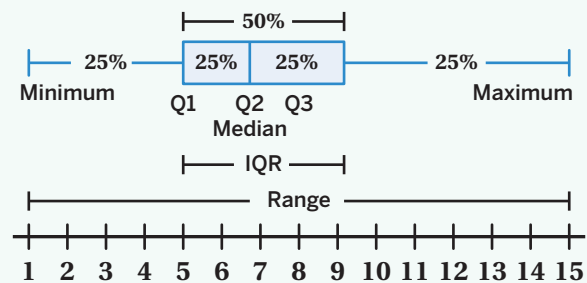
- The **range** represents the difference between the *maximum* and *minimum* values of a data set. It describes the overall spread of the data.

$$\text{Range: } 15 - 1 = 14$$

- The **interquartile range (IQR)** represents the range of the middle 50% of the data (between Q3 and Q1). It describes how spread out the middle of the data is.

$$\text{IQR: } 9.25 - 5 = 4.25$$

Min.	Q1	Median	Q3	Max.
1	5	6.75	9.25	15



Box plots do not show how many data points are in each set, or the values of any individual data points, except the minimum and maximum.

Things to Remember:

Lesson Practice

ACC6.8.08

Name: _____ Date: _____ Period: _____

Problems 1–3: Each student in a class recorded how many books they read in a school year.

- What is the greatest number of books that a student read this year?

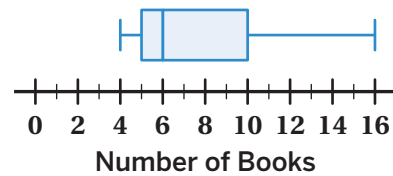
16 books

- What is the median of this data?

6 books

- What is the range of this data?

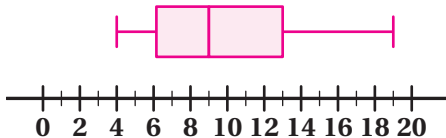
12 books



Problems 4–5: Here are five statistics about a data set.

- Create a box plot that represents this data set.

Minimum	Q1	Median	Q3	Maximum
4	6	9	13	19



- What is the IQR of this data set?

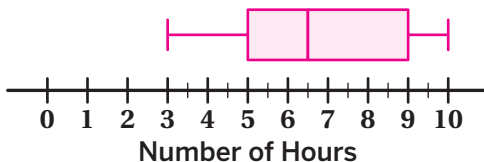
7

Problems 6–9: A group of seventh-grade students recorded the number of hours they spent doing homework in one week. Here is their data: 3, 4, 5, 5, 6, 7, 7, 9, 9, 10.

- Determine each of the statistics.

Minimum	Q1	Median	Q3	Maximum
3	5	6.5	9	10

- Create a box plot of this data.



- Based on the data above, is the following statement true or false? Most seventh-graders do less than 5 hours of homework in a week. **False**

- Based on the data above, is the following statement true or false? 25% of seventh-grade students spent between 9 and 10 hours doing homework in a week. **True**

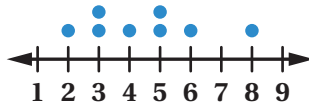
Lesson Practice

ACC6.8.08

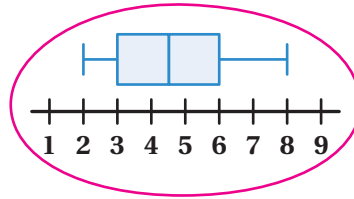
Name: _____ Date: _____ Period: _____

Problems 10–11: Here are three representations of the same data set.

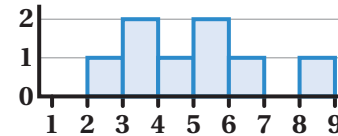
Dot Plot



Box Plot



Histogram



10. One representation has a mistake. Circle the incorrect representation.

11. Describe the mistake made in creating the representation you chose in the previous problem.

Q3 should be 5.5, not 6.

Spiral Review

12. Which of these has the largest value?

A. 25% of 80

B. 100% of 60

C. 75% of 100

D. 50% of 120

13. Nasir surveyed his classmates to determine how many miles they travel to school each day. This histogram shows the results. Select *all* the true statements.

☐ A. Most of the data is between 4 and 7 miles.

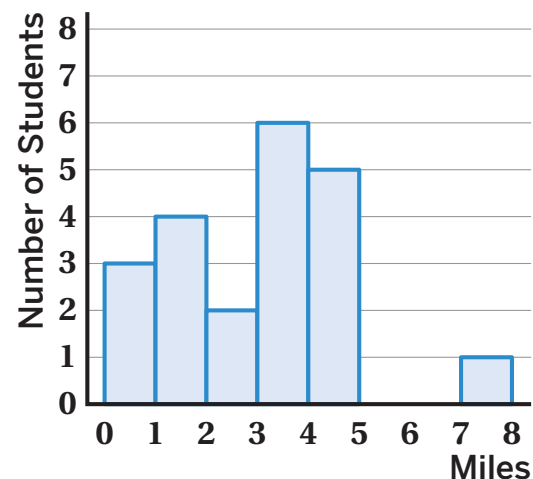
☒ B. The median of the data values is between 3 and 4 miles.

☐ C. The interquartile range of the data values is greater than 5.

☐ D. No students travel between 2 and 3 miles.

☒ E. Most of the data is between 3 and 5 miles.

Distance Traveled to School



Reflection

1. Circle the problem you think will help you most on the Sub-Unit Quiz.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

To answer a question about a **population**, it is sometimes not realistic to collect data from the entire population. Instead, you can collect data from a **sample** of the population.

- A *population* is a set of people or things that we want to study.
- A *sample* is part of a population.

The sample you choose should be large enough to be able to draw conclusions about the population.

Here are some examples of populations and samples.

Population	Sample
All of the people who watch basketball	The people at a basketball game
All 7th grade students in your school	The 7th graders in your school who are in a band
All oranges grown in the U.S.	The oranges in your local grocery store

Things to Remember:

Lesson Practice

ACC6.8.10

Name: _____ Date: _____ Period: _____

Problems 1–3: Aniyah wonders: How much time do 7th graders at my school spend outdoors on a typical day?

1. What is the population for Aniyah's question?

7th graders at Aniyah's school

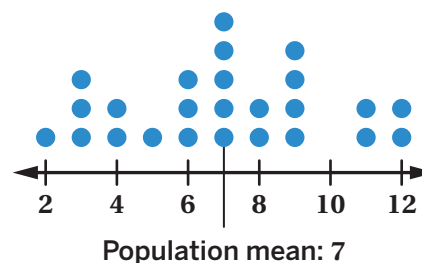
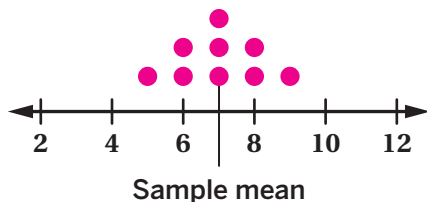
2. Select *all* possible samples for Aniyah's question.

- ☒ **A.** The 20 students in a 7th grade math class
- ☐ **B.** The first 20 people to arrive at Aniyah's middle school on a particular day
- ☐ **C.** The 7th graders participating in a science fair with students from four middle schools
- ☒ **D.** The 10 7th graders on the school soccer team
- ☐ **E.** The students on the high school debate team

3. Select two samples and list a different possible population each sample could belong to. **Responses vary.**

Sample	Possible Population
The 20 students in a 7th grade math class	All of the students in all math classes
The first 20 people to arrive at Aniyah's middle school on a particular day	All of the people who visit the middle school
The 7th graders participating in a science fair with students from four middle schools	All of the students participating in the science fair
The 10 7th graders on the school soccer team	All of the students on the school soccer team
The students on the high school debate team	All of the students at the high school

4. Here is a dot plot of a population. Create a dot plot of a sample with a mean that's the same as the population mean. Your sample should have more than 6, but fewer than 20 data points.



Lesson Practice

ACC6.8.10

Name: _____ Date: _____ Period: _____

Spiral Review

5. Select *all* the measures of center.

☒ A. Mean

☐ B. IQR (interquartile range)

☐ C. Range

☒ D. Median

☐ E. MAD (mean absolute deviation)

Problems 6–8: Solve each equation.

6. $9u = 1$

$u = \frac{1}{9}$

(or equivalent)

7. $1.5w = 7.5$

$w = 5$

8. $2 = x + \frac{7}{10}$

$x = 1.3$

(or equivalent)

Problems 9–11: Calculate the mean of each data set.

9. 8, 9, 9, 9, 10

9

10. 2, 6, 12, 16

9

11. 5, 6, 12, 13

9

Problems 12–13: A bookstore has a 15% discount on all books.

12. How much money is the 15% discount worth on a book that normally costs \$18?

\$2.70

13. After the discount, how much would the book cost?

\$15.30

Reflection

1. Put a star next to a problem where you revised your thinking.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

Samples are useful when a population is too large to survey or measure. Depending on the strategy you use to sample, your sample might or might not be representative of the population. Some samples are not good representations of the population.

A representative sample has a distribution that closely resembles the distribution of the population. Representative samples are useful for making predictions about the whole population.

For example, if you were curious about all middle school students' favorite sport to play, the population would be all middle schoolers. A representative sample of this population might be randomly selecting 5 students from each class or 15 students from each grade to ask. A sample that is *not* representative of this population would be asking students in the tennis club because their responses might lead someone to believe that tennis is the favorite sport among all middle schoolers.

Things to Remember:

Lesson Practice

ACC6.8.11

Name: _____ Date: _____ Period: _____

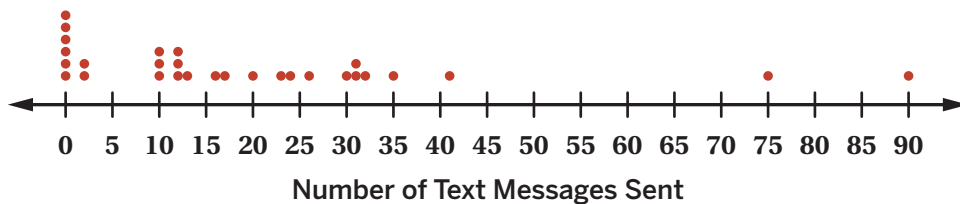
1. Select *all* the reasons why random samples tend to produce a more representative sample.
 - ☐ A. You can determine how many people you want in the sample.
 - ☐ B. It's the easiest way to select a sample from a population.
 - ☒ C. It avoids the bias that can occur with other sampling methods.
 - ☒ D. Each person in the population has an equal chance of being selected.
 - ☐ E. The sample mean will always be the same as the population mean.

2. Jada wants to learn about the percentage of students who like the food in the cafeteria. Jada asks the first 25 students who purchase lunch at the cafeteria if they like the food.

Is Jada's method likely to produce a representative sample? Explain your thinking.

No. Explanations vary. It is likely that the first 25 students like the food at the cafeteria and are an unfair representation of the population. Jada is also not asking any students who didn't want to buy lunch from the cafeteria.

Problems 3–4: This dot plot shows the number of text messages sent on one day for a sample of students at a high school. 29 random students were sampled.



3. What do the six dots at 0 on the dot plot represent?

Students who did not send any text messages on one day

4. Because this sample is representative of the population, describe what a dot plot for the entire high school might look like.

Responses vary. The dot plot would have a similar shape to the sample except with many more points. There would be many students sending fewer than 15 text messages and a small number of students sending more than 35 text messages.

Lesson Practice

ACC6.8.11

Name: _____ Date: _____ Period: _____

Problems 5–8: Think of a new situation.

5. Write a question you're interested in finding the answer to.

Responses vary. What do people in my neighborhood prefer as a pet: cats or dogs?

6. What is the population for your question?

Responses vary. People in my neighborhood

7. Describe a strategy that is unlikely to produce a representative sample to answer your question.

Responses vary. Ask people at a dog park.

8. Describe a strategy to get a representative sample to answer your question.

Responses vary. Ask one house on each block in my neighborhood.

Spiral Review

9. Here are the scores for Sahana's history homework.

100	100	100	100	95	100	90	100	0
-----	-----	-----	-----	----	-----	----	-----	---

The history teacher uses the mean to calculate an overall grade for homework.

How might Sahana argue that the median is a better way to measure her work?

Responses vary. Sahana could argue that the median more accurately reflects the center of her data. The one score of 0 decreases the mean by a lot when the rest of her grades are 90 to 100.

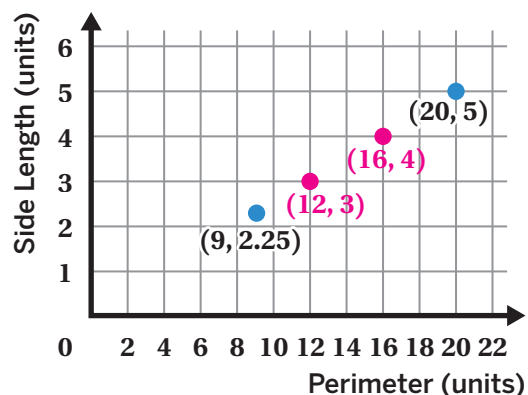
Problems 10–11: This graph shows the side length of a square and its perimeter.

10. Plot and label two more points on the graph.

Responses vary. Any points where the y -coordinate is $\frac{1}{4}$ of the x -coordinate.

11. Is there a proportional relationship between the perimeter and side length? Explain your thinking.

Yes. Explanations vary. The side length of a square is always one-quarter the length of its perimeter.



Reflection

1. Put a question mark next to a problem you're feeling unsure of.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

How can we use data from a sample to make claims about a population? One way is to use *proportional* reasoning.

Let's say someone is wondering how many students at their school might vote for a candidate for student council. It would be challenging and time consuming to ask all 500 students at the school, so they collect a sample of 25 students. It is important to gather the sample in a way that makes sure the sample is likely to be representative, like asking one student from each homeroom or asking 25 students at random.

If 10 out of the 25 students in the sample said they would vote for this candidate, there are several strategies for making a prediction about the population.

Strategy A

10 out of 25 is equal to $\frac{10}{25}$ or 40% of the sample.

40% of the population (500 students) would be $0.4 \cdot 500 = 200$ students.

Strategy B

The population is $\frac{500}{25} = 20$ times as large as the sample, so multiply the number of votes by 20 to determine the number of students in the population who would vote for them.

Votes	Total Students
10	25
200	500

Things to Remember:

Lesson Practice

ACC6.8.12

Name: _____ Date: _____ Period: _____

Problems 1–3: Faaria and Ariel wondered what percentage of students at school would dye their hair blue. They each surveyed a different random sample of the students at school.

1. Faaria asked 10 students, and 2 of them said they would. Based on Faaria's sample, what percent of the students would dye their hair blue?

20%

2. Ariel asked 100 students, and 17 of them said they would. Based on Ariel's sample, what percent of the students would dye their hair blue?

17%

3. Whose percentage is likely to be closest to the percentage of all the students? Explain how you know.

Ariel. Explanations vary. Ariel's sample is much larger than Faaria's. The larger the random sample, the closer it usually is to representing the actual population.

Problems 4–6: In a school of 580 students, one class was asked which hand they write with. "L" means they use their left hand, and "R" means they use their right hand.

Here are the results: L, R, R, R, R, R, R, R, R, L, R, R, R, R, R

4. Based on this sample, estimate the percentage of students at the school who write with their left hand.

About 13.3%

5. Estimate the number of students at the school who write with their left hand.

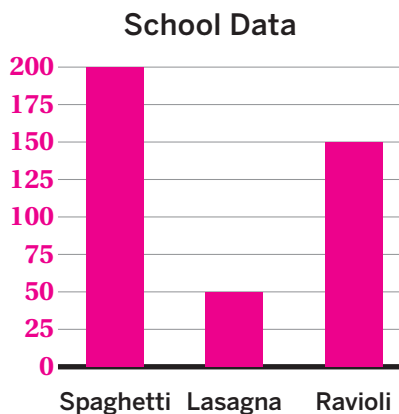
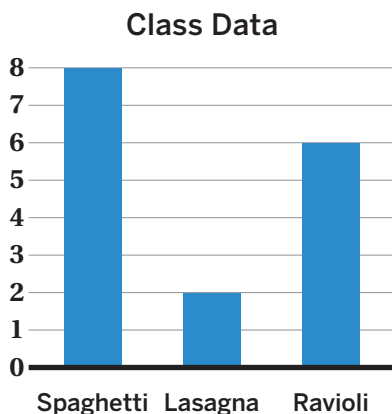
77 students

6. A different class of 18 students is surveyed. Estimate how many write with their left hand. Explain your thinking.

Responses and explanations vary. I estimate that 2 students write with their left hand because 13.3% of 18 is about 2.4 students and that number rounds down to 2.

7. 16 students at a school were asked about their favorite pasta dish. The results are shown in the bar graph. Create a new bar graph showing the possible results for all 400 students in the school. Be sure to scale and label the vertical axis.

Sample shown on graph.



Lesson Practice

ACC6.8.12

Name: _____ Date: _____ Period: _____

Spiral Review

8. Select *all* the expressions that are equivalent to 48.

☒ A. $3 \cdot 4^2$

☐ B. $6 + 6 \cdot 4$

☒ C. 50% of 96

☐ D. $(3 \cdot 4)^2$

☒ E. $4(6 + 6)$

9. What is the greatest common factor of 48 and 64?

16

10. What is the least common multiple of 4 and 6?

12

11. Complete the table so that each column has an equivalent fraction, decimal, and percent.

Fraction	$\frac{2}{5}$	$\frac{3}{4}$ (or equivalent)	$\frac{3}{10}$	$\frac{1}{8}$ (or equivalent)
Decimal	0.4	0.75	0.3	0.125
Percent	40%	75%	30%	12.5%

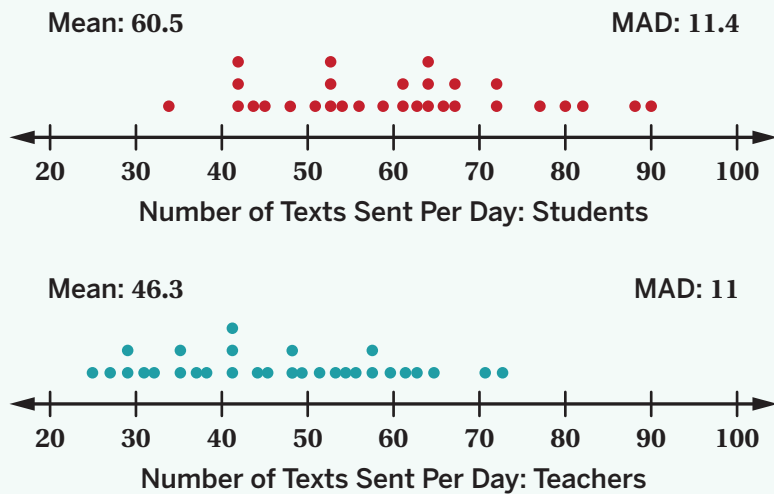
Reflection

- Put a star next to a problem that looked more difficult than it really was.
- Use this space to ask a question or share something you're proud of.

Lesson Summary

In general, it is easier to compare two individuals or objects than it is to compare two populations. For example, you can answer the question, “Which 7th grader is taller?” by measuring the heights of two 7th grade students and comparing them directly. However, to answer the question, “Do middle school students send more texts per day than their teachers?” you need to collect samples and analyze the measures of center and variability.

Here are the results of a random sample of 30 students and 30 teachers surveyed about the number of texts they send per day. To decide if the data sets are very different from each other, we can calculate the difference in their means and compare it to the larger MAD. The difference is $60.5 - 46.3 = 14.2$, which is about 1.25 times the MAD of 11.4. When the difference is more than 1 times the larger MAD, the data sets are very different. This suggests that students do send more texts than teachers.



Things to Remember:

Lesson Practice

ACC6.8.13

Name: _____ Date: _____ Period: _____

Problems 1–3: A school's art club held a fundraiser to raise money for art supplies. This data shows the number of T-shirts sold each week during each season.

1. Determine the mean number of T-shirts sold in the fall and in the spring.

Fall: 23 T-shirts; Spring: 24 T-shirts

2. Calculate how many MADs apart the means are. Use the larger MAD in your calculation.

0.36 times the larger MAD; $24 - 23 = 1$, so the difference in means is 1. $1 \div 2.8 \approx 0.36$.

3. Based on this data, are sales generally higher in the spring than in the fall? Explain your thinking.

No. Explanations vary. There isn't much of a difference between the spring and fall sales because 0.36 is less than 1.

Fall

20 26 25 24 29
20 19 19 24 24

MAD: 2.8

Spring

19 27 29 21 25
22 26 21 25 25

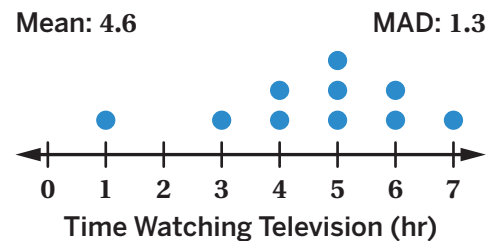
MAD: 2.6

Problems 4–5: Abdullah compared the time that students and teachers spent watching television over the weekend. Abdullah took a random sample of 10 students and 10 teachers and made a dot plot of their responses.

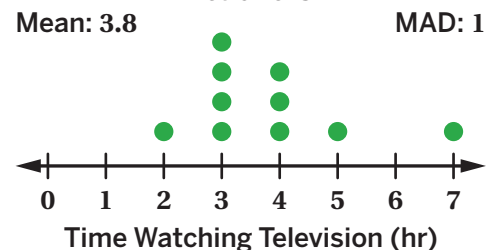
4. Is there a big difference between the students' data and the teachers' data? Explain your thinking.

No. Explanations vary. Teachers do watch less television than students, but the difference in their means is 0.8 hours, which is less than 1 MAD.

Students



Teachers



5. Abdullah then took a random sample of 10 parents and found that they watched a mean of 2.5 hours of television, with a MAD of 1 hour.

Is there a big difference between how much television parents and students watch? Explain your thinking.

Yes. Explanations vary. The difference in the means was 2.1 hours, which is way more than 1 MAD.

Lesson Practice

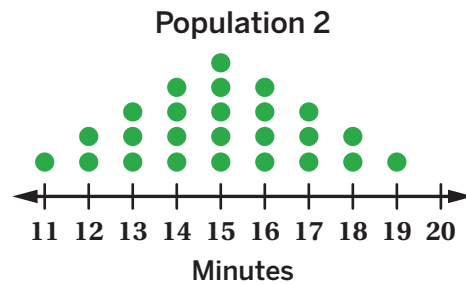
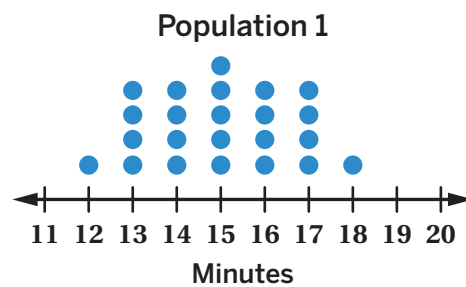
ACC6.8.13

Name: _____ Date: _____ Period: _____

6. These dot plots represent the time that it took new runners to run a mile.

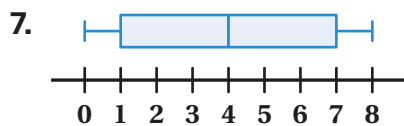
Which statement is *best* supported by the information in the dot plots?

- A. The two populations have an equal number of runners.
- B. The two populations have the same range.
- C. The two populations have different medians.
- D.** The two populations have the same mean.

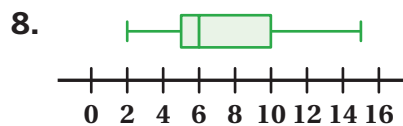


Spiral Review

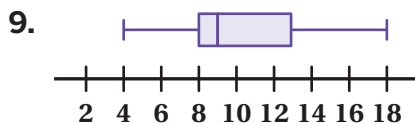
Problems 7–9: Determine the interquartile range (IQR) for each box plot.



6



5



5

Problems 10–12: Calculate each percentage.

10. 50% of 17

8.5

11. 150% of 17

25.5

12. 75% of 17

12.75

Reflection

- Star the problem you spent the most time on.
- Use this space to ask a question or share something you're proud of.

Lesson Summary

The **probability** of an **event** is a number that represents how likely the event is to occur. One way to calculate the probability is to look at all of the possible outcomes for an experiment, which is known as the **sample space**.

When all of the outcomes are equally likely, the probability of an event is a ratio.

$$\frac{\text{number of favorable outcomes}}{\text{total possible number of outcomes}}$$

Probabilities are numbers between 0 and 1 written as fractions, decimals, or *percentages*. A probability of 1 means the event will always happen. A probability of 0 means the event will never happen.

Here are some examples of events and their probabilities.

Example	Probability
Picking a green marble out of a bag that contains only red and yellow marbles	0
Rolling a 1 on a number cube	$\frac{1}{6}$ (or equivalent)
Flipping a coin and it landing heads up	50% (or equivalent)
Picking a yellow marble from a bag of 10 marbles, where 8 of the marbles are yellow	0.8 (or equivalent)
Picking a green marble in a bag that only contains green marbles	1

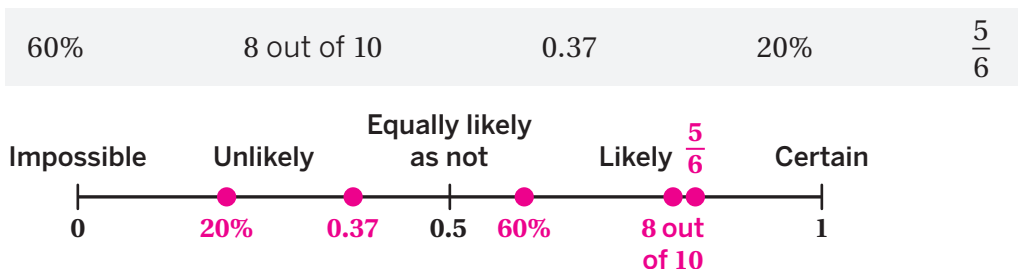
Things to Remember:

Lesson Practice

ACC6.8.15

Name: _____ Date: _____ Period: _____

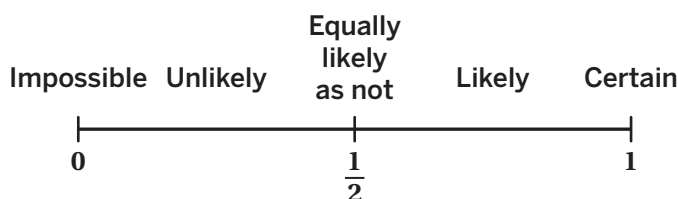
- Place a point on the line to show each probability's likelihood from impossible to certain.



Problems 2–4: List the sample space for each experiment. For example, the sample space of flipping one fair coin is heads or tails.

- Selecting a random season of the year
Winter, spring, summer, fall
- Selecting a random day of the week
Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday
- Rolling a number cube
One, two, three, four, five, six

Problems 5–10: Determine the probability of each event. Use the line if it helps with your thinking.



- You read this sentence.
1
- A fair coin lands heads up when flipped.
 $\frac{1}{2}$ (or equivalent)
- A weekend day is selected randomly from the days of the week.
 $\frac{2}{7}$ (or equivalent)
- You see a live unicorn outside today.
0
- A spinner with equal parts red, yellow, and green lands on green on the next spin.
 $\frac{1}{3}$ (or equivalent)
- You pick a green block when you randomly pick one block from a bag with 7 green blocks and 3 blue blocks.
 $\frac{7}{10}$ (or equivalent)

Lesson Practice

ACC6.8.15

Name: _____ Date: _____ Period: _____

Problems 11–13: A computer randomly selects a letter from the English alphabet.

11. How many different outcomes are possible?

26 outcomes

12. What is the probability the computer selects a vowel (A, E, I, O, or U)?

$\frac{5}{26}$

13. What is the probability the computer selects the first letter of your first name?

$\frac{1}{26}$

14. This table shows the number of bags of different flavors of crackers in a large box. A bag will be selected at random from the box.

Flavor of Cracker	Number of Bags
Plain	12
Barbeque	16
Ranch	8
Cheddar Cheese	20

Which statement about the flavor of the crackers chosen is best supported by the information in the table?

- A. Plain is the least likely flavor.
- B.** Choosing barbecue is twice as likely as choosing ranch.
- C. There is an equal chance of choosing plain, barbeque, ranch, or cheddar cheese.
- D. Choosing cheddar cheese is twice as likely as choosing ranch.

Spiral Review

Problems 15–16: e represents an object's weight on Earth and m represents that same object's weight on the Moon. The equation $m = \frac{1}{6}e$ represents the relationship between these quantities.

15. What does $\frac{1}{6}$ represent in this situation?

Responses vary. $\frac{1}{6}$ is the constant of proportionality from an object's weight on Earth to its weight on the Moon. Something that weighs 1 pound on Earth weighs $\frac{1}{6}$ pounds on the Moon.

16. If a person weighs 24 pounds on the Moon, how many pounds would they weigh on Earth?

144 pounds

Reflection

1. Star a problem you're still feeling confused about.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

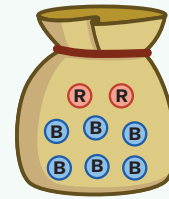
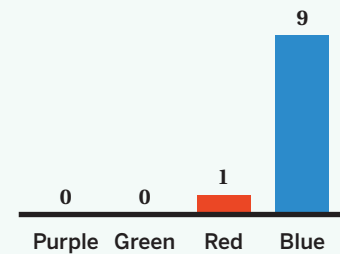
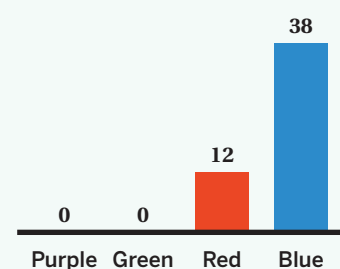
In situations where you don't know the sample space, you can use data from experiments and proportional reasoning to predict what the sample space is. Repeating an experiment more times can help make your prediction more accurate.

For example, here are the results from picking a marble out of a mystery bag 10 times. The bag has 8 marbles in it.

Only 1 out of the 10 marbles was red, so the *constant of proportionality* is 0.1. Multiplying 0.1 times the number of marbles in the bag (8), may lead someone to predict that there is only $0.1 \cdot 8 = 0.8$ or 1 red marble in the bag.

After 50 picks, the constant of proportionality (0.24) times the number of marbles in the bag is $0.24 \cdot 8 = 1.92$, which is close to 2.

This is a more accurate prediction of the number of red marbles in the bag.

**Total Picks: 10****Total Picks: 50****Things to Remember:**

Lesson Practice

ACC6.8.16

Name: _____ Date: _____ Period: _____

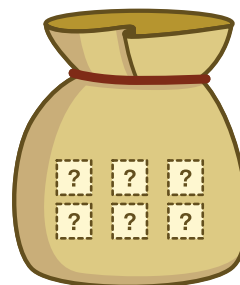
Problems 1–2: A month is chosen at random from the months of the year.

1. What is the probability of getting a month that starts with the letter “A?” Consider listing the sample space to show your thinking.
 $\frac{1}{6}$ (or equivalent)
2. If you repeat this experiment 600 times, how many times do you expect to get a month that starts with “A?”

About 100 times

3. A bag has 6 blocks in it. Joel picks a block out of the bag 60 times and gets a green block 43 times. Based on these results, how many blocks do you expect to be green? Explain your thinking.

4 blocks. *Explanations vary.* Joel picked a green block about 72% of the time because $\frac{43}{60} \approx 0.7167$. If the proportion of green blocks in the bag matched this, there would be 4.3 green blocks. Because 4.3 is closer to 4 than 5, it's likely that there are 4 green blocks in the bag.

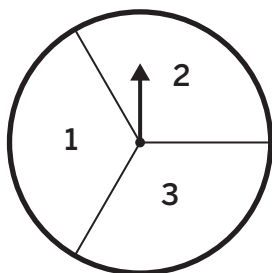


4. Miko spun a spinner with numbered sections 15 times. Here are the results. Which spinner is most likely the one Miko used?

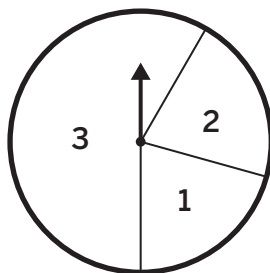
Miko's results:

3, 3, 3, 1, 3, 2, 2, 3,
3, 1, 3, 1, 3, 3, 2

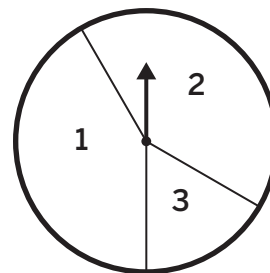
A.



B.



C.



Explain your thinking.

Explanations vary. Miko spun a 3 nine times, which is three times as many times that Miko spun a 1 or a 2. Because Spinner B has a large section for 3 and smaller equal sections for 1 and 2, this spinner seems to match Miko's results more than the other two spinners.

Problems 5–7: A textbook has 428 numbered pages, starting with 1. You are equally likely to stop on any of the pages if you flip through the book randomly.

5. What is the probability that you turn to page 45?
 $\frac{1}{428}$ (or equivalent)
6. What is the probability that you turn to an even-numbered page?
 $\frac{1}{2}$ (or equivalent)
7. If you repeat this experiment 50 times, about how many times do you expect to turn to an even-numbered page?

About 25 times

Lesson Practice

ACC6.8.16

Name: _____ Date: _____ Period: _____

8. The table shows the number of different colors of crayons in a case. A 7th grader will randomly choose one crayon from the case.

Based on the information in the table, which statement is true?

Color	Number of Crayons
Blue	2
Red	8
Orange	4
Yellow	5

- A. The crayon is least likely to be orange.
- B.** The crayon is 4 times as likely to be red as it is to be blue.
- C. The crayon is equally likely to be orange or yellow.
- D. The crayon is more likely to be red than all other colors combined.

Spiral Review

Problems 9–11: For each word, determine the probability of selecting the letter “A” at random.

9. LAMB

$\frac{1}{4}$ (or equivalent)

10. SAFETY

$\frac{1}{6}$ (or equivalent)

11. ALABAMA

$\frac{4}{7}$ (or equivalent)

12. Select *all* the situations in which the surface area would be useful to know.

- ☒ A. Ordering tiles to replace the roof of a house
- ☒ B. Estimating how long it will take to clean the windows of a greenhouse
- ☐ C. Deciding whether leftover soup will fit in a container
- ☐ D. Estimating how long it will take to fill a swimming pool with a garden hose
- ☒ E. Buying fabric to sew a couch cover

Reflection

1. Circle a problem you are still curious about.
2. Use this space to ask a question or share something you're proud of.

Lesson Summary

Repeated experiments can help you decide if an object is fair.

If an experiment is repeated only a few times, the results may not be what you expect, even if the object is fair. The more times you repeat the experiment (i.e., hundreds or thousands of times), the closer the relative frequency should get to the probability. This allows you to make a better decision about whether the object is fair.

For example, here is a fair coin. The probability of this coin landing heads up is $\frac{1}{2}$.



If you flip the coin only 3 times, it may land heads up all 3 times.

You may think the coin is unfair, but continuing to repeat the experiment can change your perspective.

If you flipped the coin 1,000 times, it would land heads up about half of the time because the sample space of this event is “heads” and “tails.”

Things to Remember:

Lesson Practice

ACC6.8.17

Name: _____ Date: _____ Period: _____

Problems 1–3: Deja has a six-sided number cube.

1. If this were a standard number cube, what would be the probability that the cube lands on a five?
 $\frac{1}{6}$ (or equivalent)
2. Deja suspects the six-sided number cube is not standard.
 - Deja rolled a five 40 times out of 100.
 - Amor rolled a five 21 times out of 50.
 - Santino rolled a five 11 times out of 30.

Based on these results, estimate the probability of rolling a five.

About 40% (or equivalent)

3. Is it likely this is a standard number cube? Explain your thinking.

No. *Explanations vary.* A standard number cube should land on five $\frac{1}{6}$ of the time. After 100 rolls, it should land on five about 16 or 17 times. All three people had it land on five more than twice as often. With this many rolls, there is strong evidence that this cube is not standard.

Problems 4–5: Santino flips a coin 10 times to see if it is fair. It lands heads up 3 times and tails up 7 times.

4. Are these results enough to determine if the coin is fair? Explain your thinking.

No. *Explanations vary.* The relative frequency from experiments is usually a good approximation for the probability of an event, but only when the experiment is done many, many times. 10 flips may not be enough for the relative frequency to be close to the $\frac{1}{2}$ probability.

5. What could Santino do to be more sure about these results?

Santino should flip the coin many more times.

Problems 6–9: A game is played with two tetrahedral (4-sided) dice. The dice are in the shape of a pyramid with the numbers 1, 2, 3, and 4 written on each triangular side. After the dice are rolled, the numbers on the sides facing down are added together.

6. What are all of the possible sums for rolling two tetrahedral dice?
2, 3, 4, 5, 6, 7, 8
7. What sum do you think you will be most likely to get when rolling these dice? Least likely? Explain your thinking.

A sum of 5 is most likely and a sum of 2 or 8 are least likely. *Explanations vary.* I think rolling a sum of 5 will be the most likely because there are lot of ways to get 5 : 1 and 4, 2 and 3, 3 and 2, and 4 and 1. I think rolling a sum of 2 or 8 will be the least likely because there is only one way to roll each of these: 1 and 1 and 4 and 4.

8. Create rules for a fair two-player game that uses these dice. *Responses vary.*

Player A wins if... the sum of the dice is 4.

Player B wins if... the sum of the dice is 6.

Lesson Practice

ACC6.8.17

Name: _____ Date: _____ Period: _____

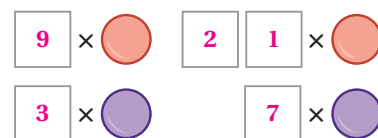
9. Explain why you think your game in Problem 8 is fair.

Responses vary. I think my game is fair because rolling a sum of 4 or a sum of 6 have the same likelihood of occurring. There are three ways to roll a sum of 4: 1 + 3, 2 + 2, and 3 + 1. There are three ways to roll a sum of 6: 2 + 4, 3 + 3, and 4 + 2. So, I assigned them the same number of points.

10. A random number generator selects a digit from 1 to 5. Troy uses the generator 1,500 times. Which statement best predicts how many times the digit 3 will appear among the 1,500 results?
- A. It will appear exactly 300 times. **B.** It will appear close to 300 times but probably not exactly 300 times.
- C. It will appear exactly 340 times. D. It will appear close to 340 times but probably not exactly 340 times.

11. Fill in each blank using the digits 0 to 9 only once so that the probability of drawing a purple marble is the same for each bag.

Responses vary.



Spiral Review

Problems 12–14: There is a proportional relationship between sugar measured in cups and sugar measured in tablespoons. For example, 48 tablespoons of sugar is equivalent to 3 cups of sugar.

12. Plot and list the coordinates of two more points that represent the relationship.

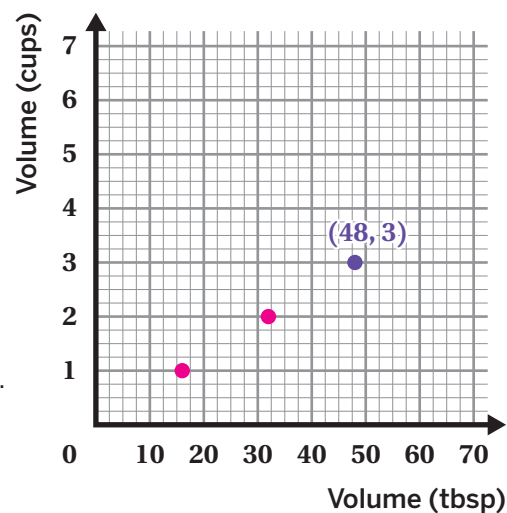
Points where the y -coordinate is $\frac{1}{16}$ of the x -coordinate are considered correct.

13. What is the constant of proportionality for this relationship?

$\frac{1}{16}$ or 16

14. Write an equation representing this relationship. Use c for cups and t for tablespoons.

$c = \frac{1}{16}t$ or $t = 16c$



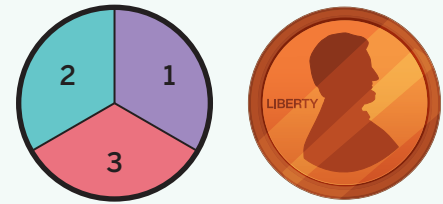
Reflection

- Put a question mark next to a problem you were feeling stuck on.
- Use this space to ask a question or share something you're proud of.

Lesson Summary

There are several different ways to make sense of **compound events**, or events that involve multiple steps.

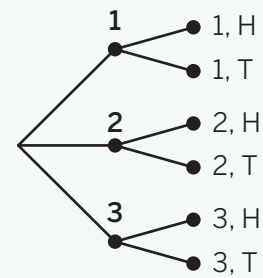
Here is one example: Let's spin a spinner and flip a fair coin. There are 6 outcomes in the *sample space* of this multistep event, which you can see in a list, a table, and a **tree diagram**.

**List**

1, Heads	1, Tails
2, Heads	2, Tails
3, Heads	3, Tails

Table

	H	T
1	1, H	1, T
2	2, H	2, T
3	3, H	3, T

Tree Diagram

Things to Remember:

Lesson Practice

ACC6.8.18

Name: _____ Date: _____ Period: _____

Problems 1–5: Farah made a tree diagram to help her choose a theme, location, and day of the week for her birthday party.

1. How many themes is Farah considering?

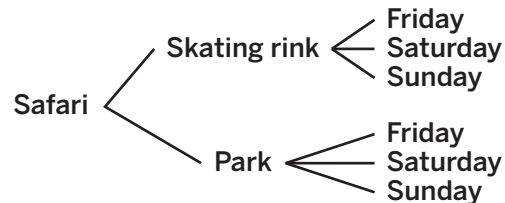
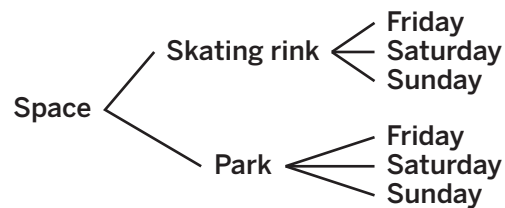
2 themes

2. How many locations is Farah considering?

2 locations

3. How many days is Farah considering?

3 days



4. One party Farah is considering is a space theme at the skating rink on Sunday. Write two other possible parties Farah could have.

Responses vary. Farah could have a safari-themed party at the skating rink on Friday, and a space-themed party at the park on Saturday.

5. If Farah lets her sibling choose a theme, location, and day at random, what is the probability that Farah's birthday will be a safari at the park on Saturday? Explain your thinking.

$\frac{1}{12}$. Explanations vary. There are 12 possible parties in the sample space.

Problems 6–7: Isabella selects one type of lettuce and one dressing to make a salad.

- Lettuce types: iceberg, romaine
- Dressings: ranch, Italian, Caesar

6. Create a list, table, or tree diagram to represent all the possible combinations of choices. Then determine the number of possible outcomes.

Representations vary. 6 possible outcomes: iceberg and ranch, iceberg and Italian, iceberg and Caesar, romaine and ranch, romaine and Italian, romaine and Caesar

7. If Isabella picks a type of lettuce and dressing at random, is getting iceberg lettuce or Caesar dressing more likely? Explain your thinking.

Iceberg lettuce. Explanations vary. The probability of choosing iceberg lettuce is $\frac{1}{2}$ and the probability of picking Caesar dressing is $\frac{1}{3}$.

Lesson Practice

ACC6.8.18

Name: _____ Date: _____ Period: _____

8. Twelve students play a game. Each student is assigned a number 1 to 12. Someone rolls two six-sided number cubes. The student whose number matches the sum gets a point. After 100 rounds, which student is most likely to have the most points? Explain your thinking.

Player 7. Explanations vary. There are 36 possible outcomes that can be rolled from the sum of two six-sided number cubes and 6 possible ways to roll a 7, which is more than any other sum. There are 5 possible ways to roll a 6 or an 8, so with fewer rounds, the students assigned 6 or 8 may come close, but over more rounds, it is more likely that the person assigned 7 would win.

Problems 9–10: Juan and Neo play a game. Each player holds out a hand at the same moment to represent a rock, paper, or scissors. Then they each choose one of the three items.

9. Create a list, table, or tree diagram to represent all the possible combinations of choices.

Representations vary. There are 9 possible outcomes: rr, rp, rs, pr, pp, ps, sr, sp, ss.

10. What is the probability that they both choose the same object?
 $\frac{3}{9}$ (or equivalent)

Spiral Review

11. Ayaan competed in a bowling tournament. Here are his scores for each game.

Game 1	Game 2	Game 3	Game 4	Game 5	Game 6
300	250	275	120	243	290

Do you think the tournament judges should use the median or the mean to calculate Ayaan's final score? Justify your choice.

Responses vary.

- The judges should use the median because it's closer to most of Ayaan's other scores in the tournament.
- The judges should use the mean because the median would ignore Ayaan's worst game and that wouldn't be fair to the other players.

12. Select *all* the true equations.

☐ A. $8 = (8 + 8 + 8 + 8) \div 3$

☒ B. $(10 + 10 + 10 + 10 + 10) \div 5 = 10$

☒ C. $(6 + 4 + 6 + 4 + 6 + 4) \div 6 = 5$

☐ D. $4 = (4 + 2 + 4 + 2) \div 4$

☐ E. $(2 + 2 + 2 + 2 + 2) \div 2 = 2$

Reflection

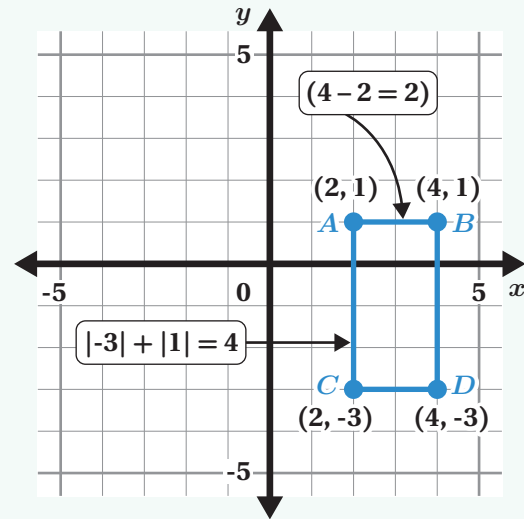
- Circle the problem you feel least confident about.
- Use this space to ask a question or share something you're proud of.

Lesson Summary

You can plot points on the coordinate plane to create *polygons*. When the vertices of a polygon are horizontally or vertically aligned on the graph, you can count the number of units between them to determine the length of that side.

You can also calculate the side lengths using the coordinates of each vertex. Here are two calculation strategies:

- If the coordinates are in the same quadrant, like points A and B , find the length by subtracting the coordinates that are different.
- If the coordinates are in different quadrants, like points A and C , use the absolute value to determine the distance each point is from the axis between them.



Things to Remember:

Lesson Practice

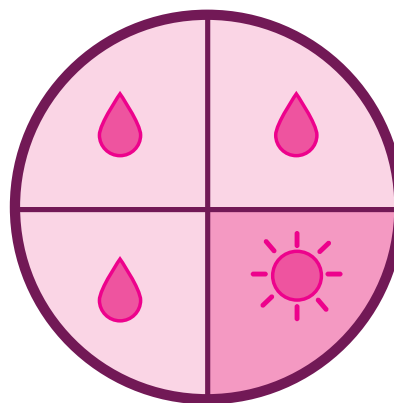
ACC6.8.19

Name: _____ Date: _____ Period: _____

Problems 1–3: The weather forecast says there is a 75% chance it will rain today.

1. Design a spinner you could use to simulate a 75% chance of rain.
2. Explain why using a number cube to simulate this probability may be less useful than using a spinner.

Responses vary. 75% corresponds to a probability of $\frac{3}{4}$. Number cubes have 6 possible outcomes making it hard to separate into fourths.



3. Describe or draw a different way you could simulate this probability.

Responses vary. I could put 4 blocks in a bag (3 to represent rain and 1 to represent sun) and then randomly pick a block from the bag.

4. Paz has 3 kittens. According to the vet, each kitten is born with blue eyes and there is a 50% chance they will change color once the kittens reach three months. Paz designs a simulation using 3 coins, where heads represent the eyes changing color.

The table shows the results of 100 experiments.

Estimate the probability that at least one of Paz's kittens will still have blue eyes at three months old. Explain your thinking.

$\frac{89}{100}$ (or equivalent). Explanations vary. Out of 100 experiments, 89 resulted in at least one kitten with blue eyes.

Experiments with:	Count
No blue-eyed kittens	11
1 blue-eyed kitten	32
2 blue-eyed kittens	43
3 blue-eyed kittens	14

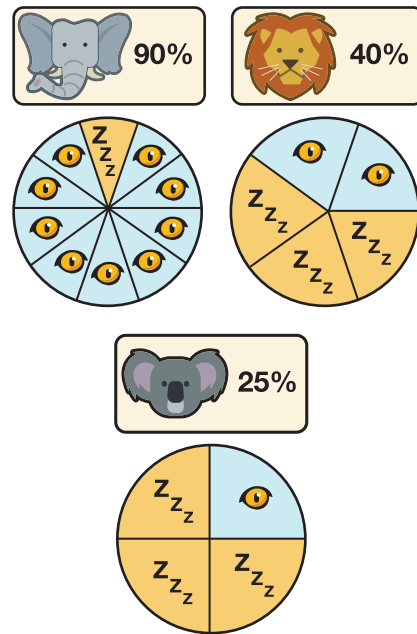
Lesson Practice

ACC6.8.19

Name: _____ Date: _____ Period: _____

Problems 5–6: Brianna designed and used a simulation to help her estimate the probability of seeing her three favorite animals awake when she visits the zoo. She records the results of 300 experiments.

Experiments with:	Count	Relative Frequency
No animals awake	12	4%
1 animal awake	171	57%
2 animals awake	105	35%
3 animals awake	12	4%



- Estimate the probability that all 3 of her favorite animals will be awake when she visits the zoo.
4%
- Estimate the probability that at least 2 of her favorite animals will be awake when she visits the zoo.
39%

Spiral Review

- The instructions for making pink paint say to mix 3 parts red paint with 2 parts white paint. Complete the table to show how much red and white paint you would need to make different amounts of pink paint.

Pink Paint (gal)	Red Paint (gal)	White Paint (gal)
50	30	20
15	9	6

Problems 8–12: Write each fraction as a percent.

8. $\frac{3}{5}$
60%

9. $\frac{1}{50}$
2%

10. $\frac{9}{10}$
90%

11. $\frac{9}{5}$
180%

12. $\frac{18}{60}$
30%

Reflection

- Put a smiley face next to the problem you learned from most.
- Use this space to ask a question or share something you're proud of.